Summary Results, v2

2026

1 Setup

1.1 Suite and pools

- 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Each adapter reads the official test or dev split and emits items with a question, a context (table, document, or record pair), a gold answer with aliases, a list gold where the answer is a set, an answer type (numeric / boolean / text / free-form) and the dataset's native metric.
- Pools: 150 items per dataset by seeded stratified sampling (strata: question type, sub-task, database, product category, match label); 100 flagged for answering (OfficeQA Pro V2: all 90; Alaska schema: 93). The remaining 50 are a seeded reserve and are not used by any step. Total answered items: 3,083 per model.
- Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % match ratio). Schema matching is framed as one question per source column with a candidate list that always contains the gold. BIRD is graded on the executed result of the gold SQL, not on SQL text.
- Official census: every dataset's official population is counted by the same strata, so that pool numbers can be re-weighted to the official split (Section 1.5).

1.2 Collecting answers

- One prompt per item: context, question, and an answer-format line requesting `FINAL ANSWER: <answer>`.
- Models (eleven): Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.1 Pro, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b (Together), Llama 3.3 70B (Together). Gemini 3.1 Pro was collected over three weeks under Google's daily quota (a scheduled GitHub Actions workflow plus a final local run) and entered the study once it passed the 90 % coverage rule.
- Decoding: temperature 0 where the API accepts it (Gemini, Grok, DeepSeek, Llama, gpt-oss). Claude 5 reports the parameter deprecated and GPT-5.5 rejects any non-default value; those ran at provider defaults. Every cached row records which applied.
- Output budget 700 tokens. A reply cut off for length with no final-answer line is re-asked at 2,500 tokens, then 8,000, then 16,000, then 32,000. The trigger is mechanical (finish reason plus absence of the final line), never correctness. Originals are archived with a reason; replacement rows carry `attempt` = 2 to 5. The budget is counted in each provider's own tokens and, for the reasoning models, includes their chain of thought, so it is a termination test rather than an equal text allowance. Over the suite 1,080 answers were re-asked at 2,500, 279 at 8,000, 118 at 16,000

and 124 at 32,000; every model except DeepSeek-R1 finished every answer, and 55 DeepSeek-R1 answers (35 OfficeQA Pro V2, 19 OfficeQA, 1 BIRD) do not terminate at 32,000 tokens and are scored as given (effectively failures).
- Every answer is cached with tokens, latency, cost, finish reason and decoding; re-runs skip cached items. Cost: USD 178 for the answers in the final cache (Claude Opus 5: USD 107 including its re-asks; Gemini 3.1 Pro: USD 12; DeepSeek-R1: USD 19 including four re-ask stages), USD 235 over the whole project including the archived truncated originals.

1.3 Scoring an answer

- Extraction: the `FINAL ANSWER:` line; else a boxed expression; else the last short line of the reply.
- Normalisation: Unicode NFKC, lower-case, canonical numbers ("1,577.00" → "1577", "(12.6)" → "−12.6"), currency symbols and punctuation removed, articles dropped, whitespace collapsed.
- Units and scale: every number is parsed with sign, percent marker (%, "percent", "pp") and scale word (thousand / million / billion / trillion and abbreviations). A parsed number exposes its equivalent readings (raw, scaled, ÷100 if marked percent, ×100 if an unmarked fraction); the relative error between prediction and gold is the minimum over all pairs of readings. Real-unit conversions (km ↔ miles) are not attempted.
- Aliases: each item carries the primary gold plus acceptable alternatives supplied by the adapter (e.g. *yes* ≡ *match* ≡ 1; several legitimate target columns in schema matching; comma- and pipe-joined forms of list golds). Every component is computed against each alias and the best-scoring alias is kept. Booleans are canonicalised before comparison.
- Components per answer: token F1, numeric decay $e^{-2.5 \cdot \text{rel. err}}$ (error clipped at 2; falls back to exact match for non-numeric golds), token precision, token recall; also exact match, numeric tolerance (1 %), ROUGE-L, and a hedge flag (several distinct numeric candidates, "A or B", or an enumeration of short candidates).
- Native metric per dataset (baseline column): numeric tolerance for numeric golds, exact match for text and boolean golds, ROUGE-L for free-form (FeTaQA).
- Composite under weighting w on the simplex: $R_w = \sum_k w_k m_k$, $k \in \{\text{F1, decay, precision, recall}\}$. v1 is $w = (0.35, 0.35, 0.15, 0.15)$.
- Dataset score of a model: mean R_w over the items that every model answered. A model is in the study only if it covered ≥ 90 % of the suite; a dataset is used only with ≥ 20 answered items.

1.4 Evaluating a weighting

1. **Agreement $J(w)$.** Rank the eleven models inside each dataset under w; pool the 31 rankings by Borda; J is the mean Kendall τ_b between each dataset's ranking and the pooled one. J measures consistency between datasets, not correctness. Computed for all 1,771 weightings of the step-0.05 simplex lattice (v1, the four corners and the uniform mix are lattice points), and also under Kemeny, Copeland and Bradley–Terry pooling.
2. **Correctness $\rho(w)$.** 31,380 synthetic answers of known utility generated from the golds of the answered items by typed operators (identity, alias, unit variant, typo, rounding, magnitude error, sign flip, wrong value, partial list, superset, hedge, verbose, no answer, ...). Reported: Spearman ρ between R_w and utility, AUC correct vs wrong, mean score of wrong answers, hedge gap; the (J, ρ) Pareto front and its knee.
3. **Honest selection.** Nested item bootstrap (B = 1,000): choose the best w on in-bag items, score it out of bag. Reported: the optimism gap, the paired out-of-bag difference to v1 with a 95 % interval, and the plateau (weightings not significantly below the best, paired bootstrap

$p > .05$) plus a practical plateau (within 0.02 of the best).

4. **Transfer.** Leave-one-dataset-out and leave-one-family-out: re-fit w on the rest, measure τ of the held-out dataset's ranking to the pooled rest; per-family optima; sensitivity to one lattice step (0.05).
5. **Designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples of 100 weightings, 20 seeds each, compared with the exhaustive optimum (regret, rank of the found best, centred L_2 discrepancy, minimum pairwise L_1).

1.5 From scores to a leaderboard

- Rules reported: exact match only, v1, the five best by J, the best by ρ, the (J, ρ) Pareto knee.
- Pooling: Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, standard errors from the observed information) as the headline; Borda, mean-z, Kemeny (with cost), Copeland, reciprocal-rank fusion and family-balanced z as checks.
- Uncertainty: 1,000 item bootstraps of the Borda and Bradley–Terry rankings (rank intervals, probability of first place), saved in full.
- Significance: Friedman with the Iman–Davenport correction across datasets, Nemenyi critical difference, Holm-corrected pairwise Wilcoxon; paired permutation and Wilcoxon per dataset and model pair (Holm within dataset); Kendall's W among datasets; majority-class baselines for boolean sets.
- Official-split estimates: the same answers re-weighted by official stratum share / sample share (post-stratification) with a weighted item bootstrap; positive-class precision, recall and F1 at the official match ratio for the matching sets.
- Diagnostics: pairwise τ between dataset rankings; floor, ceiling and self-stability flags per dataset.
- Reporting: one script regenerates the report from the cached answers in minutes, so text and numbers cannot disagree.

2 Results

All numbers are from the run of 2026-10-25 (31 datasets × 11 models, 3,083 answered items per model, truncation protocol complete through 32,000 tokens). Table 1 gives the audit in one place; the subsections that follow give one figure or table per row.

2.1 The objective is flat

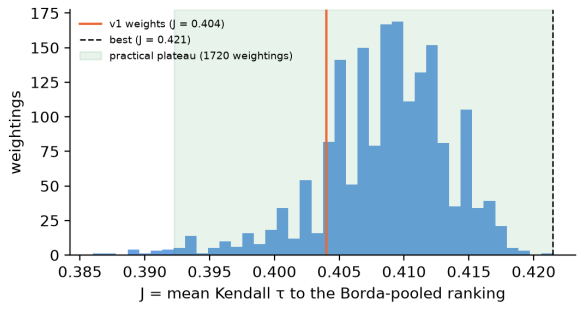
In one line: Every weighting of the four ingredients gives nearly the same agreement; v1's low rank is a rank inside noise.

Figure 1 shows J for all 1,771 weightings. The whole lattice spans 0.392–0.425. The best weighting (F1 0.35, decay 0.60, precision 0.05) reaches 0.425 and v1 reaches 0.411; 1,128 weightings are nominally better than v1, but the entire spread is only slightly larger than the bootstrap noise of a single weighting (about 0.026). No single weight has a slope worth acting on (Figure 2); the four ingredients correlate at 0.92–0.99 with one another and in 82 % of answers all four equal exact match, which is the mechanical reason.

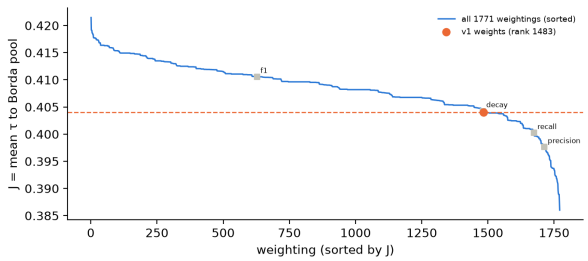
Notable. (a) The histogram is narrow: 1,771 weightings fit between 0.392 and 0.425, and v1 (0.411) sits in the main body, just below the median. (b) The sorted curve is flat for 1,700 weightings and only drops at the precision-dominated corner; v1's rank of 1,130 is a position on a plateau, which is why the report never quotes rank as a result.

Table 1: Audit summary. J = agreement objective (mean Kendall τ_b of each dataset's model ranking to the Borda pool); ρ = Spearman correlation of the composite with known utility on 31,380 synthetic anchors.

Exhaustive grid	1,771 weightings (step 0.05); best F1 .35 / decay .60 / precision .05 / recall 0, $J = 0.425$; v1 $J = 0.411$ (rank 1,130 of 1,771)
J over the lattice	range 0.392–0.425; median 0.413, 90th percentile 0.418, 99th percentile 0.422: a flat landscape
Plateau (paired bootstrap $p > .05$ vs best)	1,754 of 1,771 (v1 inside, $p = 0.66$); practical plateau within 0.02: 1,728
Nested selection ($B = 1{,}000$)	in-bag J of the selected weighting 0.379, out-of-bag 0.332 (optimism gap 0.046); v1 out-of-bag 0.339
Selected vs v1, out of bag	-0.006, 95 % CI $[-0.032, +0.011]$, $p = 0.71$; 617 distinct weightings selected across resamples
Sampled designs (100 points, 20 seeds)	Dirichlet / LHS / uniform all reach $0.422–0.423 \pm 0.001$, regret ≤ 0.004: sampling would have sufficed
Objective under other pooling rules	Kemeny and Copeland pick different "best" weightings from Borda and Bradley–Terry; v1's rank swings from 319 (Bradley–Terry) to 1,163 (Kemeny): rank is noise
Transfer (leave-one-dataset-out)	held-out τ: re-fitted weights +0.36, full best +0.40, v1 +0.38
Leave-one-family-out	v1 within 0.08 of the held-out-fitted weights in every family (EM +0.32 vs +0.33, SM +0.28 vs +0.36, TQA +0.36 vs +0.37, FIN +0.56 vs +0.50)
Anchors (31,380 synthetic answers)	v1 $\rho = 0.878$, best-ρ weighting 0.883, best-J weighting 0.876, exact match 0.828; AUC correct-vs-wrong 0.994 for every rule
J vs ρ across the lattice	-0.14 (was -0.23 before the final re-asks, -0.34 at ten models and +0.38 on the 2,500-token cache): unstable because J is flat; Pareto front of 17, knee = best J
Sensitivity	one 0.05 step moves J by at most 0.012 around v1 and 0.012 around the best
Leaderboard	Friedman $p < 0.001$ under every rule; Kendall's W among datasets 0.26–0.28
Significant pairs (Holm, v1)	15 of 55: nine models > Llama 3.3 70B; Opus, GPT-5.5, Gemini 3.1 Pro, Gemini 3.8 Flash, DeepSeek-R1 and Grok 4 > gpt-oss-120b
First place	Bradley–Terry: Gemini 3.8 Flash under v1 and every calibrated rule (P(first) 0.92–0.97), Claude Opus 5 under exact match (0.78); Borda: GPT-5.5 under v1 and the calibrated rules, Opus under exact match
Gemini 3.1 Pro and DeepSeek-R1	Gemini 3.1 Pro enters sixth by Bradley–Terry under v1 (fifth under exact match); DeepSeek-R1, after its 16,000- and 32,000-token re-asks, is fourth under v1 (third under two calibrated rules)
Official-split comparability	accuracy moves by ≤ 0.09 after post-stratification (OpenSanctions); positive-class F1 on the balanced pools overstates official-ratio F1 by up to 0.19 (Amazon–Google)
Datasets at ceiling	Valentine, WDC LSPC, WDC Products (means 0.98): flagged; still inside the objective
Cost	USD 178 for the final cache (Opus 107, DeepSeek-R1 19); USD 235 over the project including archived truncated originals



(a) Distribution of J over the lattice with v1, the best and the plateau band.



(b) Sorted J with v1 and the four single-ingredient corners.

Figure 1: The agreement landscape over all 1,771 step-0.05 weightings.

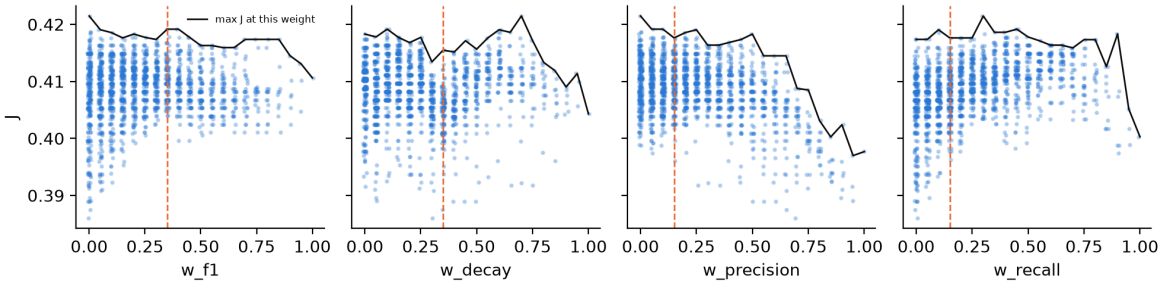


Figure 2: J against each of the four weights. Corners: F1 only 0.412, decay only 0.415, precision only 0.393, recall only 0.395; uniform 0.418.

Notable. No panel has a slope worth acting on: the mean line moves by at most 0.021 across the full range of any weight (precision, falling as it approaches 1). The vertical scatter at every weight level, caused by the other three weights, is wider than any slope, so no ingredient can be tuned in isolation.

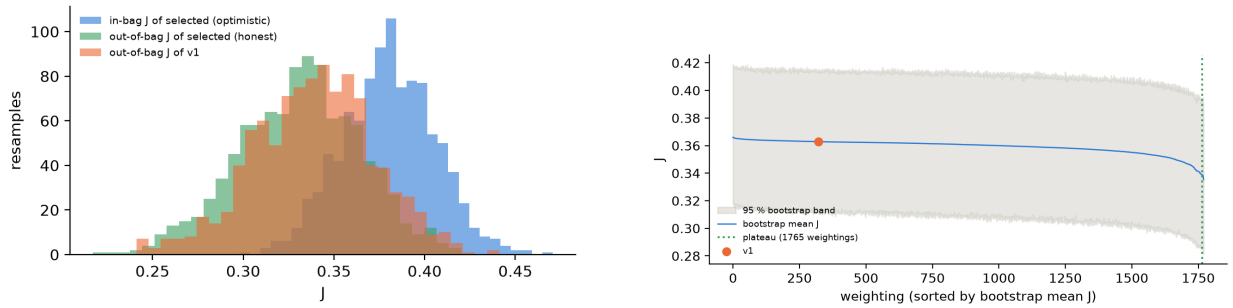
2.2 The best weighting does not survive honest selection

In one line: Chosen honestly, the best weighting is no better than v1; 617 different weightings win depending on which items are drawn.

Figure 3 reports the nested item bootstrap. The weighting chosen on in-bag items scores 0.379 in bag but 0.332 out of bag (optimism gap 0.046), below v1's out-of-bag 0.339. The paired out-of-bag difference to v1 is -0.006 with a 95 % interval of $[-0.032, +0.011]$ ($p = 0.71$); the selected weighting is ahead of v1 in 29 % of resamples, and 617 distinct weightings were selected across the 1,000 resamples. The plateau of weightings not significantly below the best contains 1,754 of 1,771 points and v1 is inside it ($p = 0.66$).

Notable. (a) The in-bag winner looks 0.046 better than it is; out of bag it scores 0.332 against v1's 0.339, and the paired difference of -0.006 has a 95 % interval through zero. (b) 1,754 of 1,771 weightings are inside the plateau and v1 is one of them ($p = 0.66$); the 17 outside are precision-heavy corners with no recall and almost no F1.

Notable. All three 100-point designs reach 0.422–0.423 with a worst-seed regret of 0.004 against the exhaustive 0.425. With four weights the exhaustive grid costs nothing, so the table matters only as evidence that sampling would have sufficed in the larger sibling searches.



(a) In-bag vs out-of-bag J of the selected weighting against v1.

(b) Bootstrap band over sorted weightings; the plateau boundary.

Figure 3: Honest selection: the optimism gap and the plateau.

Table 2: Sampled designs at a budget of 100 weightings, 20 seeds each, against the exhaustive optimum.

design	best J (mean)	sd	regret mean	regret max
dirichlet_lattice	0.4225	0.0006	0.0024	0.0037
lhs_simplex	0.4229	0.0011	0.0020	0.0041
uniform_lattice	0.4226	0.0007	0.0023	0.0037

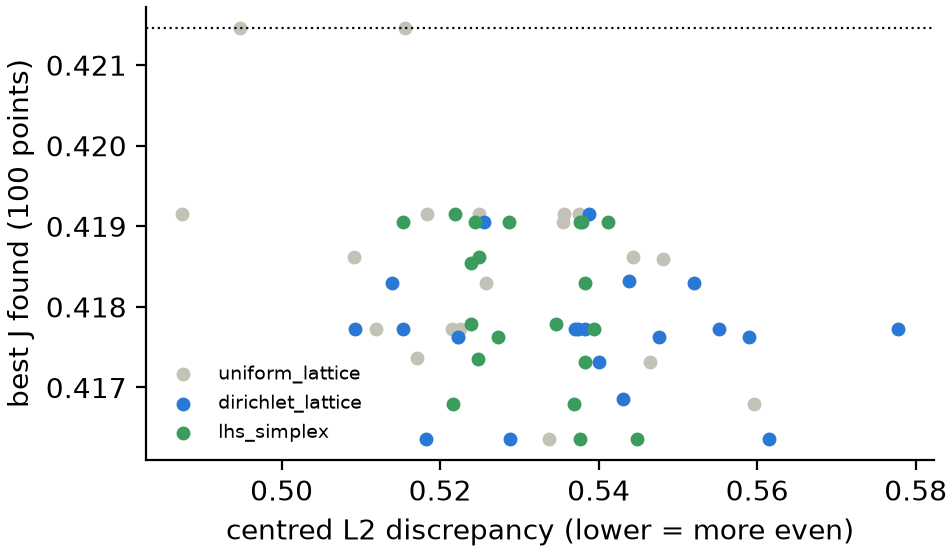


Figure 4: Any 100-point design finds the plateau; regret against the exhaustive optimum is at most 0.004.

2.3 The optimum does not transfer, and neither does the rank

In one line: Re-fitted weights predict a held-out dataset no better than v1, and families want different weights.

Leave-one-dataset-out: weights re-fitted without the held-out dataset predict its ranking with mean $\tau = +0.36$, the full-data best with $+0.40$, v1 with $+0.38$ (Table 31 gives the family-level version). Per-family optima pull in different directions (Figure 5): decay-only for schema matching, recall-heavy for table QA, F1-only for finance, F1-and-decay for entity matching, yet none is far from the plateau. Changing only the pooling rule inside the objective moves v1's rank from 319 (Bradley–Terry) to 1,163 (Kemeny) and changes which weighting is "best" under Kemeny and Copeland, so rank inside a flat objective carries no information.

Table 3: Leave-one-family-out: weights re-fitted without the family (F1 / decay / precision / recall) and the held-out family's τ to the pooled rest under those weights, the full-data best and v1.

family	LOFO weights	tau: LOFO weights	full best	v1
EM	0.05 / 0.60 / 0.25 / 0.10	0.327	0.318	0.324
SM	0.10 / 0.15 / 0.10 / 0.65	0.358	0.290	0.277
TQA	0.30 / 0.60 / 0.10 / 0.00	0.367	0.367	0.357
FIN	0.10 / 0.85 / 0.00 / 0.05	0.500	0.576	0.563

Notable. Held-out τ under re-fitted weights is within 0.08 of v1 in every family (EM $+0.33$ vs $+0.32$, SM $+0.36$ vs $+0.28$, TQA $+0.37$ vs $+0.36$, FIN $+0.50$ vs $+0.56$), and v1 is ahead on finance. The re-fitted weights themselves swing from decay-heavy (EM, FIN) to recall-heavy (SM) to decay-and-F1 (TQA): there is no single weighting the families agree on.

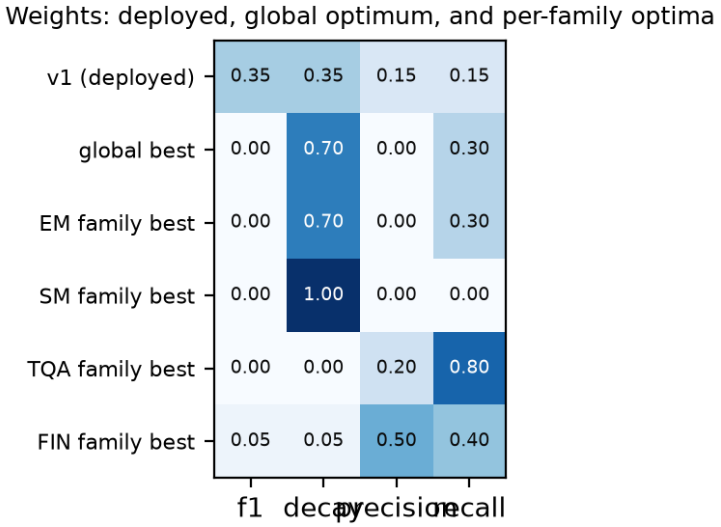


Figure 5: Deployed weights (v1), the global optimum and the per-family optima.

Notable. Schema matching wants decay only, table QA wants recall, finance wants F1 only, entity matching wants F1 and decay; the global optimum is a compromise between them, and v1 is another. The suite-level flatness is partly these pulls cancelling, which is also why leave-one-family-out re-fits land far from the global best without doing better on the held-out family.

2.4 Agreement and correctness do not pull the same way

In one line: Agreement among datasets and correctness on known answers are different criteria; v1 is near the correctness optimum.

On 31,380 synthetic answers of known utility, v1 scores $\rho = 0.878$; the best weighting by ρ reaches 0.883 and the best by J 0.876; exact match 0.828. Every rule separates correct from wrong answers with AUC 0.994. Across the lattice J and ρ correlate -0.14 on this cache, -0.23 before the final re-asks, -0.34 at ten models and $+0.38$ on the earlier 2,500-token cache, which is what a flat J predicts. With eleven models the Pareto knee coincides with the best-J weighting (Table 9, Figure 6).

Table 4: Anchor correctness for the labelled rules.

label	rank	w__f1	w__decay	w__precision	w__recall	J	tau__native	rho	rho__ds	auc	wrong__mean	hedge__gap
uniform	220	0.250	0.250	0.250	0.250	0.418	0.901	0.875	0.866	0.994	0.033	-0.480
only__decay	601	0.000	1.000	0.000	0.000	0.415	0.915	0.772	0.804	0.991	0.040	-0.595
only__f1	974	1.000	0.000	0.000	0.000	0.412	0.877	0.856	0.846	0.989	0.030	-0.496
v1__original	1130	0.350	0.350	0.150	0.150	0.411	0.915	0.878	0.868	0.994	0.034	-0.506
only__recall	1763	0.000	0.000	0.000	1.000	0.395	0.831	0.736	0.741	0.989	0.030	-0.244
only__precision	1767	0.000	0.000	1.000	0.000	0.393	0.845	0.848	0.839	0.989	0.031	-0.587

Notable. v1 scores $\rho = 0.878$, within 0.005 of the best attainable 0.883; the best-by-J weighting is 0.876 and exact match 0.828. Every rule has AUC 0.994: all separate right from wrong, and they differ only in partial credit for near misses, unit variants and partial lists.

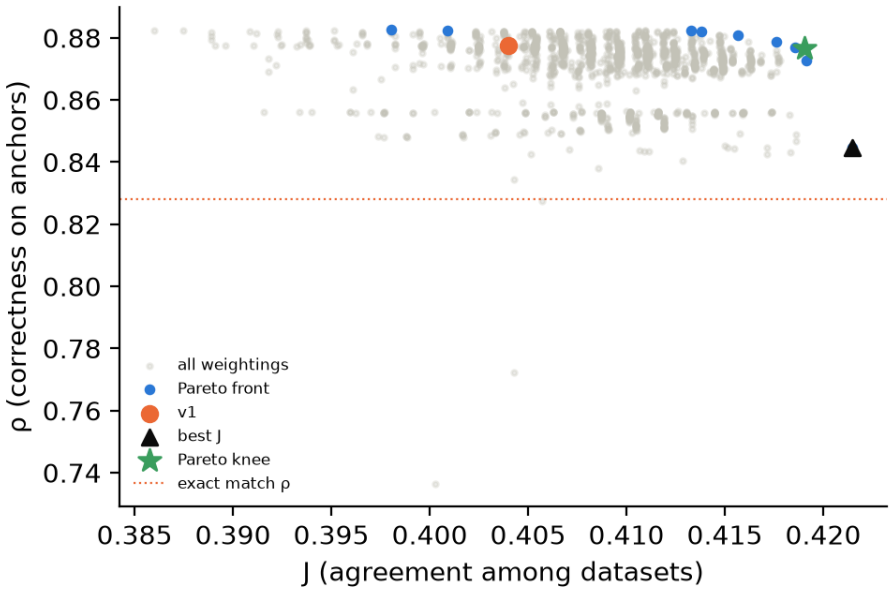


Figure 6: Agreement J against anchor correctness ρ for every weighting; v1, the best-J, best-ρ and knee points marked.

Notable. The cloud is wide in ρ (0.74–0.88) and narrow in J (0.39–0.43), and the two correlate -0.14: agreement is not a proxy for correctness. v1 sits on the upper edge near the ρ optimum; recall-only and decay-only, the corners that score well on J, are the worst by ρ.

2.5 Leaderboard

In one line: The ends of the table are certain, the top three reorder with the rule, DeepSeek-R1 rises to fourth once allowed to finish, and Gemini 3.1 Pro lands sixth.

Under every rule the Friedman test rejects equal models ($p < 0.001$) while datasets agree only moderately (Kendall's W 0.26–0.28). The ends of the ranking are stable: Llama 3.3 70B is last and gpt-oss-120b tenth under every rule and every pooling method, with Holm-corrected significance against the top group. The order inside the top three depends on the rule and the pooling method (Tables 21 and 58, Figure 7): Bradley–Terry puts Gemini 3.8 Flash first under v1 and all calibrated rules, Borda puts GPT-5.5 first under those rules, and exact match puts Claude Opus 5 first under both. Gemini 3.1 Pro lands sixth by Bradley–Terry under v1 and fifth under exact match. DeepSeek-R1 moved from seventh to fourth once its unfinished answers were re-asked at 16,000 and 32,000 tokens: the earlier position was a budget artefact on the two OfficeQA sets.

Table 5: Leaderboard under the deployed rule (v1 weights): Bradley–Terry strength ± SE vs reference with the 95 % bootstrap rank interval, Borda rank with its interval, and the mean composite over the suite.

rank	model	BT strength	SE	BT 95%	P(first) BT	Borda rank	Borda 95%	P(first) Borda	mean score
1	Gemini 3.8 Flash	0.112	0.015	[1, 2]	0.920	3	[1, 4]	0.060	0.812
2	Claude Opus 5	0.097	0.000	[1, 2]	0.080	2	[1, 3]	0.500	0.812
3	GPT-5.5	0.042	0.015	[3, 5]	0.000	1	[1, 3]	0.430	0.808
4	DeepSeek-R1	0.037	0.015	[3, 6]	0.000	4	[3, 7]	0.000	0.801
5	Claude Sonnet 5	0.023	0.015	[3, 7]	0.000	5	[4, 7]	0.000	0.796
6	Gemini 3.1 Pro	0.021	0.015	[4, 7]	0.000	6	[4, 7]	0.000	0.799
7	Grok 4	0.016	0.015	[4, 7]	0.000	7	[4, 7]	0.000	0.797
8	GPT-5.4-mini	-0.022	0.015	[8, 9]	0.000	9	[7, 9]	0.000	0.790
9	DeepSeek-chat	-0.031	0.015	[8, 9]	0.000	8	[7, 9]	0.000	0.784
10	gpt-oss-120b	-0.111	0.015	[10, 10]	0.000	10	[10, 11]	0.000	0.767
11	Llama 3.3 70B	-0.183	0.015	[11, 11]	0.000	11	[10, 11]	0.000	0.738

Notable. Gemini 3.8 Flash is first by Bradley–Terry with $P(\text{first}) = 0.92$ because it wins many items by small margins; by Borda, which counts dataset-level ranks, GPT-5.5 is first and Flash third. The bottom two (gpt-oss-120b, Llama 3.3 70B) have Bradley–Terry rank intervals of width zero; ranks 3–7 overlap (DeepSeek-R1, Sonnet, Gemini 3.1 Pro and Grok 4 all span 3–7) and are not established.

Table 6: Bradley–Terry rank (Borda rank in brackets) under each reported rule, with the Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
exact match	3 (3)	1 (1)	2 (2)	4 (4)	7 (5)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	2.7e-15	0.28
v1	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	1.3e-14	0.27
top1	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	7.5e-15	0.28
top2	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	8.0e-15	0.28
top3	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (5)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	5.9e-15	0.28
top4	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (5)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	5.9e-15	0.28
top5	1 (3)	2 (2)	4 (1)	3 (4)	5 (5)	6 (6)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	6.1e-15	0.28
best-rho	1 (3)	2 (2)	4 (1)	3 (4)	5 (5)	6 (6)	8 (7)	7 (8)	9 (9)	10 (10)	11 (11)	7.2e-14	0.26
knee	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	7.5e-15	0.28

Notable. Rows 10 and 11 never move; the top three permute with the rule (exact match favours Opus, every graded rule favours Flash by Bradley–Terry and GPT-5.5 by Borda). DeepSeek-R1 is third or fourth under every graded rule and fourth under exact match now that its reasoning is allowed to finish; Gemini 3.1 Pro is sixth under every graded rule and fifth under exact match. Kendall's W among datasets is 0.26–0.28 under every rule.

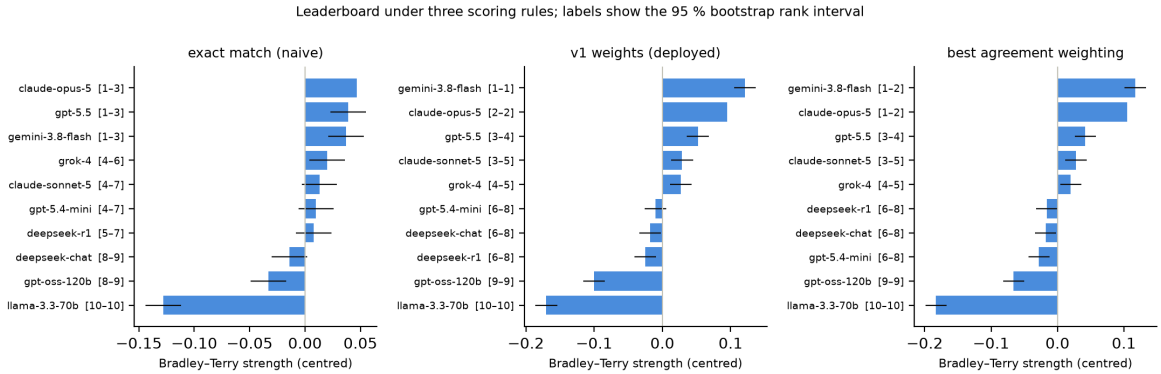


Figure 7: Bradley–Terry strengths ± SE with bootstrap rank intervals under exact match, v1 and the best-J weighting. The ends are stable; the top order is rule-dependent.

Notable. The strengths of ranks 3–7 lie within 0.03 of each other with overlapping error bars; the gap from rank 9 to rank 10 and from 10 to 11 is larger than the whole middle band. The figure is the visual form of the Holm result: 15 of 55 pairs significant, all of them against the bottom two models.

Notable. Blue blocks sit inside families and the finance and table-QA sets agree with most of the suite; Alaska Camera and HiTab are red against almost everyone. The leaderboard is an average over datasets that disagree about the middle of the table, which is what $W = 0.28$ says in one number.

2.6 Comparability with published numbers

In one line: Accuracy survives re-weighting to the official splits; balanced-pool F1 on sparse matching sets does not.

Post-stratification to the official split changes dataset accuracy by at most 0.09 (OpenSanctions pairs). For the matching sets, positive-class F1 computed on the balanced 50/50 pool overstates F1 at the official match ratio wherever the official split is sparse, by up to 0.19 on Amazon–Google (Table 34, Figure 9).

Notable. Accuracy moves by at most 0.09 (OpenSanctions, whose official class ratio is 0.77 rather than 0.5). Positive-class F1 drops by 0.19 on Amazon–Google, 0.17 on Papadakis–Christen and 0.16 on Machamp, all through precision: at a 2–10 % official match rate a 1 % false-positive rate costs more than any recall gain.

Notable. Almost every (dataset, model) point lies on the diagonal for accuracy; the sparse matching sets fall below it for F1. Model order is preserved after re-weighting; the absolute numbers are not, so the official-ratio column is the one to compare with published results.

2.7 Dataset diagnostics

In one line: Three sets are at ceiling and twelve separate no model pair; a dozen table and document sets carry the result.

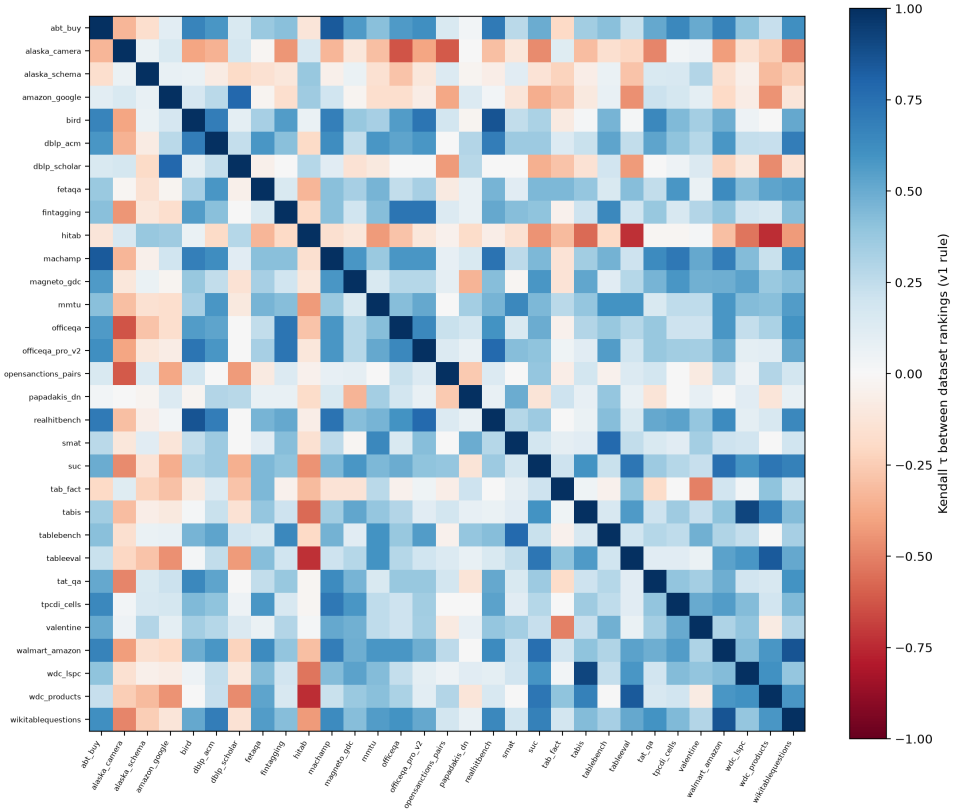


Figure 8: Kendall τ between the model rankings of each pair of datasets under v1. Datasets agree only moderately ($W \approx 0.28$).

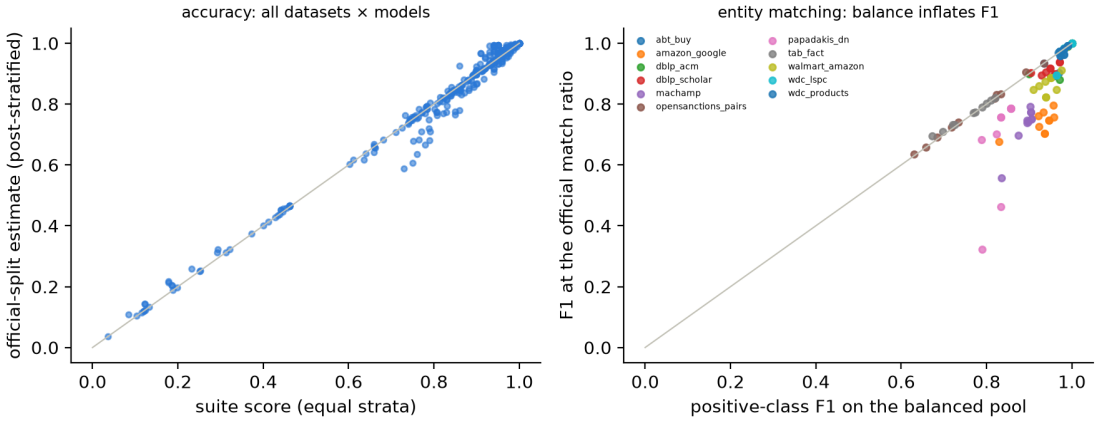


Figure 9: Suite score vs post-stratified official estimate per dataset, and matching F1 on the balanced pool vs at the official match ratio.

Table 7: Official-split estimates under v1 averaged over the ten models.

dataset	suite_ mean	official_ estimate	suite_f1_pos	official_precision_ pos	official_recall_ pos	official_f1_pos	official_positive_ share
abt_buy	0.971	0.987	0.969	0.933	0.950	0.940	0.108
alaska_camera	0.945	0.945	0.939	1.000	0.895	0.939	0.500
alaska_schema	0.929	0.929	–	–	–	–	–
amazon_google	0.935	0.932	0.928	0.609	0.941	0.738	0.102
bird	0.191	0.218	–	–	–	–	–
dblp_acm	0.987	0.992	0.986	0.981	0.980	0.978	0.180
dblp_scholar	0.948	0.966	0.947	0.901	0.922	0.909	0.186
fetaqa	0.387	0.387	–	–	–	–	–
fintagging	0.884	0.882	–	–	–	–	–
hitab	0.772	0.776	–	–	–	–	–
machamp	0.887	0.891	0.892	0.649	0.852	0.735	0.175
magneto_gdc	0.937	0.940	–	–	–	–	–
mmtu	0.874	0.837	–	–	–	–	–
officeqa	0.135	0.135	–	–	–	–	–
officeqa_pro_ v2	0.151	0.151	–	–	–	–	–
opensanctions_ pairs	0.835	0.748	0.790	0.993	0.677	0.795	0.769
papadakis_dn	0.940	0.987	0.831	0.527	0.967	0.662	0.018
realhitbench	0.677	0.676	–	–	–	–	–
smat	0.916	0.914	–	–	–	–	–
suc	0.917	0.915	–	–	–	–	–
tab_fact	0.802	0.792	0.761	0.903	0.674	0.767	0.500
tabis	0.918	0.923	–	–	–	–	–
tablebench	0.824	0.822	–	–	–	–	–
tableeval	0.755	0.758	–	–	–	–	–
tat_qa	0.804	0.759	–	–	–	–	–
tpcdi_cells	0.456	0.462	–	–	–	–	–
valentine	0.985	0.981	–	–	–	–	–
walmart_ amazon	0.959	0.974	0.949	0.824	0.931	0.872	0.094
wdc_lspc	0.982	0.984	0.983	0.968	0.978	0.972	0.264
wdc_products	0.978	0.986	0.979	0.982	0.968	0.975	0.280
wikitablequestions	0.848	0.846	–	–	–	–	–

Three datasets sit at ceiling (mean composite 0.98 with a spread of 0.02–0.04 across models): Valentine, WDC LSPC and WDC Products. They are flagged and remain inside the objective in this run. Two sit near floor for every model (OfficeQA closed-book and OfficeQA Pro V2) and one is a known hard set (BIRD); none is excluded. Self-stability τ (rank agreement between bootstrap halves of a dataset) ranges from 0.58 to 0.92. Twelve datasets, nine entity-matching sets plus Valentine, TPC-DI and FinTagging, separate no model pair at the item level; three separate twenty or more.

A All figures, with a reading guide

Every figure that can be drawn from the saved outputs, each followed by what it encodes, how to read it and what it shows. Source data for each is in `data/`.

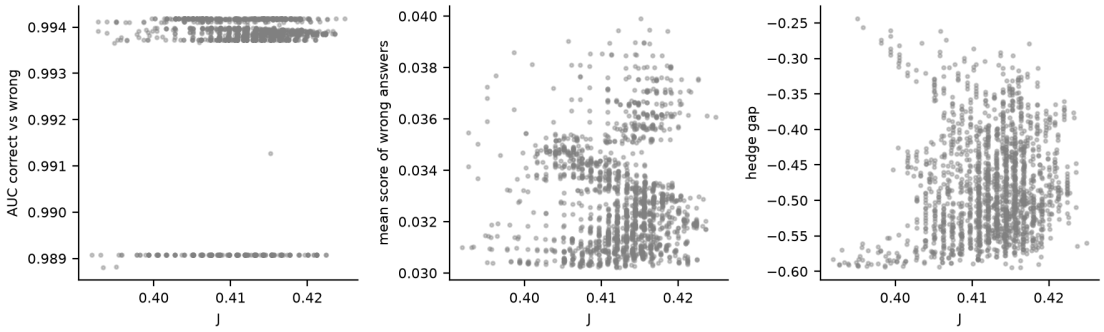


Figure 10: Three further anchor diagnostics against J for every weighting.

What it encodes. Three anchor diagnostics against J for every weighting: AUC for separating correct from wrong anchors, the mean score given to wrong anchors, and the hedge gap (score of a committed answer minus the same answer hedged). **How to read it.** Flat clouds mean the diagnostic does not depend on the weighting; a slope means agreement trades against it. **What it shows.** AUC is 0.989–0.994 for every weighting: all separate right from wrong. Wrong answers score below 0.06 everywhere. The differences between weightings are in partial credit, not in discrimination.

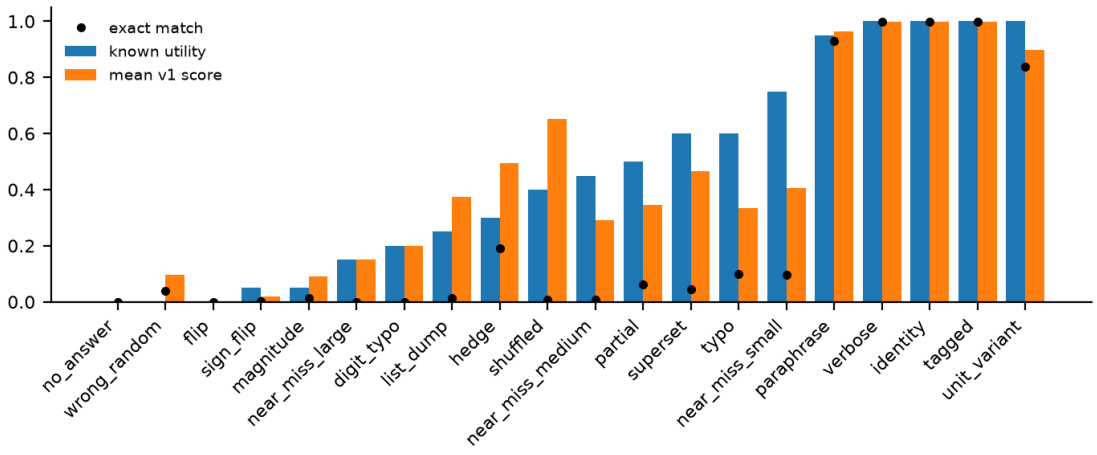


Figure 11: Synthetic anchors: known utility of each perturbation operator against the mean v1 score and exact match it receives.

What it encodes. Each anchor operator (a typed perturbation applied to gold answers) with its assigned utility, the mean v1 score it receives, and its exact-match rate. **How to read it.** A perfect judge would put the blue point on the grey one for every operator. Blue above grey is over-generosity, blue below is over-severity. **What it shows.** The ends are right (identity, verbose and tagged score 1; flips and non-answers score 0). The judge is generous to shuffled lists, supersets and list dumps (token overlap rewards extra correct tokens) and severe on typos, digit typos and medium near misses. These operators are where a human label would disagree with any lexical rule.

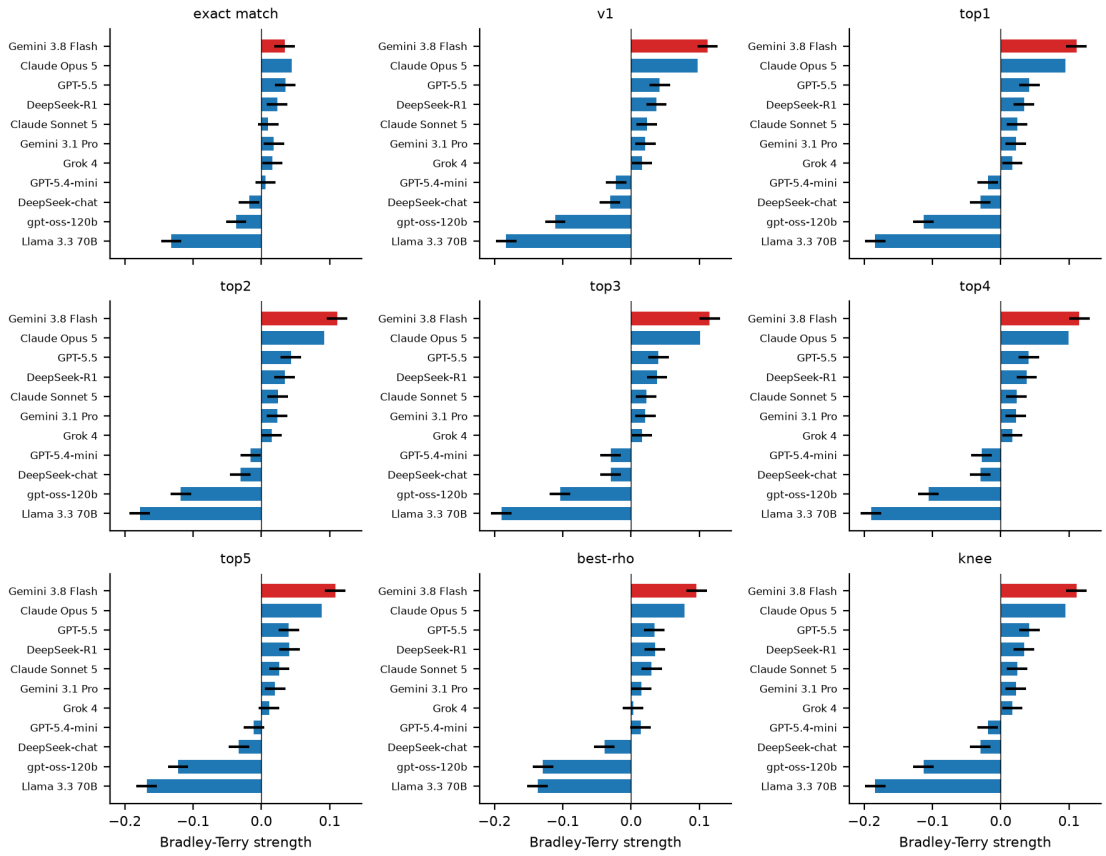


Figure 12: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in v1 order.

What it encodes. Nine small panels, one per reported rule (exact match, v1, the five best by J, the best by ρ, the Pareto knee), each showing Bradley–Terry strength ± SE for the ten models in a fixed order; the first-placed model is orange. **How to read it.** Scan a row across panels to see whether a model's strength is stable; scan the orange markers to see who is first under which rule. **What it shows.** Rows 4–10 barely move; first place is Opus under exact match and Gemini Flash under every graded rule. Kendall τ between the rules' rankings is 0.78–1.00.

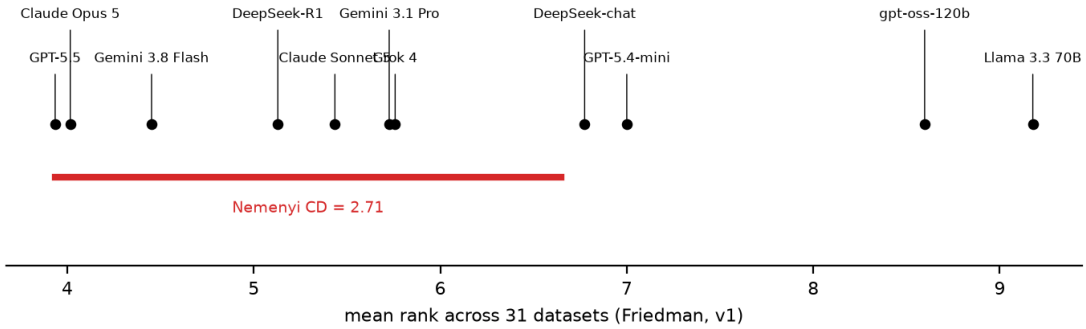


Figure 13: Critical-difference view of the Friedman test under v1: mean rank per model and the Nemenyi critical difference at $\alpha = .05$.

What it encodes. The Friedman test's mean rank of each model across the 31 datasets on a single axis, with the Nemenyi critical difference (2.71) drawn as a bar; models closer than the critical difference are not separated at $\alpha = .05$. **How to read it.** Two models are significantly different only if their positions differ by more than the bar's length. **What it shows.** The top group is separated from the bottom two; nothing inside the top group, and nothing in the middle, is separated. With 31 datasets and eleven models this is the resolution the cross-dataset test can offer.

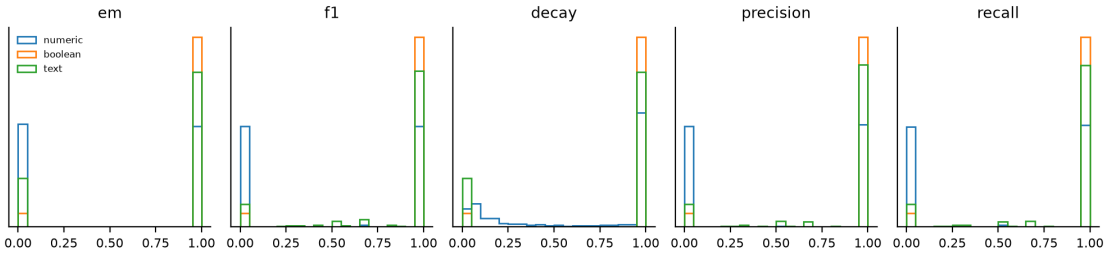


Figure 14: Distribution of each v1 ingredient over all answers, by answer type.

What it encodes. Five panels (exact match, F1, decay, precision, recall); within each, one density curve per answer type (numeric, boolean, text, free-form). **How to read it.** A curve concentrated at 0 and 1 means the ingredient behaves like a binary score on that answer type; mass in between is graded credit. **What it shows.** On boolean and text golds every ingredient is binary and identical to exact match. Decay is graded only on numeric golds; F1, precision and recall are graded only on free-form and some text golds. This is the mechanical reason the weights barely matter: the ingredients differ on a minority of answers.

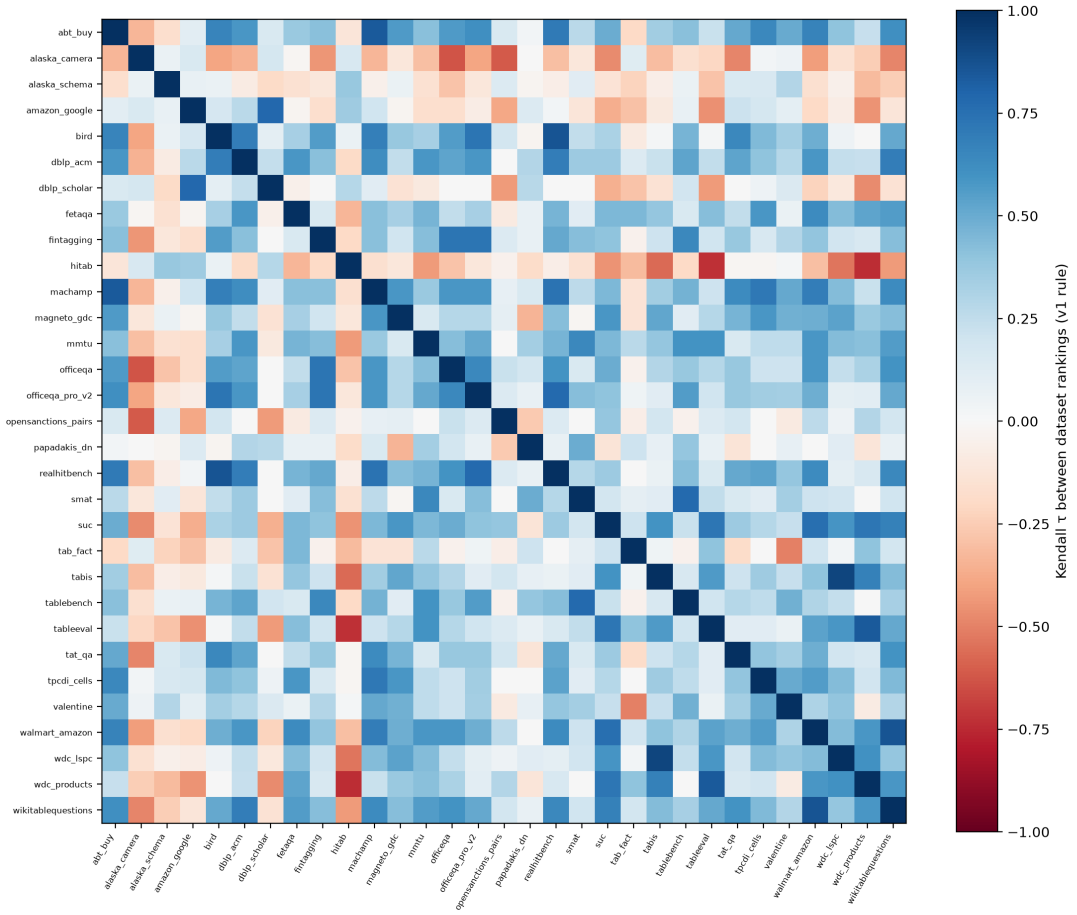


Figure 15: Kendall τ between dataset rankings under v1.

What it encodes. Kendall τ between the model rankings of every pair of datasets under v1 (blue agreement, red disagreement). **How to read it.** Blocks of blue are groups of datasets that rank the models the same way; red rows are datasets that disagree with most others. **What it shows.** Mean pairwise τ is $+0.21$ and Kendall's W is 0.27: datasets agree about the ends of the table and disagree about the middle. Alaska Camera and HiTab are red against most of the suite.

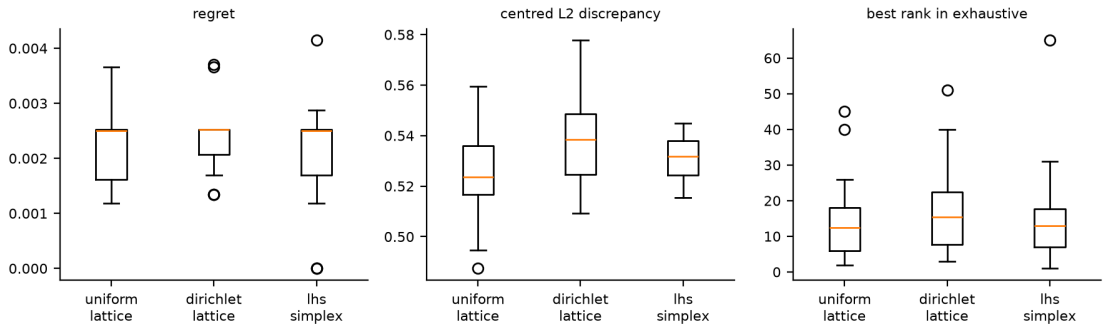


Figure 16: Sampled designs over 20 seeds: regret against the exhaustive optimum, centred L_2 discrepancy and rank of the found best.

What it encodes. Box plots over 20 seeds of regret, centred L_2 discrepancy (coverage of the simplex, lower is more even) and the rank of the found best within the exhaustive grid, per design. **How to read it.** Lower discrepancy means the sample is spread more evenly; lower regret and rank mean the sample found a better point. **What it shows.** The three designs are interchangeable at this budget; the uniform lattice has marginally better coverage and the Latin hypercube marginally lower regret, and all three differ by less than the bootstrap noise of J.

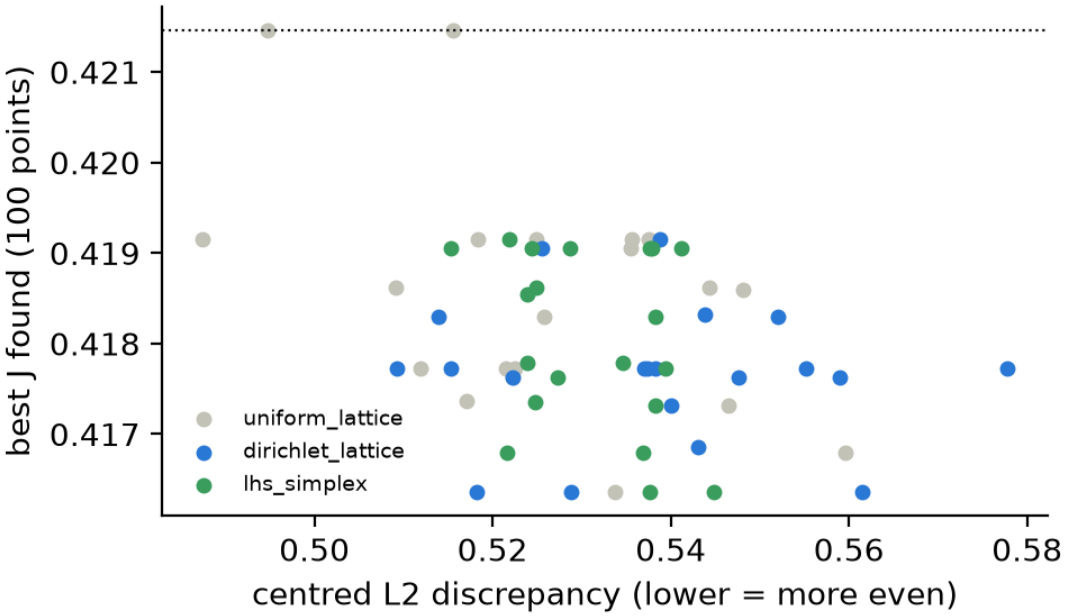


Figure 17: Sampled designs against the exhaustive optimum.

What it encodes. For three sampling designs of 100 weightings (uniform lattice, Dirichlet-snapped, Latin hypercube) and 20 seeds each, the best J found, against the exhaustive optimum. **How to read it.** Regret is the vertical distance to the exhaustive best; the spread across seeds is the risk of an unlucky sample. **What it shows.** Every design and every seed lands within 0.004 of the optimum; the worst seed of any design still finds a top-65 weighting. With four weights the exhaustive grid is free, so the designs serve only to show that sampling would have sufficed.

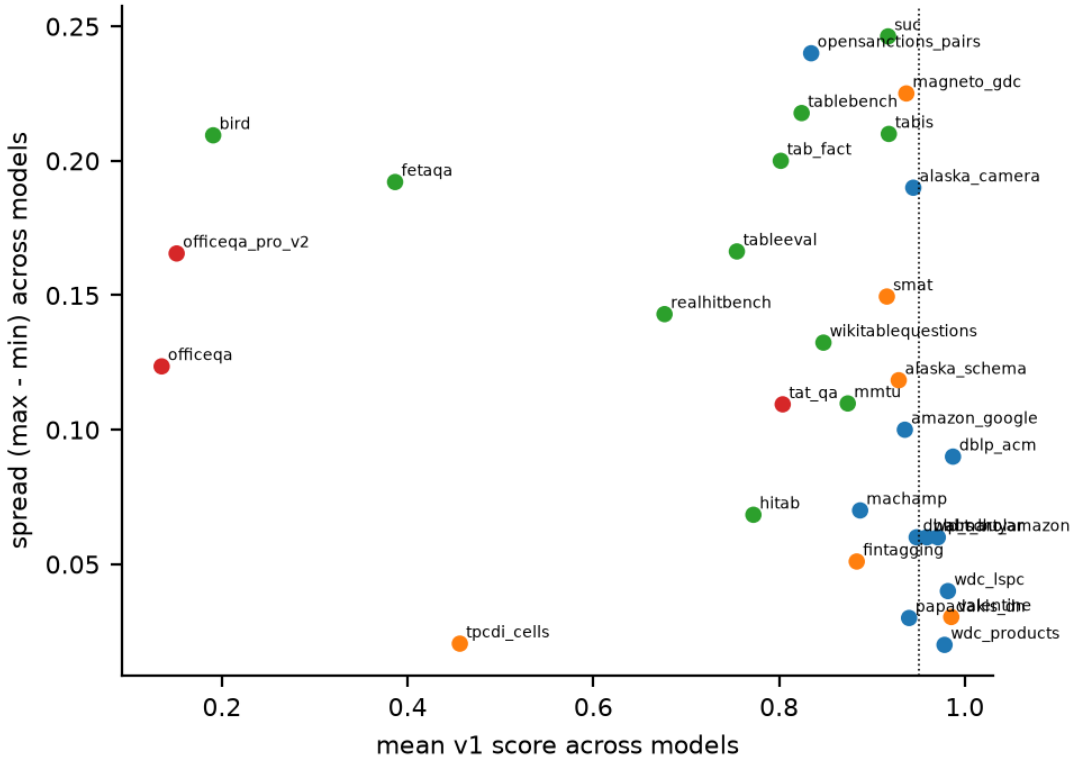


Figure 18: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

What it encodes. Each dataset placed by its mean v1 score across models (horizontal) and the spread between the best and worst model (vertical); the dotted line is the ceiling flag. **How to read it.** Datasets at the right with low spread are at ceiling and cannot rank models; datasets with high spread are the discriminating ones. **What it shows.** 3 datasets are flagged at ceiling (mean ≥ 0.95, spread ≤ 0.04). They remain inside the objective in this run and are listed as a limitation.

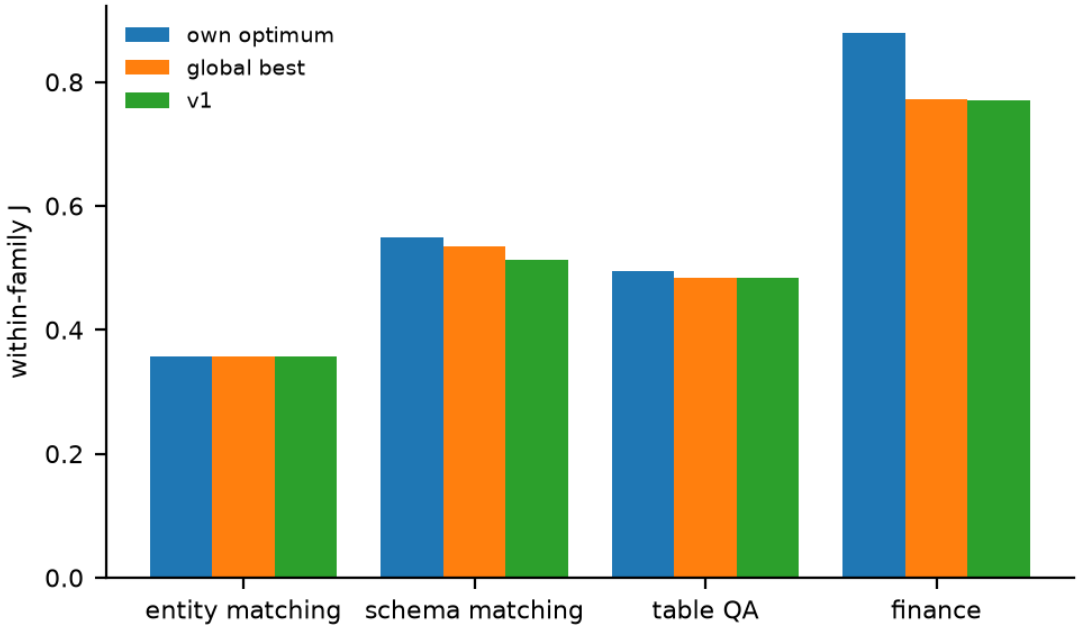


Figure 19: Within-family agreement J under each family's own optimum, the global best and v1.

What it encodes. Within-family agreement J (computed over that family's datasets only) under the family's own optimum, the global best and v1. **How to read it.** The gap between the first bar and the other two is what family-specific weights would gain for that family. **What it shows.** Within a family the datasets agree far more than across the suite (finance highest, entity matching lowest), and the family optimum beats the global best by a visible margin only for table QA and finance.

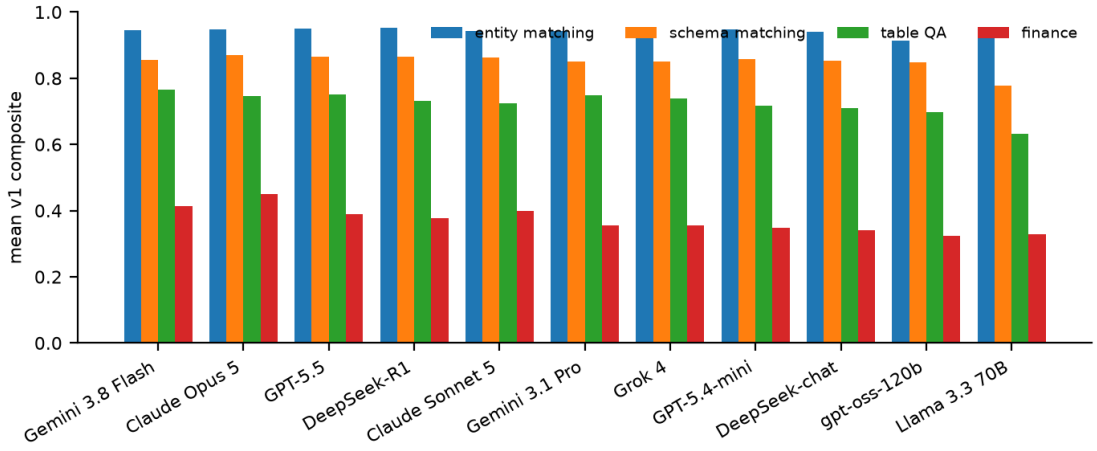


Figure 20: Mean v1 composite per model within each task family.

What it encodes. For each model, four bars: its mean v1 composite within entity matching, schema matching, table QA and finance. **How to read it.** Compare bars of the same colour across models to see the family-specific ranking; compare the four bars of one model to see which family drives its overall mean. **What it shows.** Entity matching is near ceiling for every model, finance is near floor for every model, and the differences between models live in table QA and schema matching. A model's overall position is therefore decided by a minority of the suite.

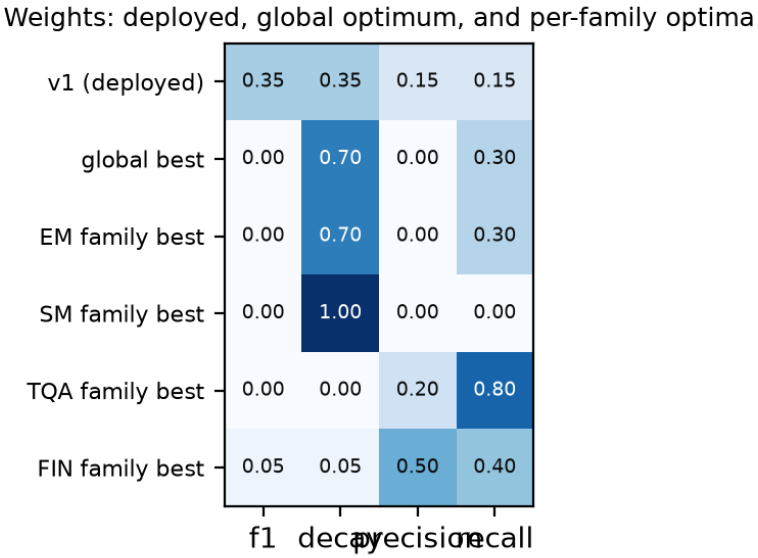


Figure 21: Deployed, global-optimum and per-family weights.

What it encodes. The four weights of v1, of the global optimum, and of the optimum fitted inside each task family. **How to read it.** Compare the family profiles with each other and with the global point. Agreement would show the same shape in every row. **What it shows.** The families disagree: decay dominates for schema matching, recall for table QA, precision for finance. The suite-level flatness is partly these pulls cancelling, which also explains why leave-one-family-out re-fits land far from the global best.

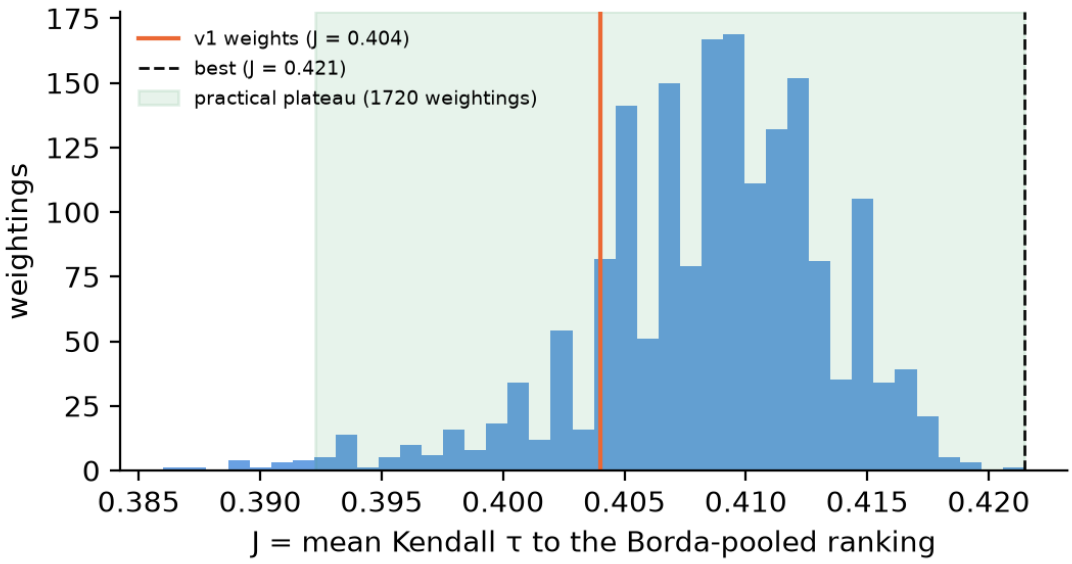


Figure 22: Distribution of J over the 1,771 weightings with v1, the best and the plateau band.

What it encodes. The horizontal axis is the agreement objective J (mean Kendall τ between each dataset's model ranking and the Borda-pooled ranking); the bars count how many of the 1,771 step-0.05 weightings fall in each bin. The vertical lines mark the deployed weights (v1) and the best weighting; the shaded band is the plateau of weightings statistically indistinguishable from the best. **How to read it.** If the weights mattered, the histogram would be wide and v1 would sit in a low tail far from the best. Instead the whole distribution spans 0.392–0.425, v1 sits at 0.415 and the best at 0.425, and the plateau covers 1,754 of 1,771 points. **What it shows.** The objective is flat: moving weight between F1, decay, precision and recall changes J by less than the bootstrap noise of a single point, so no weighting is distinguishable from v1 on this suite.

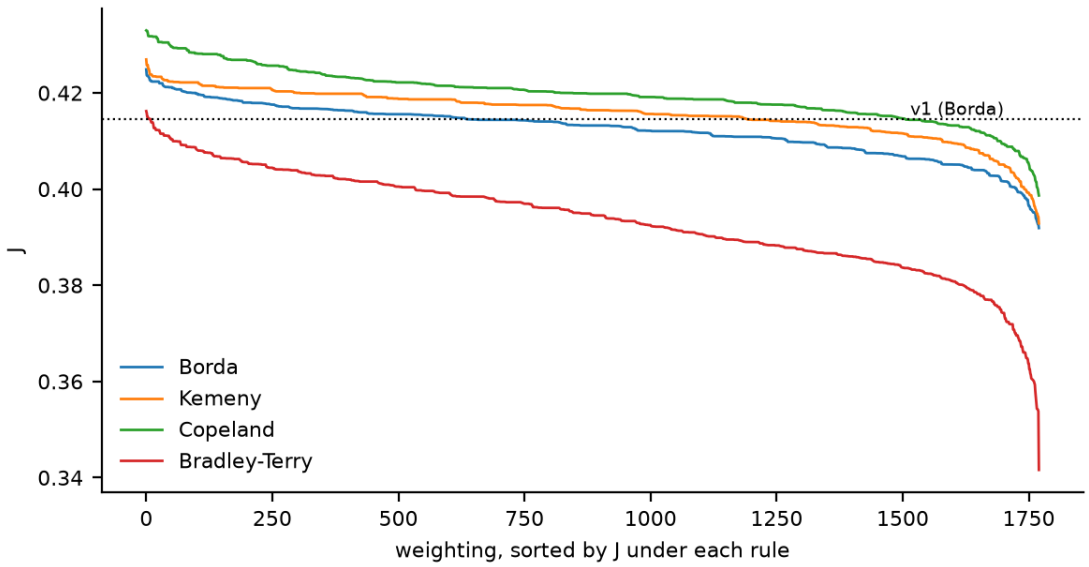


Figure 23: Agreement objective J over all 1,771 weightings under four pooling rules (each curve sorted separately). The landscape is flat under every rule.

What it encodes. The objective recomputed under four ways of pooling the 31 per-dataset rankings into one (Borda, Kemeny, Copeland, Bradley–Terry), each curve sorted separately from best to worst. **How to read it.** The curves answer whether the flatness is an artefact of Borda pooling. If another rule produced a steeper curve or a clear winner, the choice of pooling rule would matter for the audit. **What it shows.** All four curves are flat and sit within a few hundredths of each other; which weighting is "best" changes with the rule (v1's rank moves from 144 under Bradley–Terry to 607 under Kemeny), confirming that rank inside the lattice is noise.

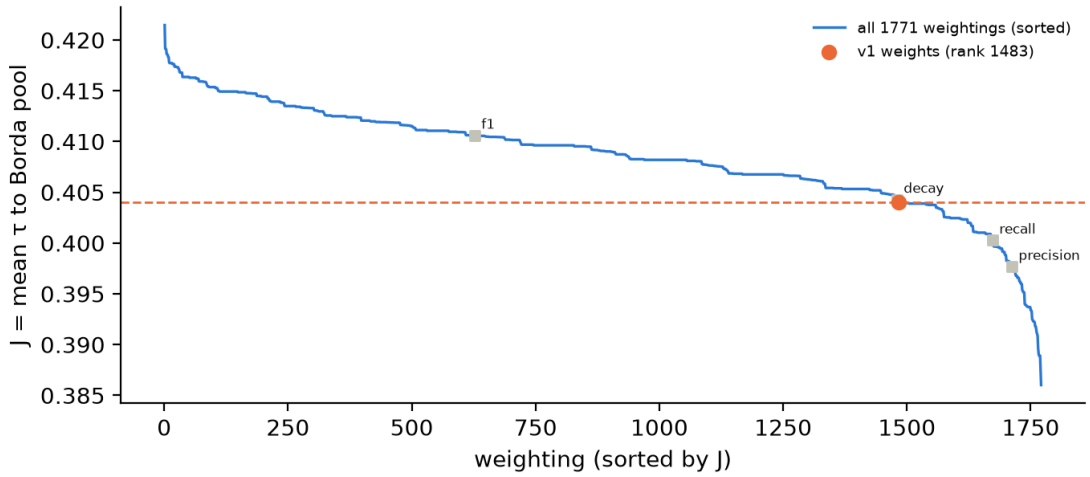


Figure 24: Sorted J with v1 and the four single-ingredient corners.

What it encodes. Every weighting sorted from best (left) to worst (right) by J; the curve is J itself, and the marked points are v1 and the four single-ingredient corners (F1 only, decay only, precision only, recall only). **How to read it.** The height of the curve at any rank is the J of that weighting; the vertical drop from left to right is the whole range the search could ever exploit. A rank far to the right looks bad until the vertical scale is read: v1 is at rank 639 of 1,771, but the curve drops only 0.033 from first to last. **What it shows.** The curve is nearly horizontal until the last few dozen weightings, which are the extreme corners. Rank inside such a curve carries no information, which is why the report quotes the plateau and the out-of-bag difference instead of v1's rank.

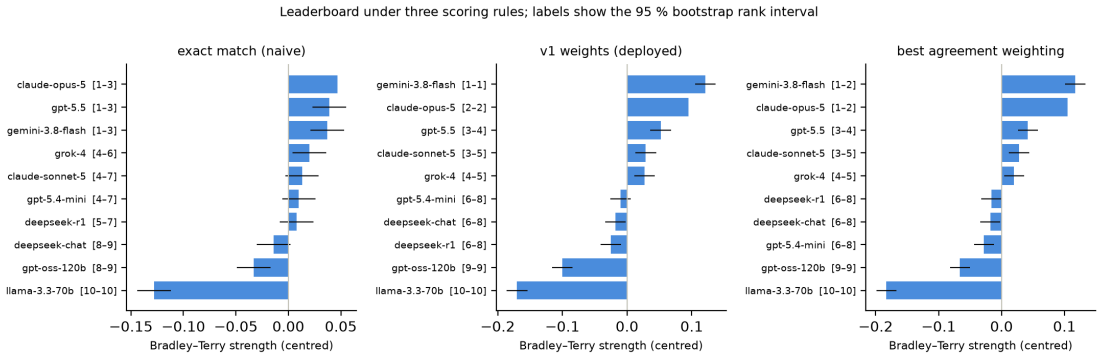


Figure 25: Bradley–Terry strengths with rank intervals under exact match, v1 and the best-J weighting.

What it encodes. Bradley–Terry strengths (log-odds of winning an item against the reference model, with standard errors) and bootstrap rank intervals under exact match, v1 and the best-J weighting. **How to read it.** Overlapping intervals mean the order is not established; a model whose point moves between panels is one whose position depends on the scoring rule. **What it shows.** The ends are fixed under every rule (Llama 3.3 70B last, gpt-oss-120b tenth); the top three (Gemini 3.8 Flash, Claude Opus 5, GPT-5.5) reorder with the rule. Exact match favours Opus; the graded rules favour Gemini Flash.

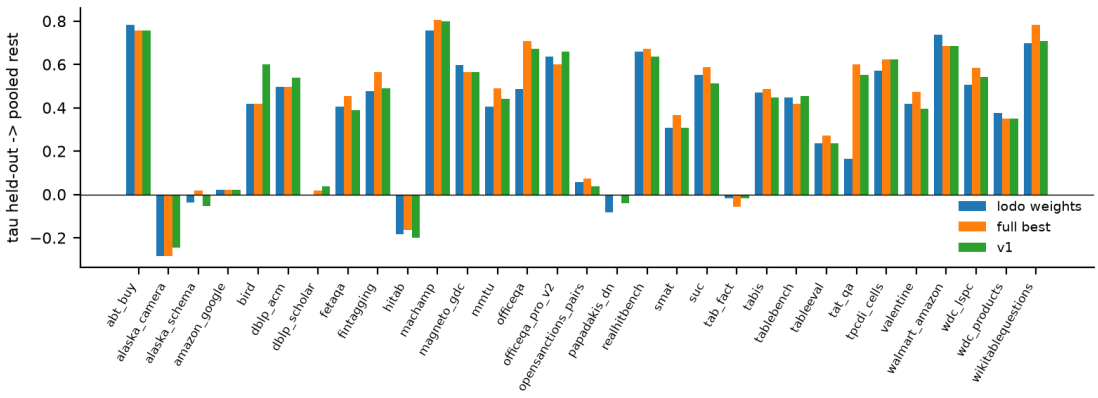


Figure 26: Leave-one-dataset-out: agreement of the held-out dataset with the pooled ranking of the rest, under weights fitted without it, the full-data best and v1.

What it encodes. For each dataset held out in turn: the Kendall τ between its model ranking and the pooled ranking of the other 30, under weights re-fitted without it, the full-data best, and v1. **How to read it.** If weight search generalised, the re-fitted bars would beat v1's on most datasets. Negative bars are datasets whose ranking opposes the consensus however the weights are set. **What it shows.** Means are +0.36 (re-fitted), +0.40 (full best) and +0.38 (v1); re-fitting helps on 12 datasets and hurts on 15. The same handful of sets (Alaska Camera, HiTab, Alaska schema) are negative under every rule.

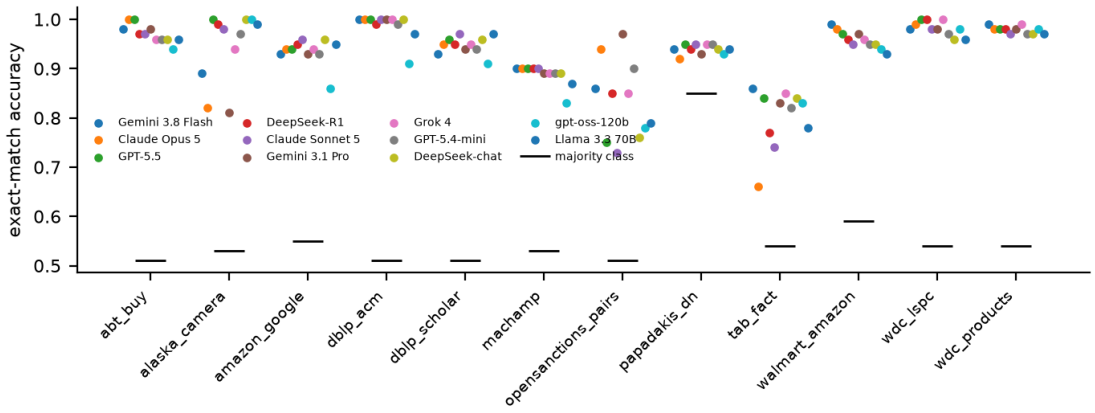


Figure 27: Exact-match accuracy of every model on the boolean datasets against the majority-class baseline of each pool.

What it encodes. For the boolean datasets, every model's exact-match accuracy against the majority-class baseline of the pool (the accuracy of always answering the more common label). **How to read it.** A model below the baseline line is doing worse than a constant answer on that dataset. **What it shows.** Every model clears the baseline on every boolean set, usually by a wide margin; the balanced pools make the baseline about 0.5 so this is a weak test, which is why the official-ratio F1 is reported beside it.

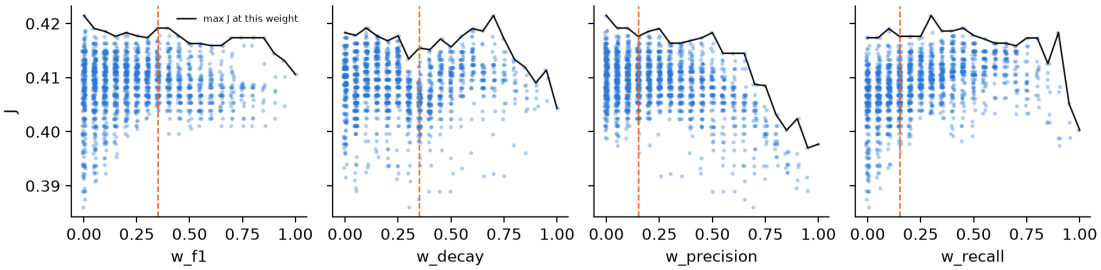


Figure 28: J against each of the four weights.

What it encodes. Four panels, one per ingredient; each point is one weighting placed by the weight it gives that ingredient (horizontal) and its J (vertical); the line is the mean J at each weight level. **How to read it.** A sloping mean line would say that more (or less) of that ingredient helps; a flat line says the ingredient's weight is irrelevant. The vertical scatter at each level is the effect of the other three weights. **What it shows.** The mean lines drift by at most 0.021 of J across the whole range (precision is the steepest, falling as its weight approaches 1); no ingredient has a slope worth acting on, and the scatter around each line is larger than any slope.

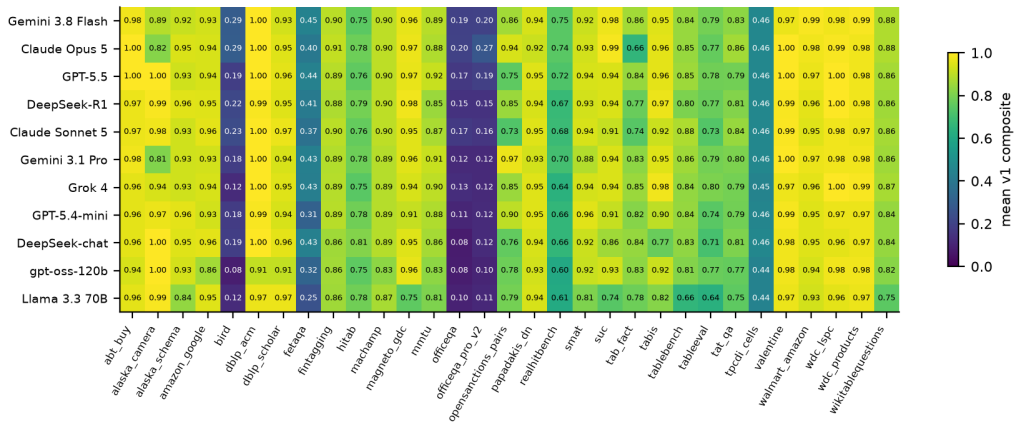


Figure 29: Mean v1 composite score per model and dataset (models in Bradley–Terry order under v1).

What it encodes. Rows are the ten models in Bradley–Terry order under v1; columns are the 31 datasets; each cell is the model's mean v1 composite on that dataset (dark = high). **How to read it.** Columns show how hard a dataset is for everyone (BIRD and the two OfficeQA sets are dark for all models); rows show a model's profile. A leaderboard is a summary of this matrix, so any disagreement between datasets is visible here as a row that is high in one column and low in another. **What it shows.** Most columns are near ceiling with tiny spreads (the matching sets); the spread that separates models comes from a dozen table and document sets. No row dominates every column.

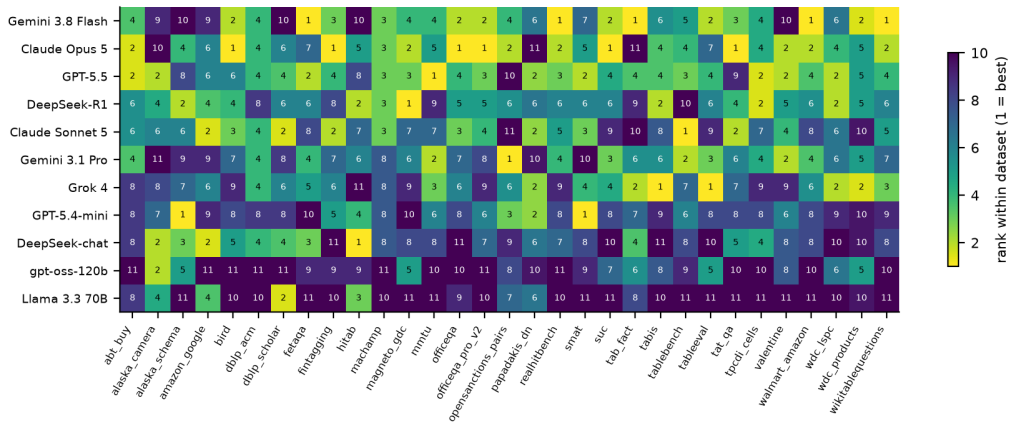


Figure 30: Rank of each model inside each dataset under v1 (ties by average rank).

What it encodes. The same matrix as the score heatmap, but each column is replaced by the within-dataset rank (1 = best, ties averaged). **How to read it.** A perfectly consistent suite would show each row as a single colour. Colour changes along a row are datasets that rank that model differently from the rest. **What it shows.** Only the bottom rows are nearly uniform; the top three rows trade first place from column to column (no model is first on more than 6 datasets), which is the picture behind Kendall's W of 0.27 among datasets.

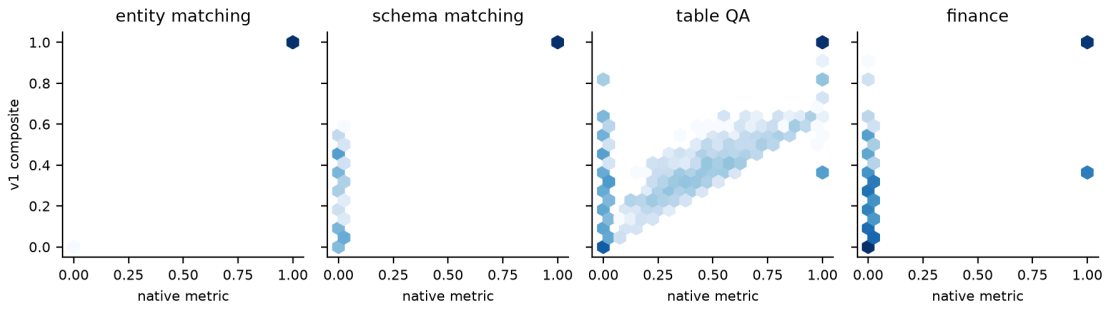


Figure 31: Per-answer native metric against the v1 composite, by family (log-count hexbins).

What it encodes. One panel per family; each answer placed by its dataset's native metric (horizontal) and its v1 composite (vertical), with cell colour showing how many answers fall there (log scale). **How to read it.** Mass on the diagonal means the composite agrees with the native metric; mass off the diagonal is where the four ingredients give partial credit (above) or penalise (below) relative to the dataset's own rule. **What it shows.** Entity and schema matching sit almost entirely on the diagonal corners (0,0) and (1,1): the composite is exact match there. Table QA and finance show the off-diagonal cloud of partial credit, which is the only place the weights can act.

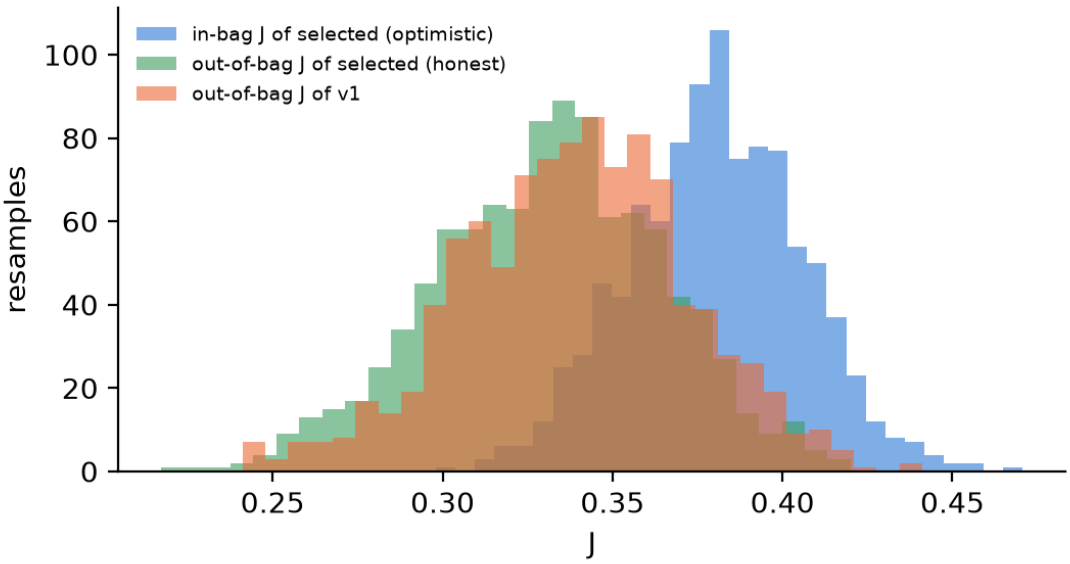


Figure 32: In-bag vs out-of-bag J of the selected weighting against v1.

What it encodes. From 1,000 item bootstraps: the in-bag J of the weighting selected as best on the in-bag items, the same weighting's out-of-bag J, and v1's out-of-bag J. **How to read it.** The gap between the first two is the optimism of selection (the winner's curse); the comparison between the second and the third is the honest test of whether searching found anything. **What it shows.** The selected weighting looks 0.046 better in-bag than out of bag, and out of bag it is -0.006 from v1 with a 95 % interval of [-0.032, +0.011]. Searching the weights buys nothing that survives held-out items.

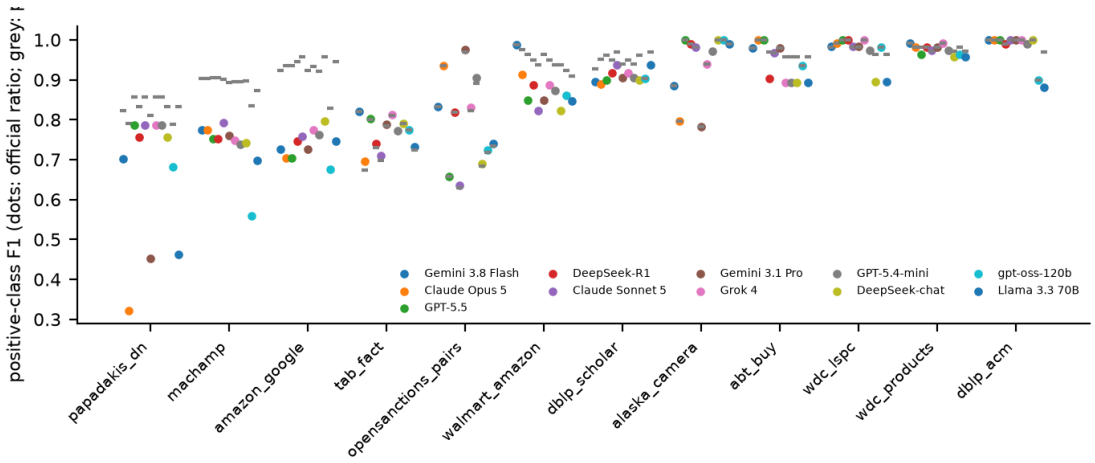


Figure 33: Positive-class F1 on the balanced pool (grey) and at the official match ratio (coloured) per model and matching dataset.

What it encodes. For each matching dataset and model, positive-class F1 on the balanced pool (grey ticks) and at the official match ratio (coloured dots). **How to read it.** The vertical gap between tick and dot for a model is how much its balanced-pool F1 overstates its official-ratio F1; the spread of dots is the between-model difference that survives re-weighting. **What it shows.** The gap is large and similar for all models on Amazon–Google, Machamp and Papadakis–Christen, and negligible on the dense sets. Model order is mostly preserved; the absolute numbers are not.

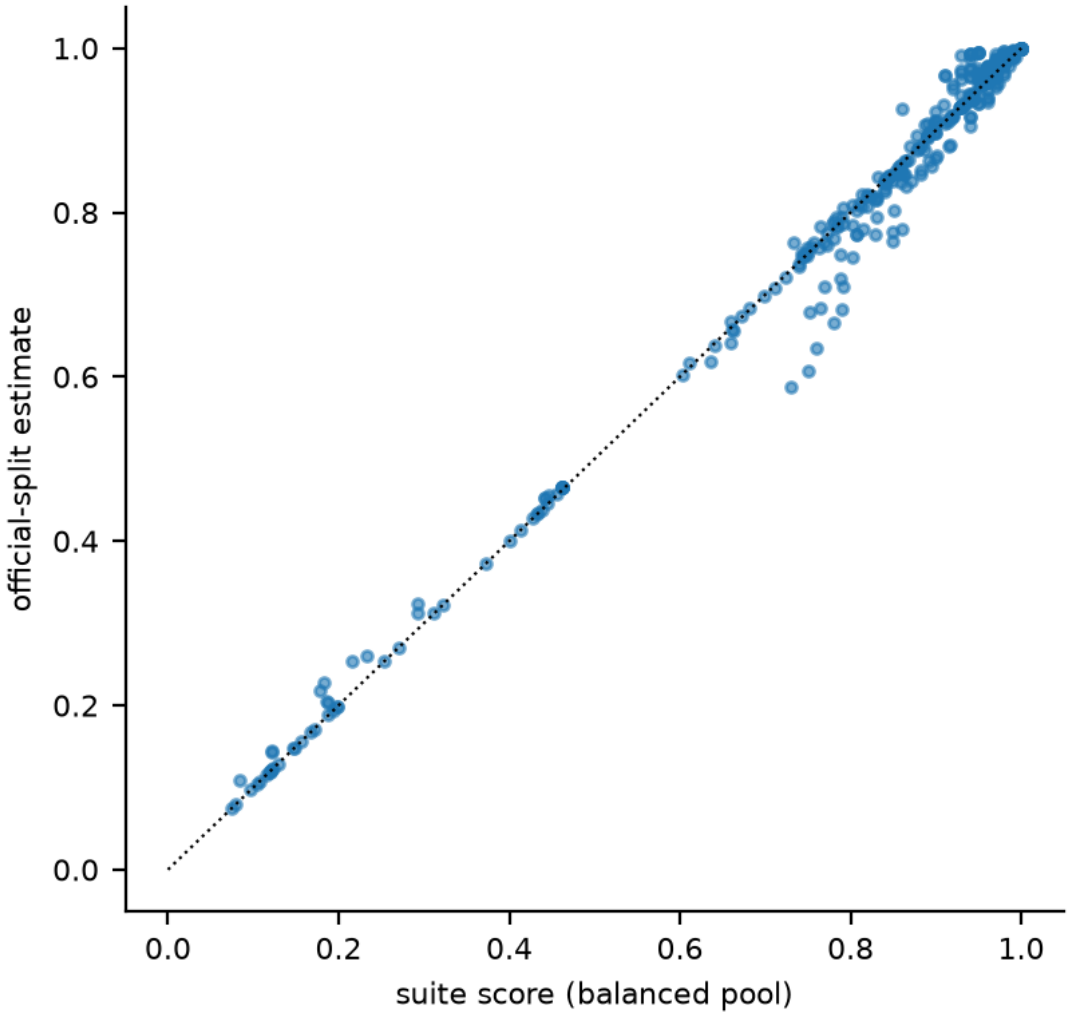


Figure 34: Suite score against post-stratified official-split estimate for every (dataset, model) pair under v1.

What it encodes. Every (dataset, model) pair placed by its suite score and its post-stratified official estimate under v1. **How to read it.** Distance from the diagonal is the effect of re-weighting for that cell; the cloud's tightness is how comparable the suite numbers are to published ones. **What it shows.** Almost every cell is within a few hundredths of the diagonal; the exceptions are the sparse matching sets and OpenSanctions, where the official class ratio differs most from 50/50.

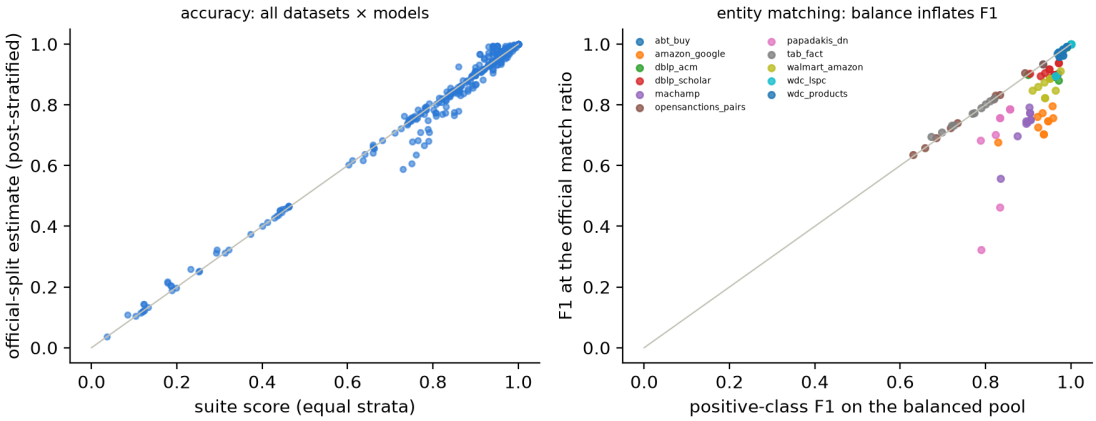


Figure 35: Suite score vs post-stratified official estimate; matching F1 at the official ratio.

What it encodes. Per dataset, the suite score (mean over the balanced, stratified pool) against the same answers re-weighted to the official split's stratum or class proportions; for the matching sets, positive-class F1 on the pool against F1 at the official match ratio. **How to read it.** Points on the diagonal are datasets whose pool already matches the official split; points below it are where the balanced pool overstates performance. **What it shows.** Accuracy moves by at most 0.09; F1 on the matching sets drops by up to 0.19 where matches are rare, through precision: at a 2 % match rate a 1 % false-positive rate halves precision.

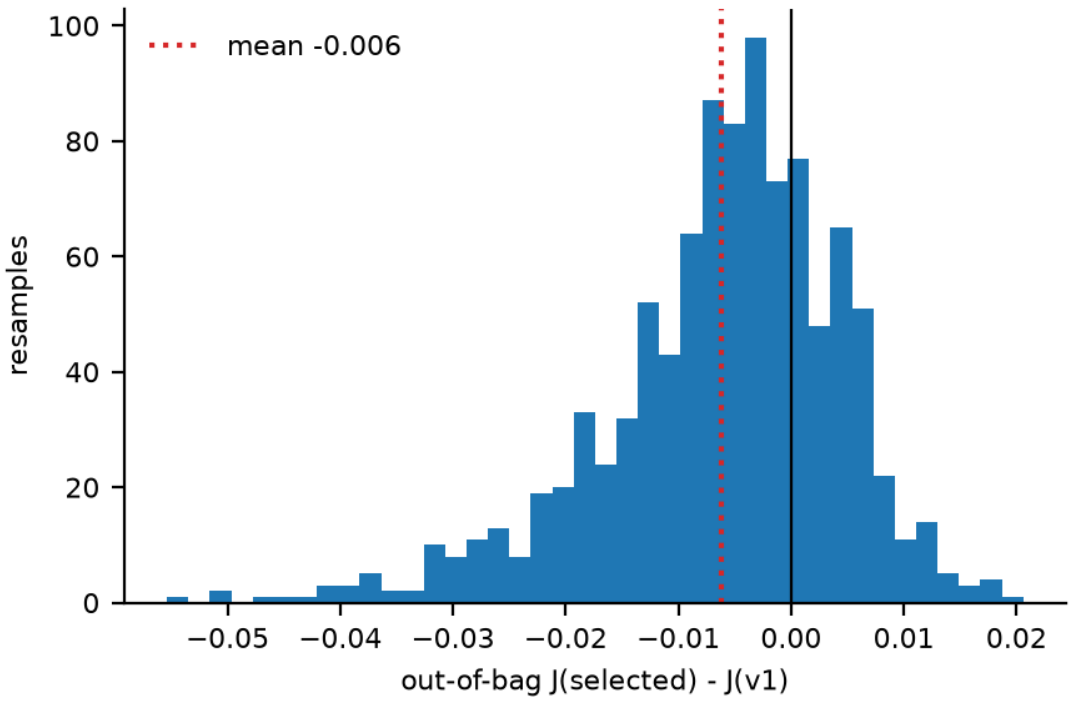


Figure 36: Paired out-of-bag difference between the selected weighting and v1 over 1,000 resamples.

What it encodes. For each of the 1,000 resamples, the out-of-bag J of the selected weighting minus the out-of-bag J of v1; the vertical line is zero. **How to read it.** Mass to the right of zero means the search beat v1 on held-out items in that resample. The mean and the 2.5–97.5 percentile interval are the paired test. **What it shows.** The distribution is centred at -0.006 with the interval straddling zero; the selected weighting is ahead in only 29 % of resamples.

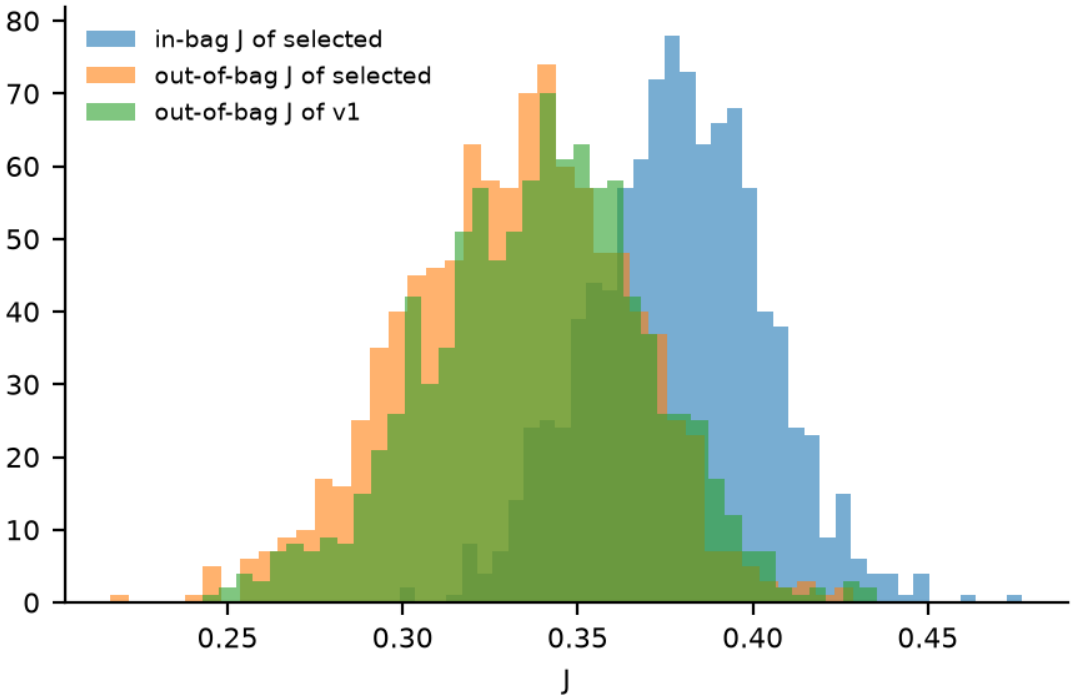


Figure 37: Distributions behind the optimism gap: in-bag J of the selected weighting, its out-of-bag J, and v1's out-of-bag J.

What it encodes. The three bootstrap distributions behind the optimism gap, as histograms over the 1,000 resamples. **How to read it.** The in-bag distribution of the selected weighting (shifted right) is what an un-validated search would report; the two out-of-bag distributions overlap almost completely. **What it shows.** The shift is the selection effect itself, about 0.046 of J; the overlap of the out-of-bag distributions is why no weighting can be declared better than v1.

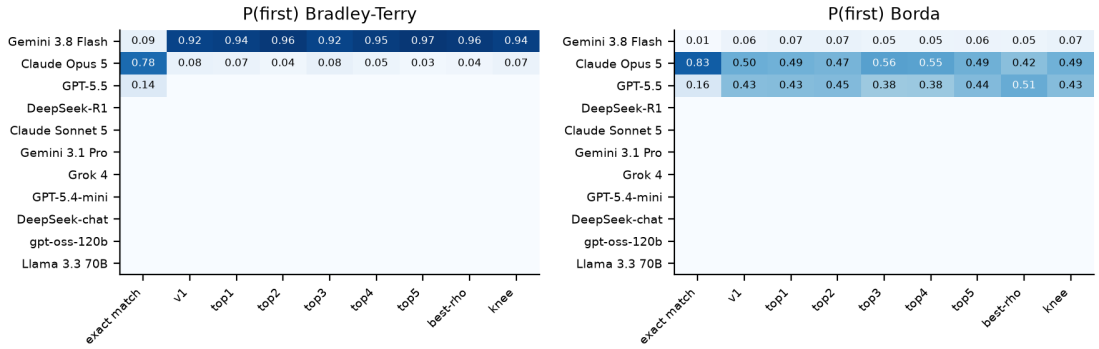


Figure 38: Bootstrap probability of first place for every model under every rule, by pooling method.

What it encodes. The bootstrap probability of first place for every model under every rule, by Bradley–Terry (left) and Borda (right) pooling. **How to read it.** Each column sums to one. A dark cell is a confident first place; a column with several mid-tone cells is a contested one. **What it shows.** Bradley–Terry is nearly certain about Gemini Flash under the graded rules and about Opus under exact match; Borda divides first place between Opus and GPT-5.5 under most rules. The pooling method, not only the scoring rule, decides the headline winner.

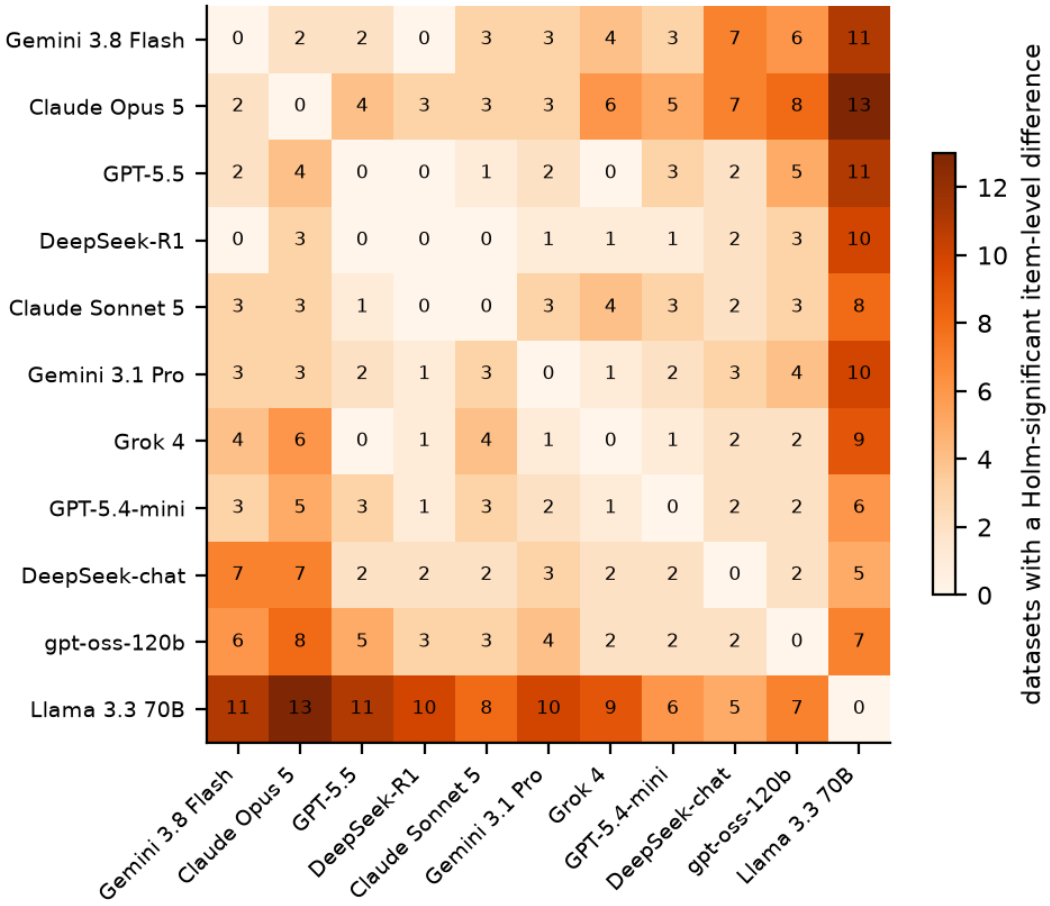


Figure 39: Number of datasets (of 31) in which the per-item paired test separates each pair of models under v1 (Holm within dataset).

What it encodes. For each pair of models, the number of datasets (of 31) in which a Holm-corrected per-item paired test separates them under v1. **How to read it.** A high count is a pair separated on many datasets; zero is a pair never separated at the item level even where the cross-dataset test suggests a difference. **What it shows.** The pairs involving the bottom two models are separated on many datasets; pairs inside the top group are separated on few. Item-level evidence agrees with the Friedman picture.

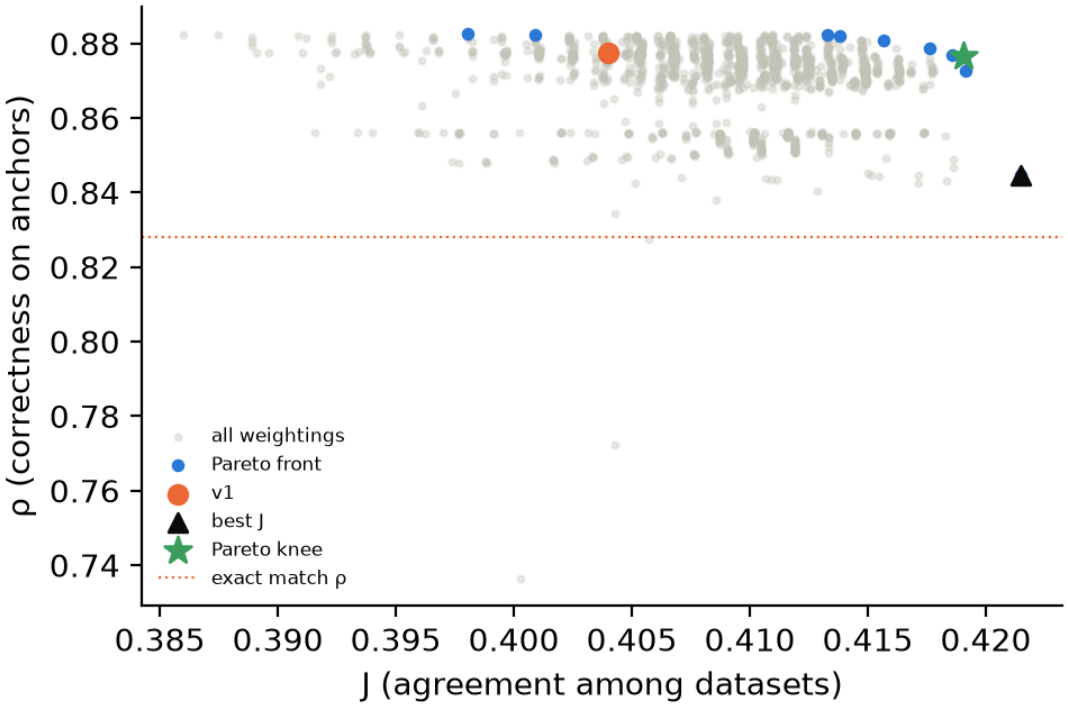


Figure 40: Agreement J against anchor correctness ρ for every weighting.

What it encodes. Every weighting placed by its agreement objective J (horizontal) and its Spearman ρ against the known utility of 31,380 synthetic answers (vertical); the Pareto front and the named points (v1, best J, best ρ, knee) are marked. **How to read it.** If agreement and correctness pulled together, the cloud would rise to the upper right. A front that bends away means the two criteria trade off. **What it shows.** The best-J weighting is among the worst by ρ (0.876) while v1 (0.878) sits near the ρ optimum (0.883); across the lattice J and ρ correlate -0.14. Agreement is not correctness, and v1 is close to the correct end.

What it encodes. Rows are the 31 datasets; columns are named weightings (v1, the best, the four corners, the uniform mix); each cell is that dataset's Kendall τ to the pooled ranking under that weighting (blue positive, red negative). J is the column mean. **How to read it.** A row that is red in every column is a dataset whose model ranking runs against the consensus whatever the weights; a column that is uniformly bluer than the others would be a weighting that genuinely improves agreement. **What it shows.** Rows differ far more than columns: the same 3 datasets are negative under every weighting and the same high-agreement datasets are blue everywhere. The weights redistribute a little τ between datasets; they do not change which datasets agree.

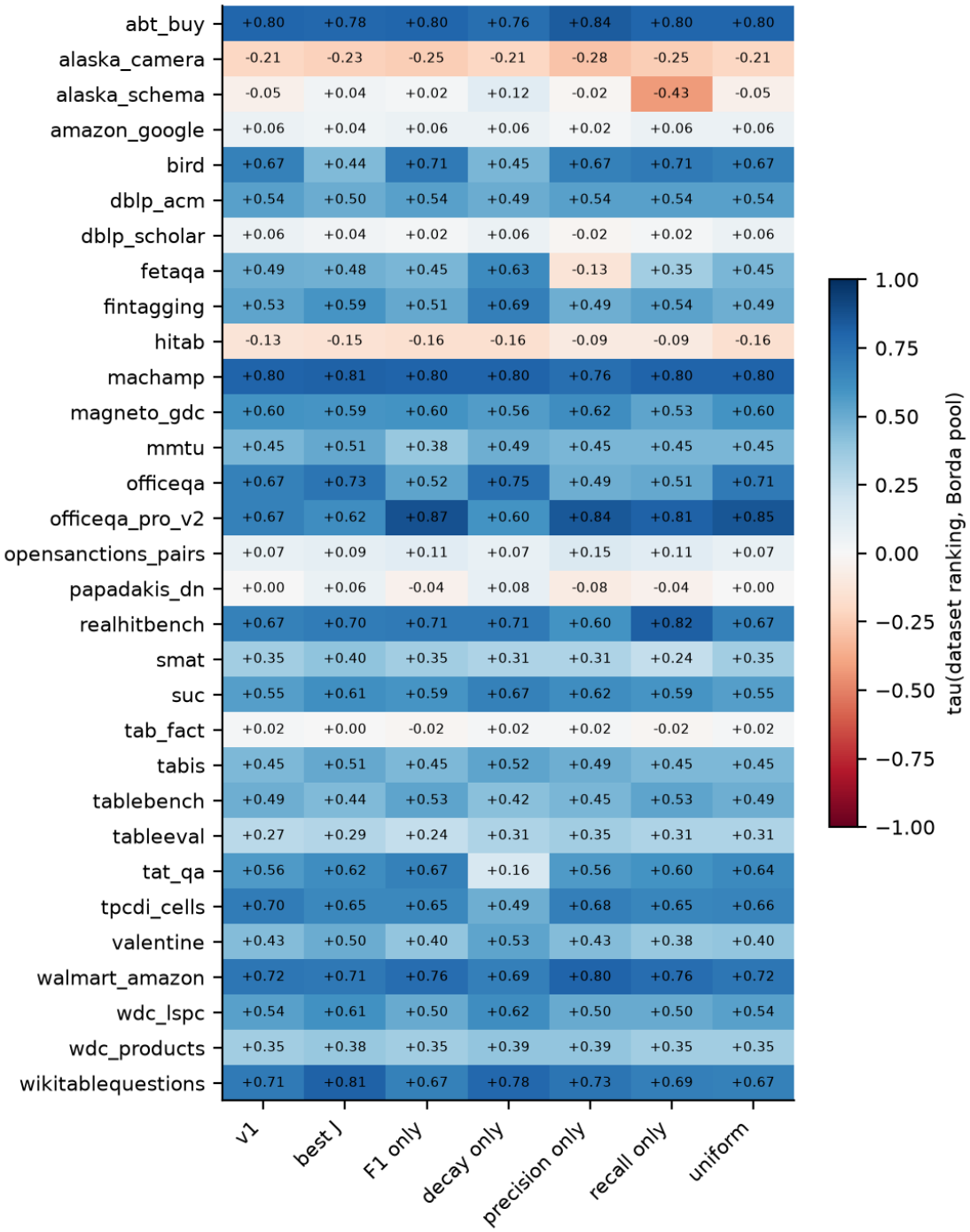


Figure 41: Per-dataset Kendall τ to the Borda pool under v1, the best weighting, the four single-ingredient corners and the uniform mix.

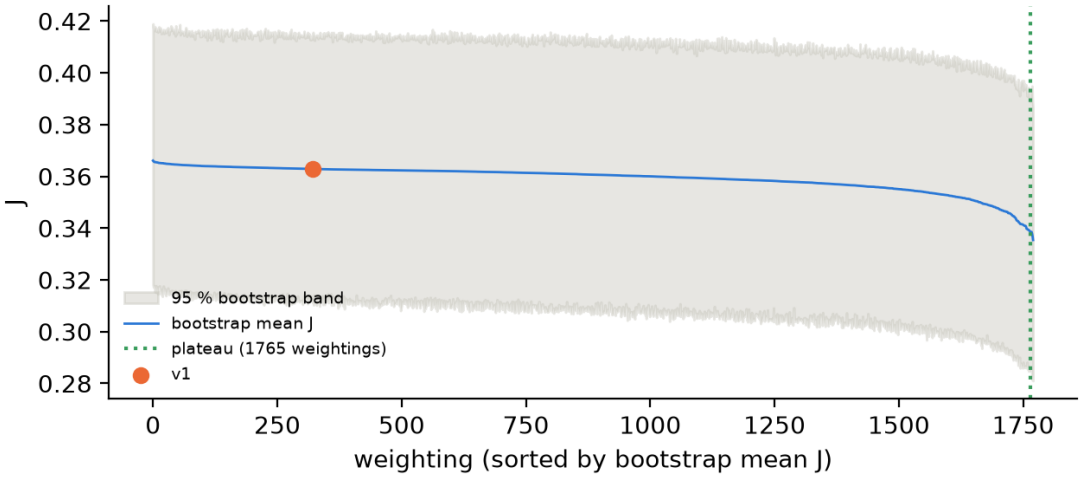


Figure 42: Bootstrap band over sorted weightings and the plateau boundary.

What it encodes. Weightings sorted by full-data J (horizontal) with the 95 % bootstrap band of each (shaded) and the plateau boundary: the set of weightings not significantly below the best in a paired bootstrap test. **How to read it.** A weighting is in the plateau when its band overlaps the best's in the paired sense; the boundary is where p drops below .05. **What it shows.** 1,754 of 1,771 weightings are in the plateau and v1 is inside it ($p = 0.66$); only the extreme precision-free, decay-dominated corner falls out. The practical plateau (within 0.02 of the best) holds 1,728 weightings.

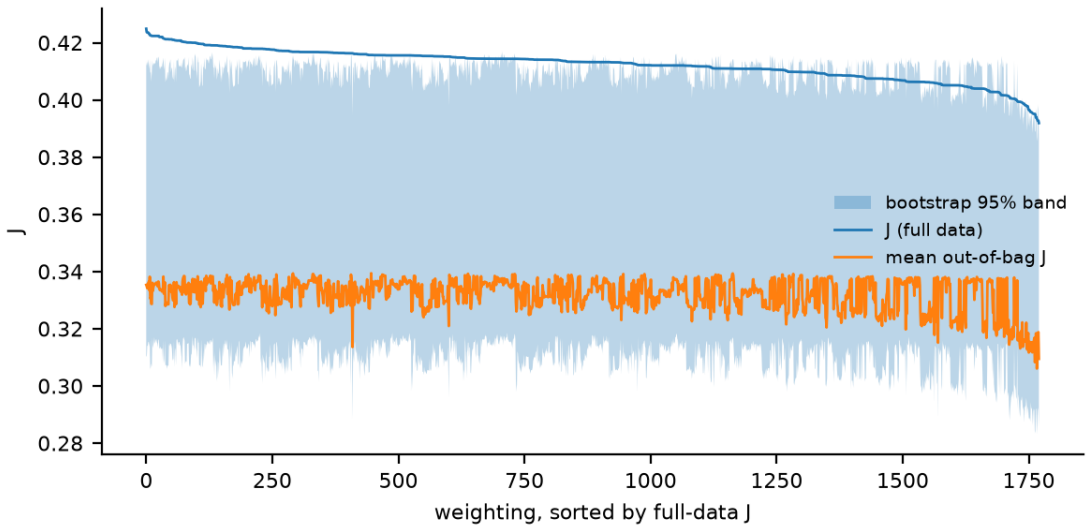


Figure 43: Full-data J, bootstrap band and mean out-of-bag J for every weighting in rank order.

What it encodes. Three curves over the sorted weightings: full-data J, the bootstrap mean with its band, and the mean out-of-bag J when each weighting is evaluated on items it was not selected on. **How to read it.** The vertical distance between the full-data curve and the out-of-bag curve is the optimism for every weighting, not only the winner. If the out-of-bag curve were steep, some weightings would be reliably better on new items. **What it shows.** The out-of-bag curve is flat and noisy across the entire lattice: the ordering of weightings on the full data does not reproduce on held-out items.

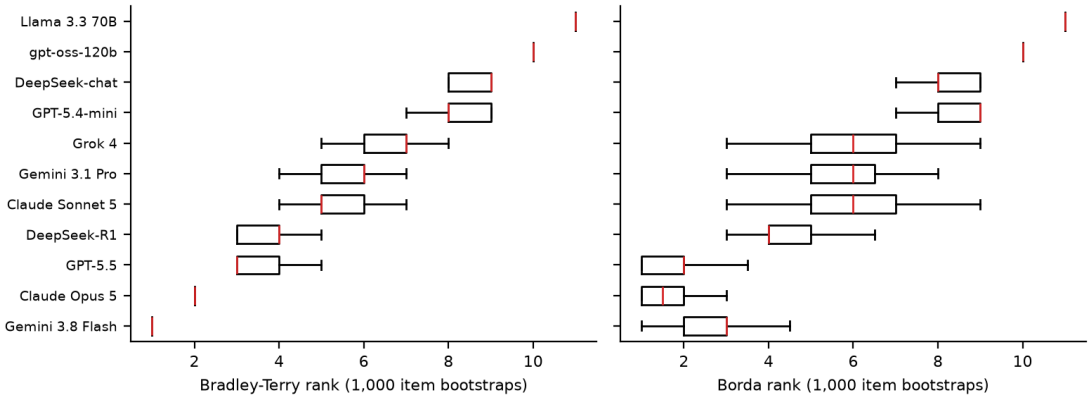


Figure 44: Bootstrap distribution of each model's rank under v1 (1,000 item resamples).

What it encodes. For each model, the distribution of its Bradley–Terry rank (left) and Borda rank (right) over 1,000 item bootstraps under v1. **How to read it.** A narrow box is a rank the data pin down; a wide box is a rank that depends on which items are drawn. Overlapping boxes are models whose order is not established. **What it shows.** Bradley–Terry pins Gemini Flash to first ($P = 0.92$) and the last two to their places; the middle four overlap. Borda spreads first place over three models, because it counts dataset-level ranks rather than item-level wins.

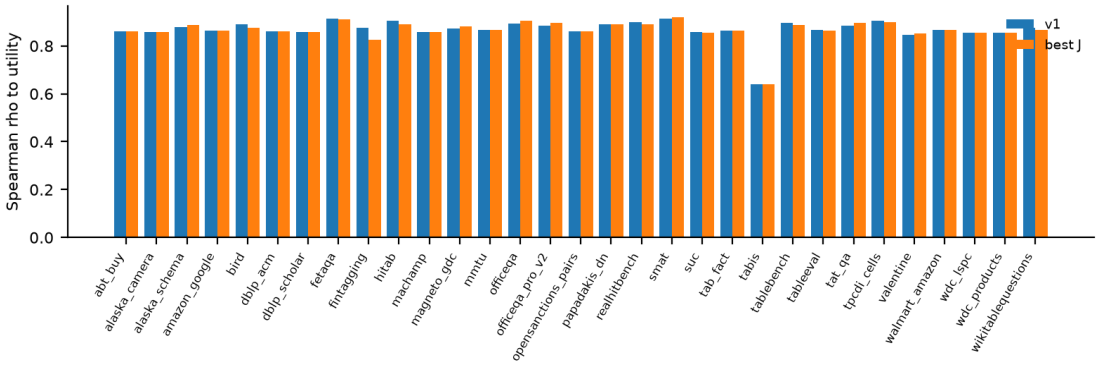


Figure 45: Anchor correctness (Spearman ρ between composite and known utility) per dataset under v1, the best-J and the best-ρ weightings.

What it encodes. Spearman ρ between score and known utility computed within each dataset's anchors, under v1, the best-J and the best-ρ weightings. **How to read it.** Datasets where the three bars differ are where the weighting changes how well the judge grades answers; equal bars mean the weighting is irrelevant there. **What it shows.** v1 and best-ρ are close everywhere; best-J falls behind on the numeric and list-answer datasets, which is where decay-only weighting loses the partial credit that token overlap provides.

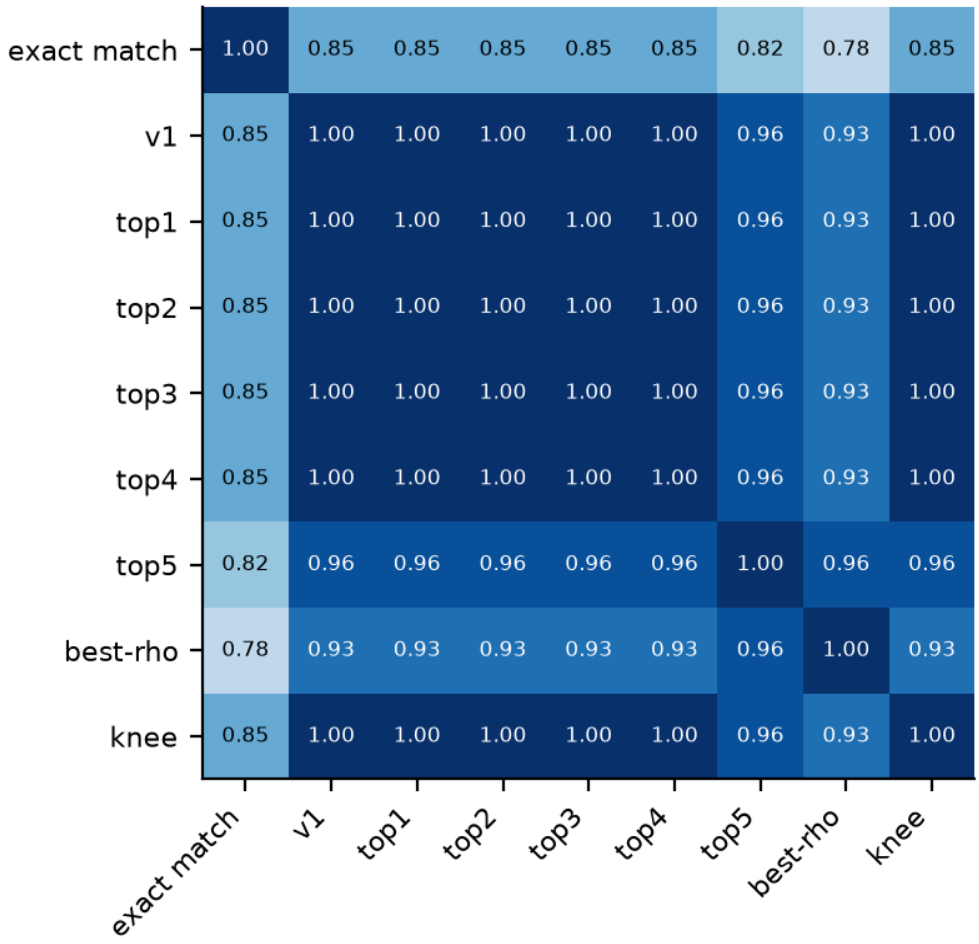


Figure 46: Kendall τ between the Bradley–Terry rankings produced by each pair of rules.

What it encodes. Kendall τ between the Bradley–Terry rankings produced by each pair of rules. **How to read it.** 1.00 means identical rankings; values near 0.8 mean a few adjacent swaps. **What it shows.** The graded rules agree with each other at 0.78 or above; exact match is the outlier because it ranks the reasoning-heavy models differently. Changing the weights inside the plateau does not change the leaderboard beyond adjacent swaps.

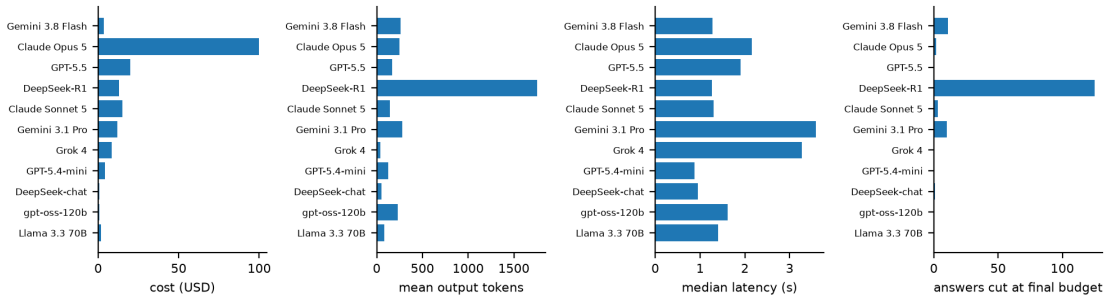


Figure 47: Run facts per model from the prediction cache: cost, output length, latency and answers still cut off at the final budget.

What it encodes. Per model: total API cost for the suite, mean output tokens, median latency, and the number of answers still cut off at the final budget. **How to read it.** Cost and output length are the price of a model's position; cut-offs are the answers scored as given because the model did not finish. **What it shows.** One model accounts for most of the bill, output length varies by more than an order of magnitude, and the residual cut-offs are concentrated in the reasoning model that writes its chain of thought into the output budget.

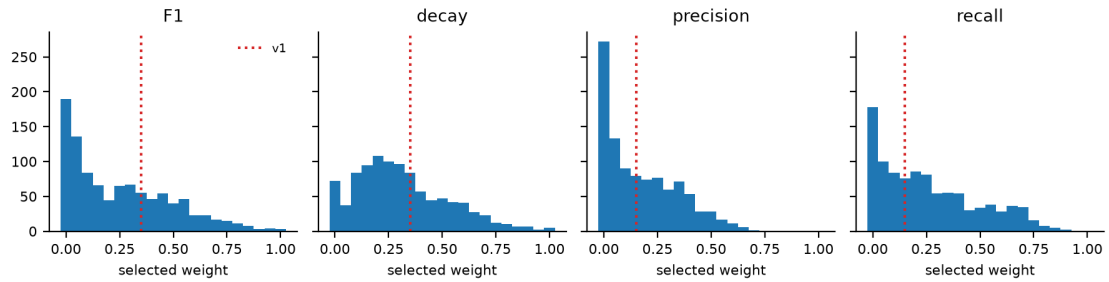


Figure 48: Weights selected as best on the in-bag items across 1,000 bootstrap resamples; dotted line = v1.

What it encodes. Four histograms: the weight on each ingredient of the weighting selected as in-bag best, over the 1,000 resamples; the dotted line is v1's value. **How to read it.** If the data pointed to a specific weighting, each histogram would be a narrow spike. Spread across the whole range means the "best" weighting depends on which items happen to be drawn. **What it shows.** 617 different weightings win across the resamples, and every weight spreads over most of its range (standard deviations f1 0.23, decay 0.21, precision 0.17, recall 0.23). The selection is close to uniform over the simplex.

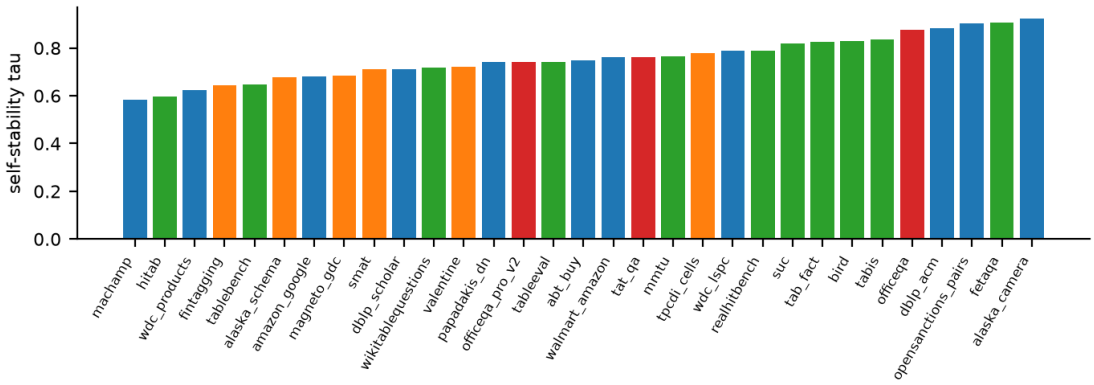


Figure 49: Self-stability of each dataset's model ranking (Kendall τ between bootstrap halves) under v1.

What it encodes. For each dataset, the Kendall τ between the model rankings obtained on two bootstrap halves of its items (self-stability). **How to read it.** A value near 1 means the dataset's ranking is reproducible from its own items; a low value means its ranking is driven by a few items. **What it shows.** Self-stability ranges from 0.58 (machamp) to 0.92 (alaska_camera). Low-stability sets are the ones whose disagreement with the consensus is partly noise.

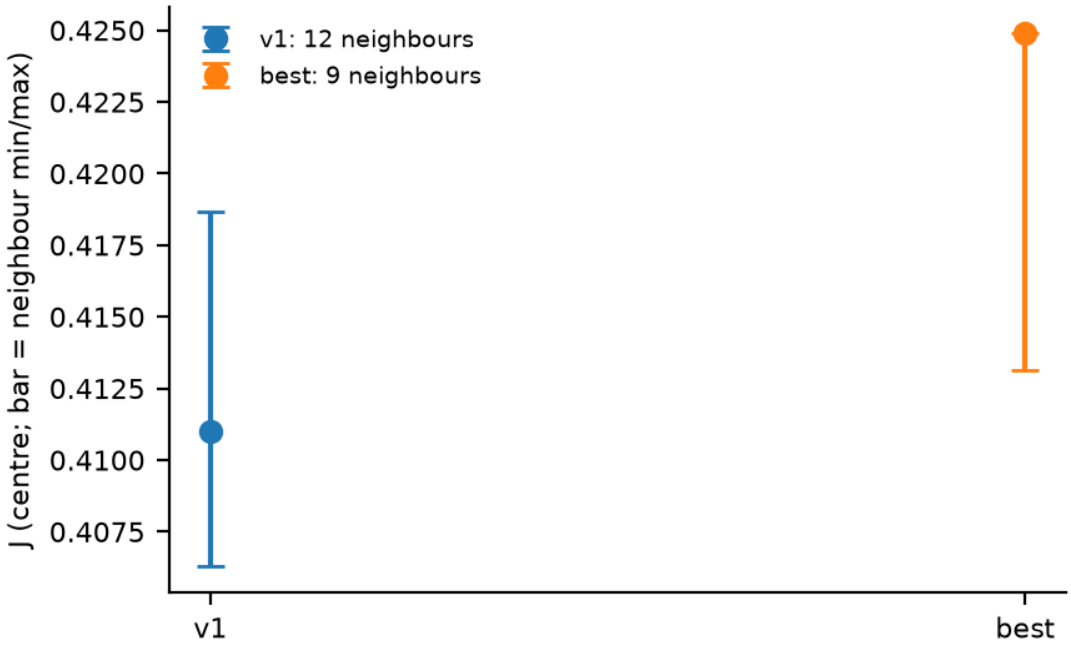


Figure 50: One-step sensitivity: J at v1 and at the best weighting with the range over all lattice neighbours one 0.05 move away.

What it encodes. For v1 and for the best weighting: the J at the point and the range of J over all neighbours one lattice step (0.05) away. **How to read it.** A long bar would mean a knife-edge optimum; a short bar means the neighbourhood is flat. **What it shows.** One step moves J by at most 0.012 around v1 and 0.012 around the best: both sit on flat ground, and the best is not a peak but a point on a plateau.

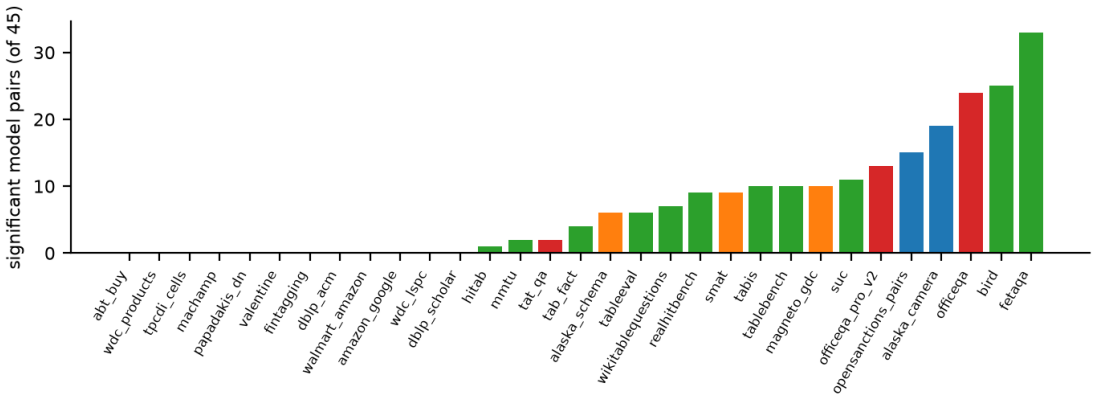


Figure 51: How many of the 45 model pairs each dataset separates at the item level under v1 (colour = family).

What it encodes. For each dataset, how many of the 45 model pairs the per-item paired test separates, coloured by family. **How to read it.** A dataset with zero separates no pair: either every model scores the same or the items are too few for the differences present. High bars are the datasets carrying the leaderboard's statistical power. **What it shows.** 12 datasets separate no pair, all of them entity-matching or ceiling sets; 3 datasets separate 20 or more, led by FeTaQA and the two OfficeQA sets. Power lives in a dozen table and document sets.

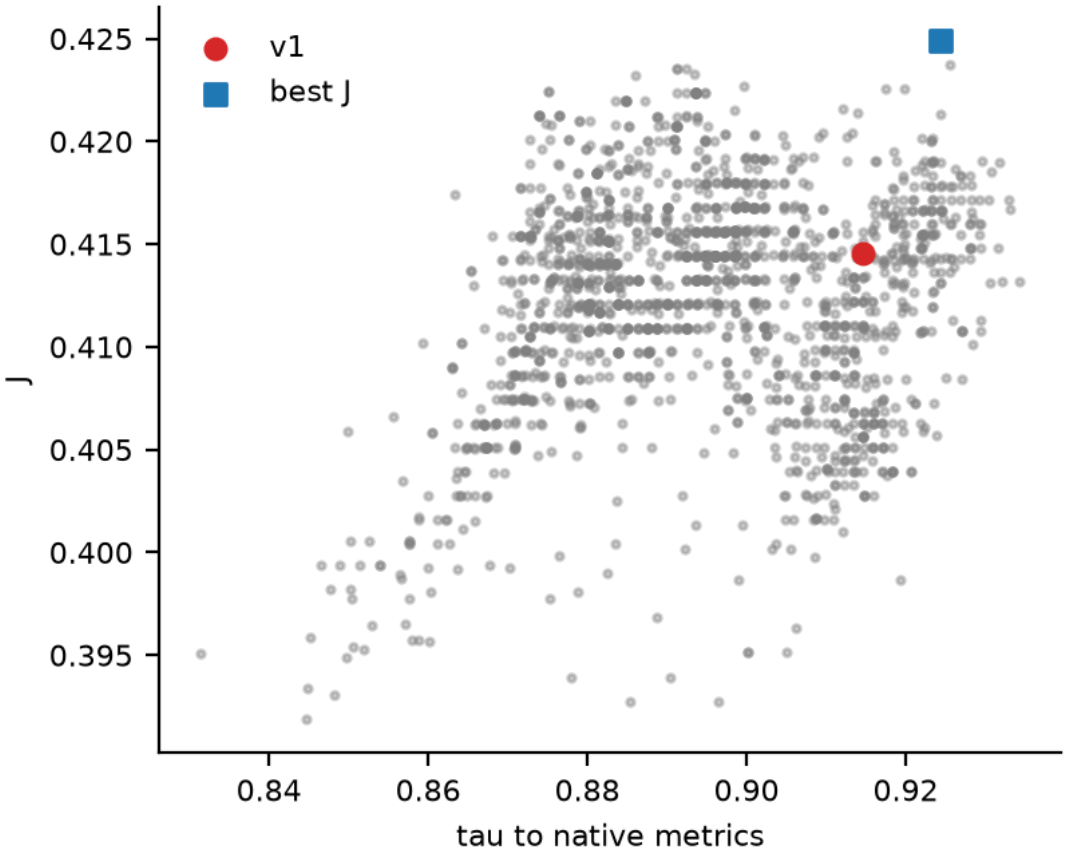


Figure 52: Agreement J against Kendall τ between each weighting's ranking and the native-metric ranking.

What it encodes. Each weighting placed by its agreement with the datasets' own native metrics (horizontal: Kendall τ between the composite's model ranking and the native-metric ranking) and its J (vertical). **How to read it.** Points to the right rank the models the way the datasets' own metrics would; points to the top make the datasets agree with each other. A strong diagonal would mean the two criteria coincide. **What it shows.** The correlation is weak (+0.12); v1 is among the most native-faithful weightings (0.915) while the best-J weighting is slightly less so. Agreement among datasets is a different criterion from fidelity to each dataset's own metric.

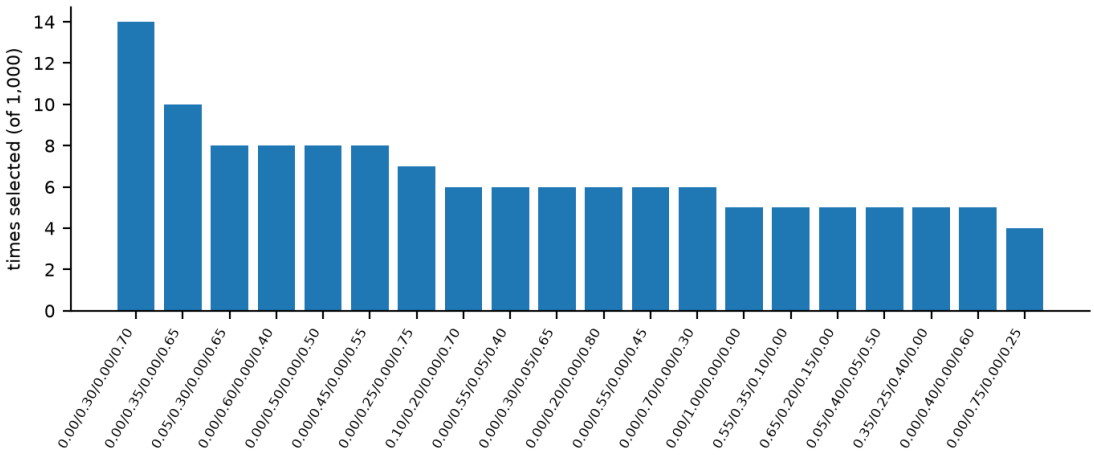


Figure 53: The twenty weightings most often selected as in-bag best (F1 / decay / precision / recall).

What it encodes. The twenty weightings most often chosen as in-bag best across the 1,000 resamples (labels give F1 / decay / precision / recall). **How to read it.** A dominant weighting would be selected hundreds of times. A flat bar chart with small counts means no weighting is consistently preferred. **What it shows.** The most-selected weighting wins only 14 of 1,000 resamples, and the winners include decay-dominated, F1-dominated and recall-dominated points alike.

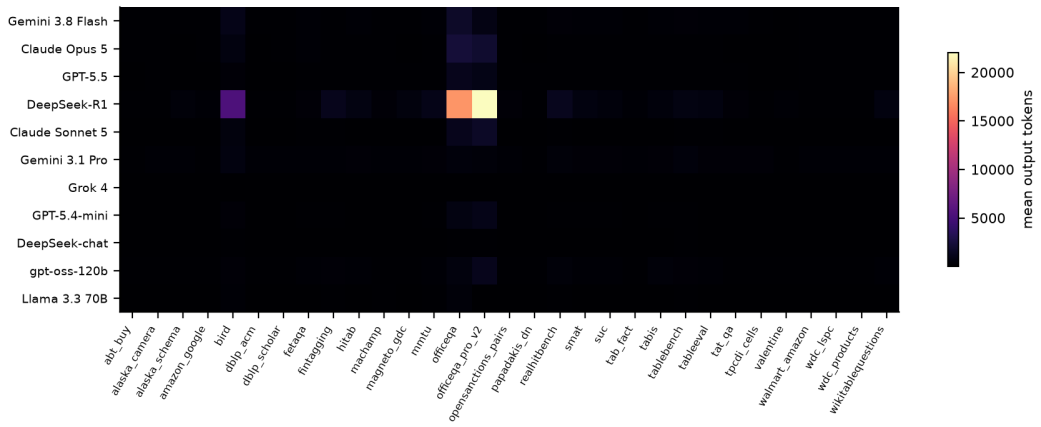


Figure 54: Mean output tokens per model and dataset.

What it encodes. Mean output tokens per model and dataset. **How to read it.** Columns show which tasks make every model write more (documents and SQL); rows show each model's verbosity. **What it shows.** The two OfficeQA sets and BIRD draw the longest replies from every model, and the reasoning models write several times more than the rest on the same items. These are the cells where the output budget bites.

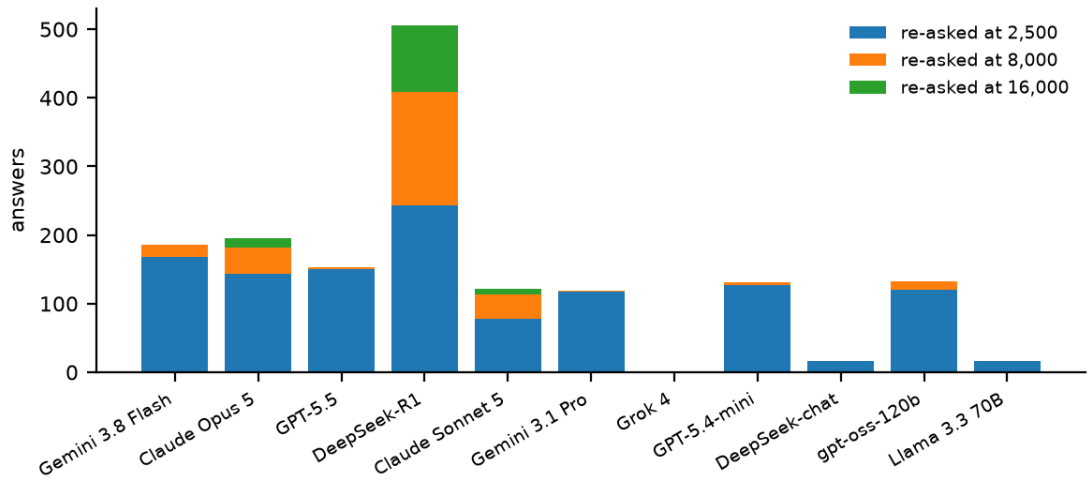


Figure 55: Truncation protocol: answers re-asked at 2,500, 8,000 and 16,000 output tokens per model.

What it encodes. Per model, how many answers were re-asked at each stage of the truncation protocol after being cut off without a final line at the previous budget. **How to read it.** The trigger is mechanical (finish reason "length" and no final-answer line), never correctness; a tall bar is a verbose or reasoning-heavy model, not a worse one. **What it shows.** The reasoning models dominate the re-asks; the short-answer models almost never hit the first budget.

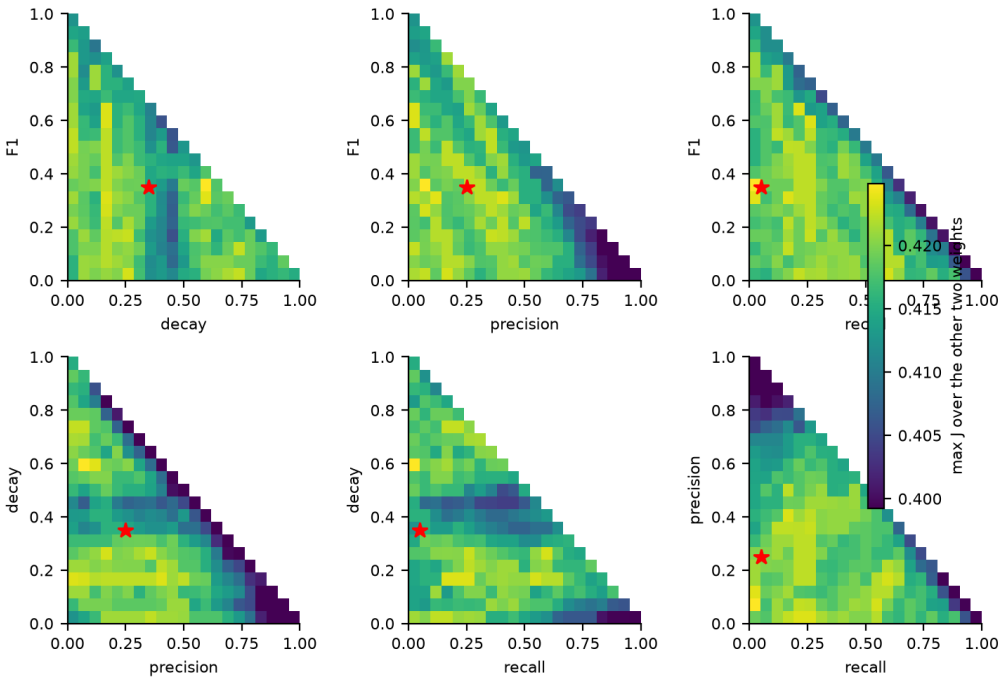


Figure 56: Best J attainable for each pair of weight values (maximised over the remaining two weights); red star = v1.

What it encodes. Six heatmaps, one per pair of ingredients; each cell is the best J attainable with those two weights fixed at the cell's values and the other two weights free. The red star is v1. **How to read it.** A ridge or a hot spot would point to a region of good weightings; a uniform colour means the pair does not constrain J. Only the lower-left triangle is feasible because the two weights cannot sum to more than one. **What it shows.** The feasible triangle is uniformly coloured except for a slightly cooler edge where precision dominates. There is no ridge to follow: the best J is reachable from almost any pair of values.

B All tables

Every saved table, exported from `data/`. Each is a separate file under `tables/`; include any of them in the main text with `\input`.

Table 8: Anchor operators: known utility, mean v1 score, exact-match rate and count.

op	utility	v1	em	n
no_answer	0.000	0.001	0.000	3083
wrong_random	0.000	0.097	0.041	1845
flip	0.000	0.000	0.000	1238
sign_flip	0.050	0.018	0.003	632
magnitude	0.050	0.091	0.013	632
near_miss_large	0.150	0.150	0.000	632
digit_typo	0.200	0.199	0.000	632
list_dump	0.250	0.373	0.013	1845
hedge	0.300	0.493	0.192	2974
shuffled	0.400	0.652	0.009	109
near_miss_medium	0.450	0.293	0.008	632
partial	0.500	0.345	0.061	1213
superset	0.600	0.467	0.045	1213
typo	0.600	0.334	0.101	1104
near_miss_small	0.750	0.405	0.097	632
paraphrase	0.950	0.965	0.930	3083
verbose	1.000	1.000	0.999	3083
identity	1.000	1.000	0.999	3083
tagged	1.000	1.000	0.999	3083
unit_variant	1.000	0.898	0.837	632

Table 9: Anchor correctness for the labelled rules.

label	rank	w_f1	w_decay	w_precision	w_recall	J	tau_native	rho	rho_ds	auc	wrong_mean	hedge_gap
uniform	220	0.250	0.250	0.250	0.250	0.418	0.901	0.875	0.866	0.994	0.033	-0.480
only_decay	601	0.000	1.000	0.000	0.000	0.415	0.915	0.772	0.804	0.991	0.040	-0.595
only_f1	974	1.000	0.000	0.000	0.000	0.412	0.877	0.856	0.846	0.989	0.030	-0.496
v1_original	1130	0.350	0.350	0.150	0.150	0.411	0.915	0.878	0.868	0.994	0.034	-0.506
only_recall	1763	0.000	0.000	0.000	1.000	0.395	0.831	0.736	0.741	0.989	0.030	-0.244
only_precision	1767	0.000	0.000	1.000	0.000	0.393	0.845	0.848	0.839	0.989	0.031	-0.587

Table 10: Coverage rule outcome per dataset.

dataset	answered items	items kept	models used	models dropped
abt_buy	100	100	11	–
alaska_camera	100	100	11	–
alaska_schema	93	93	11	–
amazon_google	100	100	11	–
bird	100	100	11	–
dblp_acm	100	100	11	–
dblp_scholar	100	100	11	–
fetaqa	100	100	11	–
fintagging	100	100	11	–
hitab	100	100	11	–
machamp	100	100	11	–
magneto_gdc	100	100	11	–
mmtu	100	100	11	–
officeqa	100	100	11	–
officeqa_pro_v2	90	90	11	–
opensanctions_pairs	100	100	11	–
papadakis_dn	100	100	11	–
realhitbench	100	100	11	–
smat	100	100	11	–
suc	100	100	11	–
tab_fact	100	100	11	–
tabis	100	100	11	–
tablebench	100	100	11	–
tableeval	100	100	11	–
tat_qa	100	100	11	–
tpcdi_cells	100	100	11	–
valentine	100	100	11	–
walmart_amazon	100	100	11	–
wdc_lspc	100	100	11	–
wdc_products	100	100	11	–
wikitablequestions	100	100	11	–

Table 11: Datasets: family, items answered, dominant answer type and native metric.

dataset	items	answer_type	native_metric	family
abt_buy	100	boolean	em	EM
alaska_camera	100	boolean	em	EM
alaska_schema	93	text	em	SM
amazon_google	100	boolean	em	EM
bird	100	numeric	em	TQA
dblp_acm	100	boolean	em	EM
dblp_scholar	100	boolean	em	EM
fetaqa	100	freeform	rouge_l	TQA
fintagging	100	text	em	SM
hitab	100	numeric	num_tol	TQA
machamp	100	boolean	em	EM
magneto_gdc	100	text	em	SM
mmtu	100	text	em	TQA
officeqa	100	numeric	num_tol	FIN
officeqa_pro_v2	90	numeric	num_tol	FIN
opensanctions_pairs	100	boolean	em	EM
papadakis_dn	100	boolean	em	EM
realhitbench	100	numeric	num_tol	TQA
smat	100	text	em	SM
suc	100	text	em	TQA
tab_fact	100	boolean	em	TQA
tabis	100	text	em	TQA
tablebench	100	numeric	num_tol	TQA
tableeval	100	numeric	num_tol	TQA
tat_qa	100	numeric	em	FIN
tpcdi_cells	100	numeric	num_tol	SM
valentine	100	text	em	SM
walmart_amazon	100	boolean	em	EM
wdc_lspc	100	boolean	em	EM
wdc_products	100	boolean	em	EM
wikitablequestions	100	numeric	em	TQA

Table 12: Sampled designs: mean and sd over 20 seeds of every coverage and performance measure.

design	best_J_found mean	best_J_found std	regret mean	regret std	centred_L2_discrepancy mean	centred_L2_discrepancy std	min_pairwise_L1 mean	min_pairwise_L1 std	share_with_a_zero_weight mean	share_with_a_zero_weight std	best_rank_in_exhaustive mean	best_rank_in_exhaustive std
dirichlet_lattice	0.4225	0.0006	0.0024	0.0006	0.5382	0.0178	0.1000	0.0000	0.2690	0.0370	17.4500	12.1373
lhs_simplex	0.4229	0.0011	0.0020	0.0011	0.5312	0.0083	0.1000	0.0000	0.2795	0.0268	14.7500	14.3522
uniform_lattice	0.4226	0.0007	0.0023	0.0007	0.5256	0.0177	0.1000	0.0000	0.4500	0.0543	15.0000	11.4340

Table 13: Sampled designs at a budget of 100 weightings, 20 seeds each, against the exhaustive optimum.

design	best J (mean)	sd	regret mean	regret max
dirichlet_lattice	0.4225	0.0006	0.0024	0.0037
lhs_simplex	0.4229	0.0011	0.0020	0.0041
uniform_lattice	0.4226	0.0007	0.0023	0.0037

Table 14: Dataset diagnostics under v1.

dataset	family	mean_score	spread_ across_models	floor	ceiling	share_ constant_ bootstrap_ rankings	ranking_self_ stability_tau	tau_to_ others_mean	n_items	exclude_ from_ objective
abt_buy	EM	0.971	0.060	no	no	0.000	0.748	0.367	100	no
alaska_camera	EM	0.945	0.190	no	no	0.000	0.924	-0.199	100	no
alaska_schema	SM	0.929	0.118	no	no	0.000	0.676	-0.022	93	no
amazon_google	EM	0.935	0.100	no	no	0.000	0.683	0.015	100	no
bird	TQA	0.191	0.209	no	no	0.000	0.830	0.325	100	no
dblp_acm	EM	0.987	0.090	no	no	0.000	0.883	0.328	100	no
dblp_scholar	EM	0.948	0.060	no	no	0.000	0.713	-0.015	100	no
fetaqa	TQA	0.387	0.192	no	no	0.000	0.906	0.264	100	no
fintagging	SM	0.884	0.051	no	no	0.000	0.644	0.290	100	no
hitab	TQA	0.772	0.068	no	no	0.000	0.597	-0.140	100	no
machamp	EM	0.887	0.070	no	no	0.000	0.584	0.399	100	no
magneto_gdc	SM	0.937	0.225	no	no	0.000	0.684	0.266	100	no
mmtu	TQA	0.874	0.110	no	no	0.000	0.765	0.274	100	no
officeqa	FIN	0.135	0.124	no	no	0.000	0.877	0.337	100	no
officeqa_pro_v2	FIN	0.151	0.166	no	no	0.000	0.741	0.354	90	no
opensanctions_pairs	EM	0.835	0.240	no	no	0.000	0.905	0.038	100	no
papadakis_dn	EM	0.940	0.030	no	no	0.002	0.741	0.025	100	no
realhitbench	TQA	0.677	0.143	no	no	0.000	0.788	0.363	100	no
smat	SM	0.916	0.150	no	no	0.000	0.710	0.185	100	no
suc	TQA	0.917	0.246	no	no	0.000	0.818	0.294	100	no
tab_fact	TQA	0.802	0.200	no	no	0.000	0.827	0.003	100	no
tabis	TQA	0.918	0.210	no	no	0.000	0.836	0.267	100	no
tablebench	TQA	0.824	0.218	no	no	0.000	0.649	0.233	100	no
tableeval	TQA	0.755	0.166	no	no	0.000	0.742	0.177	100	no
tat_qa	FIN	0.804	0.109	no	no	0.000	0.762	0.274	100	no
tpcdi_cells	SM	0.456	0.020	no	no	0.000	0.779	0.330	100	no
valentine	SM	0.985	0.030	no	yes	0.000	0.722	0.236	100	yes
walmart_amazon	EM	0.959	0.060	no	no	0.000	0.761	0.362	100	no
wdc_lspc	EM	0.982	0.040	no	yes	0.004	0.788	0.282	100	yes
wdc_products	EM	0.978	0.020	no	yes	0.006	0.623	0.199	100	yes
wikitablequestions	TQA	0.848	0.132	no	no	0.000	0.719	0.361	100	no

Table 15: Mean v1 composite per model within each family.

model	entity matching	finance	schema matching	table QA
Gemini 3.8 Flash	0.945	0.414	0.854	0.766
Claude Opus 5	0.947	0.449	0.869	0.745
GPT-5.5	0.950	0.388	0.864	0.751
DeepSeek-R1	0.953	0.378	0.865	0.733
Claude Sonnet 5	0.942	0.399	0.862	0.724
Gemini 3.1 Pro	0.944	0.355	0.852	0.749
Grok 4	0.948	0.354	0.851	0.738
GPT-5.4-mini	0.947	0.347	0.859	0.716
DeepSeek-chat	0.941	0.342	0.852	0.711
gpt-oss-120b	0.915	0.324	0.848	0.698
Llama 3.3 70B	0.936	0.328	0.777	0.633

Table 16: Quantiles of J over the lattice.

count	mean	std	min	1%	5%	10%	25%	50%	75%	90%	95%	99%	max
1771.0000	0.4124	0.0053	0.3919	0.3962	0.4028	0.4051	0.4095	0.4132	0.4157	0.4185	0.4200	0.4224	0.4249

Table 17: v1, the uniform mix and the four single-ingredient corners on the lattice.

label	F1 / decay / prec / rec	rank	J	J Kemeny	J Copeland	J BT	tau native
uniform	0.25 / 0.25 / 0.25 / 0.25	220	0.418	0.423	0.423	0.405	0.901
only_decay	0.00 / 1.00 / 0.00 / 0.00	601	0.415	0.419	0.419	0.378	0.915
only_f1	1.00 / 0.00 / 0.00 / 0.00	974	0.412	0.415	0.422	0.388	0.877
only_recall	0.00 / 0.00 / 0.00 / 1.00	1763	0.395	0.395	0.399	0.365	0.831
only_precision	0.00 / 0.00 / 1.00 / 0.00	1767	0.393	0.394	0.402	0.342	0.845

Table 18: The 25 best weightings by J with the objective under the other pooling rules.

rank	F1 / decay / prec / rec	J	J Kemeny	J Copeland	J BT	tau native	min agree
1	0.35 / 0.60 / 0.05 / 0.00	0.425	0.425	0.425	0.416	0.924	-0.229
2	0.30 / 0.60 / 0.10 / 0.00	0.424	0.424	0.424	0.415	0.926	-0.229
3	0.60 / 0.15 / 0.00 / 0.25	0.424	0.423	0.423	0.400	0.891	-0.229
4	0.65 / 0.15 / 0.00 / 0.20	0.424	0.423	0.423	0.400	0.891	-0.229
5	0.30 / 0.15 / 0.40 / 0.15	0.424	0.422	0.422	0.392	0.892	-0.229
6	0.00 / 0.25 / 0.15 / 0.60	0.423	0.421	0.421	0.388	0.893	-0.267
7	0.25 / 0.15 / 0.05 / 0.55	0.423	0.421	0.421	0.392	0.886	-0.267
8	0.15 / 0.25 / 0.35 / 0.25	0.423	0.426	0.426	0.399	0.900	-0.208
9	0.10 / 0.75 / 0.00 / 0.15	0.423	0.424	0.424	0.400	0.920	-0.245
10	0.00 / 0.75 / 0.05 / 0.20	0.423	0.424	0.424	0.400	0.917	-0.245
11	0.50 / 0.00 / 0.35 / 0.15	0.422	0.422	0.430	0.394	0.875	-0.283
12	0.55 / 0.00 / 0.35 / 0.10	0.422	0.422	0.430	0.394	0.875	-0.283
20	0.40 / 0.15 / 0.25 / 0.20	0.422	0.422	0.422	0.401	0.894	-0.229
26	0.20 / 0.15 / 0.50 / 0.15	0.422	0.421	0.421	0.391	0.891	-0.229
25	0.55 / 0.15 / 0.05 / 0.25	0.422	0.422	0.422	0.398	0.892	-0.229
24	0.50 / 0.15 / 0.15 / 0.20	0.422	0.422	0.422	0.398	0.894	-0.229
23	0.45 / 0.15 / 0.20 / 0.20	0.422	0.422	0.422	0.398	0.894	-0.229
22	0.40 / 0.15 / 0.20 / 0.25	0.422	0.422	0.422	0.398	0.894	-0.229
21	0.35 / 0.15 / 0.25 / 0.25	0.422	0.422	0.422	0.398	0.894	-0.229
18	0.30 / 0.15 / 0.35 / 0.20	0.422	0.422	0.422	0.402	0.894	-0.229
19	0.35 / 0.15 / 0.30 / 0.20	0.422	0.422	0.422	0.402	0.894	-0.229
16	0.25 / 0.15 / 0.35 / 0.25	0.422	0.422	0.422	0.401	0.894	-0.229
15	0.20 / 0.15 / 0.40 / 0.25	0.422	0.422	0.422	0.390	0.894	-0.229
14	0.45 / 0.15 / 0.15 / 0.25	0.422	0.422	0.422	0.398	0.895	-0.229
13	0.15 / 0.20 / 0.40 / 0.25	0.422	0.422	0.422	0.393	0.895	-0.229

Table 19: Full leaderboard under rule best-rho; weights F1/decay/prec/rec = 0.00 / 0.35 / 0.60 / 0.05.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.096	0.015	1.000	1.000	2.000	0.958	3.000	1.000	4.000	0.053	3.000	3.000	3.000	3.000	2.000	0.814
Claude Opus 5	0.078	0.000	2.000	1.000	2.000	0.042	2.000	1.000	3.000	0.423	1.000	1.000	1.000	2.000	1.000	0.813
DeepSeek-R1	0.035	0.015	3.000	3.000	6.000	0.000	4.000	3.000	7.000	0.003	5.000	5.000	4.000	4.000	5.000	0.804
GPT-5.5	0.034	0.015	4.000	3.000	6.000	0.000	1.000	1.000	3.000	0.509	2.000	2.000	2.000	1.000	3.000	0.810
Claude Sonnet 5	0.030	0.015	5.000	3.000	7.000	0.000	5.000	4.000	8.000	0.000	4.000	5.000	5.000	5.000	4.000	0.799
Gemini 3.1 Pro	0.015	0.015	6.000	4.000	8.000	0.000	6.000	4.000	7.000	0.000	6.000	5.000	6.000	6.000	6.000	0.801
GPT-5.4-mini	0.014	0.015	7.000	4.000	8.000	0.000	8.000	6.988	9.000	0.000	8.000	7.000	8.000	8.000	8.000	0.796
Grok 4	0.003	0.015	8.000	6.000	8.000	0.000	7.000	4.000	8.000	0.000	7.000	8.000	7.000	7.000	7.000	0.798
DeepSeek-chat	-0.039	0.015	9.000	9.000	9.000	0.000	9.000	7.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.786
gpt-oss-120b	-0.129	0.015	10.000	10.000	11.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.768
Llama 3.3 70B	-0.137	0.015	11.000	10.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.744

Table 20: Full leaderboard under rule exact match; weights F1/decay/prec/rec = –.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.044	0.000	1.000	1.000	3.000	0.777	1.000	1.000	2.000	0.826	1.000	1.000	1.000	1.000	1.000	0.768
GPT-5.5	0.035	0.015	2.000	1.000	4.000	0.136	2.000	1.000	4.000	0.156	2.000	2.000	2.000	2.000	2.000	0.764
Gemini 3.8 Flash	0.034	0.015	3.000	1.000	4.000	0.085	3.000	2.000	5.512	0.009	3.000	3.000	3.000	3.000	4.000	0.763
DeepSeek-R1	0.023	0.015	4.000	2.000	7.000	0.002	4.000	2.000	7.000	0.001	4.000	4.000	4.000	4.000	3.000	0.758
Gemini 3.1 Pro	0.018	0.015	5.000	3.000	8.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.755
Grok 4	0.016	0.015	6.000	4.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	7.000	6.000	7.000	0.754
Claude Sonnet 5	0.010	0.015	7.000	4.000	8.000	0.000	5.000	3.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.751
GPT-5.4-mini	0.006	0.015	8.000	5.000	8.000	0.000	9.000	7.988	9.000	0.000	8.000	8.000	8.000	9.000	9.000	0.749
DeepSeek-chat	-0.018	0.015	9.000	9.000	10.000	0.000	8.000	7.000	9.000	0.000	9.000	9.000	9.000	8.000	8.000	0.737
gpt-oss-120b	-0.037	0.015	10.000	9.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.728
Llama 3.3 70B	-0.132	0.015	11.000	11.000	11.000	0.000	11.000	11.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.680

Table 21: Leaderboard under the deployed rule (v1 weights): Bradley–Terry strength ± SE vs reference with the 95 % bootstrap rank interval, Borda rank with its interval, and the mean composite over the suite.

rank	model	BT strength	SE	BT 95%	P(first) BT	Borda rank	Borda 95%	P(first) Borda	mean score
1	Gemini 3.8 Flash	0.112	0.015	[1, 2]	0.920	3	[1, 4]	0.060	0.812
2	Claude Opus 5	0.097	0.000	[1, 2]	0.080	2	[1, 3]	0.500	0.812
3	GPT-5.5	0.042	0.015	[3, 5]	0.000	1	[1, 3]	0.430	0.808
4	DeepSeek-R1	0.037	0.015	[3, 6]	0.000	4	[3, 7]	0.000	0.801
5	Claude Sonnet 5	0.023	0.015	[3, 7]	0.000	5	[4, 7]	0.000	0.796
6	Gemini 3.1 Pro	0.021	0.015	[4, 7]	0.000	6	[4, 7]	0.000	0.799
7	Grok 4	0.016	0.015	[4, 7]	0.000	7	[4, 7]	0.000	0.797
8	GPT-5.4-mini	-0.022	0.015	[8, 9]	0.000	9	[7, 9]	0.000	0.790
9	DeepSeek-chat	-0.031	0.015	[8, 9]	0.000	8	[7, 9]	0.000	0.784
10	gpt-oss-120b	-0.111	0.015	[10, 10]	0.000	10	[10, 11]	0.000	0.767
11	Llama 3.3 70B	-0.183	0.015	[11, 11]	0.000	11	[10, 11]	0.000	0.738

Table 22: Full leaderboard under rule knee; weights F1/decay/prec/rec = 0.35 / 0.60 / 0.05 / 0.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.111	0.015	1.000	1.000	2.000	0.935	3.000	1.000	4.000	0.067	3.000	3.000	3.000	3.000	2.000	0.810
Claude Opus 5	0.095	0.000	2.000	1.000	2.000	0.065	2.000	1.000	3.000	0.488	1.000	1.000	1.000	2.000	1.000	0.810
GPT-5.5	0.042	0.015	3.000	3.000	5.025	0.000	1.000	1.000	3.000	0.434	2.000	2.000	2.000	1.000	3.000	0.804
DeepSeek-R1	0.034	0.015	4.000	3.000	7.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.797
Claude Sonnet 5	0.024	0.015	5.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	4.000	5.000	5.000	4.000	0.794
Gemini 3.1 Pro	0.022	0.015	6.000	3.000	7.000	0.000	6.500	4.000	7.000	0.000	7.000	7.000	6.000	6.000	6.000	0.794
Grok 4	0.017	0.015	7.000	4.000	7.000	0.000	6.500	4.000	8.000	0.000	6.000	6.000	7.000	7.000	7.000	0.792
GPT-5.4-mini	-0.019	0.015	8.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	8.000	8.000	8.000	0.788
DeepSeek-chat	-0.030	0.015	9.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.776
gpt-oss-120b	-0.113	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.762
Llama 3.3 70B	-0.184	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.730

Table 23: Full leaderboard under rule top1; weights F1/decay/prec/rec = 0.35 / 0.60 / 0.05 / 0.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.111	0.015	1.000	1.000	2.000	0.935	3.000	1.000	4.000	0.067	3.000	3.000	3.000	3.000	2.000	0.810
Claude Opus 5	0.095	0.000	2.000	1.000	2.000	0.065	2.000	1.000	3.000	0.488	1.000	1.000	1.000	2.000	1.000	0.810
GPT-5.5	0.042	0.015	3.000	3.000	5.025	0.000	1.000	1.000	3.000	0.434	2.000	2.000	2.000	1.000	3.000	0.804
DeepSeek-R1	0.034	0.015	4.000	3.000	7.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.797
Claude Sonnet 5	0.024	0.015	5.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	4.000	5.000	5.000	4.000	0.794
Gemini 3.1 Pro	0.022	0.015	6.000	3.000	7.000	0.000	6.500	4.000	7.000	0.000	7.000	7.000	6.000	6.000	6.000	0.794
Grok 4	0.017	0.015	7.000	4.000	7.000	0.000	6.500	4.000	8.000	0.000	6.000	6.000	7.000	7.000	7.000	0.792
GPT-5.4-mini	-0.019	0.015	8.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	8.000	8.000	8.000	0.788
DeepSeek-chat	-0.030	0.015	9.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.776
gpt-oss-120b	-0.113	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.762
Llama 3.3 70B	-0.184	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.730

Table 24: Full leaderboard under rule top2; weights F1/decay/prec/rec = 0.30 / 0.60 / 0.10 / 0.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.111	0.015	1.000	1.000	2.000	0.963	3.000	1.000	4.000	0.069	3.000	3.000	3.000	3.000	2.000	0.810
Claude Opus 5	0.092	0.000	2.000	1.000	2.000	0.037	2.000	1.000	3.000	0.472	1.000	1.000	1.000	2.000	1.000	0.810
GPT-5.5	0.043	0.015	3.000	3.000	5.000	0.000	1.000	1.000	3.000	0.445	2.000	2.000	2.000	1.000	3.000	0.804
DeepSeek-R1	0.034	0.015	4.000	3.000	6.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.797
Claude Sonnet 5	0.024	0.015	5.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	4.000	5.000	5.000	4.000	0.794
Gemini 3.1 Pro	0.023	0.015	6.000	3.000	7.000	0.000	6.500	4.000	7.000	0.000	7.000	7.000	6.000	6.000	6.000	0.794
Grok 4	0.015	0.015	7.000	4.000	7.000	0.000	6.500	4.000	8.000	0.000	6.000	6.000	7.000	7.000	7.000	0.793
GPT-5.4-mini	-0.016	0.015	8.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	8.000	8.000	8.000	0.789
DeepSeek-chat	-0.031	0.015	9.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.776
gpt-oss-120b	-0.118	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.762
Llama 3.3 70B	-0.178	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.731

Table 25: Full leaderboard under rule top3; weights F1/decay/prec/rec = 0.60 / 0.15 / 0.00 / 0.25.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.015	1.000	1.000	2.000	0.920	3.000	1.000	4.000	0.052	3.000	3.000	3.000	3.000	3.000	0.813
Claude Opus 5	0.101	0.000	2.000	1.000	2.000	0.080	2.000	1.000	3.000	0.559	1.000	1.000	1.000	2.000	1.000	0.812
GPT-5.5	0.040	0.015	3.000	3.000	6.000	0.000	1.000	1.000	3.000	0.378	2.000	2.000	2.000	1.000	2.000	0.810
DeepSeek-R1	0.038	0.015	4.000	3.000	6.000	0.000	4.000	3.000	7.000	0.002	4.000	4.000	4.000	4.000	5.000	0.803
Claude Sonnet 5	0.022	0.015	5.000	4.000	7.000	0.000	5.500	4.000	8.000	0.000	6.000	6.000	5.000	5.000	4.000	0.796
Gemini 3.1 Pro	0.021	0.015	6.000	4.000	7.000	0.000	5.500	4.000	7.000	0.000	5.000	5.000	6.000	6.000	6.000	0.802
Grok 4	0.016	0.015	7.000	4.000	7.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	7.000	7.000	7.000	0.799
GPT-5.4-mini	-0.030	0.015	8.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	8.000	9.000	9.000	0.787
DeepSeek-chat	-0.030	0.015	9.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	9.000	8.000	8.000	0.788
gpt-oss-120b	-0.104	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.770
Llama 3.3 70B	-0.190	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.742

Table 26: Full leaderboard under rule top4; weights F1/decay/prec/rec = 0.65 / 0.15 / 0.00 / 0.20.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.015	1.000	1.000	2.000	0.947	3.000	1.000	4.000	0.051	3.000	3.000	3.000	3.000	3.000	0.813
Claude Opus 5	0.099	0.000	2.000	1.000	2.000	0.053	2.000	1.000	3.000	0.553	1.000	1.000	1.000	2.000	1.000	0.812
GPT-5.5	0.041	0.015	3.000	3.000	6.000	0.000	1.000	1.000	3.000	0.382	2.000	2.000	2.000	1.000	2.000	0.810
DeepSeek-R1	0.038	0.015	4.000	3.000	6.000	0.000	4.000	3.000	7.000	0.001	4.000	4.000	4.000	4.000	5.000	0.803
Claude Sonnet 5	0.023	0.015	5.000	4.000	7.000	0.000	5.500	4.000	8.000	0.000	6.000	6.000	5.000	5.000	4.000	0.796
Gemini 3.1 Pro	0.022	0.015	6.000	3.975	7.000	0.000	5.500	4.000	7.000	0.000	5.000	5.000	6.000	6.000	6.000	0.802
Grok 4	0.017	0.015	7.000	4.000	7.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	7.000	7.000	7.000	0.799
GPT-5.4-mini	-0.028	0.015	8.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	8.000	9.000	9.000	0.788
DeepSeek-chat	-0.030	0.015	9.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	9.000	8.000	8.000	0.788
gpt-oss-120b	-0.106	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.770
Llama 3.3 70B	-0.190	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.741

Table 27: Full leaderboard under rule top5; weights F1/decay/prec/rec = 0.30 / 0.15 / 0.40 / 0.15.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.108	0.015	1.000	1.000	2.000	0.971	3.000	1.000	4.000	0.058	3.000	3.000	3.000	3.000	3.000	0.814
Claude Opus 5	0.088	0.000	2.000	1.000	2.000	0.029	2.000	1.000	3.000	0.489	1.000	1.000	1.000	2.000	1.000	0.813
DeepSeek-R1	0.041	0.015	3.000	3.000	6.000	0.000	4.000	3.000	7.000	0.002	4.000	4.000	4.000	4.000	5.000	0.805
GPT-5.5	0.040	0.015	4.000	3.000	6.000	0.000	1.000	1.000	3.000	0.444	2.000	2.000	2.000	1.000	2.000	0.812
Claude Sonnet 5	0.026	0.015	5.000	3.000	7.000	0.000	5.000	4.000	8.000	0.000	6.000	6.000	5.000	5.000	4.000	0.799
Gemini 3.1 Pro	0.020	0.015	6.000	4.000	7.000	0.000	6.000	4.000	7.000	0.000	5.000	5.000	6.000	6.000	6.000	0.804
Grok 4	0.011	0.015	7.000	5.000	8.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	7.000	7.000	7.000	0.801
GPT-5.4-mini	-0.011	0.015	8.000	7.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	8.000	9.000	9.000	0.793
DeepSeek-chat	-0.033	0.015	9.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	9.000	8.000	8.000	0.790
gpt-oss-120b	-0.122	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.771
Llama 3.3 70B	-0.168	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.746

Table 28: Full leaderboard under rule v1; weights F1/decay/prec/rec = 0.35 / 0.35 / 0.15 / 0.15.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.112	0.015	1.000	1.000	2.000	0.922	3.000	1.000	4.000	0.056	3.000	3.000	3.000	3.000	2.000	0.812
Claude Opus 5	0.097	0.000	2.000	1.000	2.000	0.078	2.000	1.000	3.000	0.499	1.000	1.000	1.000	2.000	1.000	0.812
GPT-5.5	0.042	0.015	3.000	3.000	5.025	0.000	1.000	1.000	3.000	0.431	2.000	2.000	2.000	1.000	3.000	0.808
DeepSeek-R1	0.037	0.015	4.000	3.000	6.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.801
Claude Sonnet 5	0.023	0.015	5.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	5.000	5.000	5.000	4.000	0.796
Gemini 3.1 Pro	0.021	0.015	6.000	4.000	7.000	0.000	6.000	4.000	7.000	0.000	6.000	5.000	6.000	6.000	6.000	0.799
Grok 4	0.016	0.015	7.000	4.000	7.000	0.000	7.000	4.000	7.000	0.000	7.000	7.000	7.000	7.000	7.000	0.797
GPT-5.4-mini	-0.022	0.015	8.000	8.000	9.000	0.000	9.000	7.000	9.000	0.000	8.000	8.000	8.000	9.000	8.000	0.790
DeepSeek-chat	-0.031	0.015	9.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	9.000	9.000	9.000	8.000	9.000	0.784
gpt-oss-120b	-0.111	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.767
Llama 3.3 70B	-0.183	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.738

Table 29: Leave-one-dataset-out: fitted weights and held-out τ for every dataset and weight source.

kind	held_out	weights_from	w_f1	w_decay	w_precision	w_recall	J_rest	tau_heldout_mean	n_heldout
dataset	abt_buy	lodo_weights	0.200	0.250	0.300	0.250	0.412	0.783	1
dataset	abt_buy	full_best	0.350	0.600	0.050	0.000	0.411	0.755	1
dataset	abt_buy	v1	0.350	0.350	0.150	0.150	0.399	0.755	1
dataset	alaska_camera	lodo_weights	0.200	0.250	0.300	0.250	0.451	-0.283	1
dataset	alaska_camera	full_best	0.350	0.600	0.050	0.000	0.446	-0.283	1
dataset	alaska_camera	v1	0.350	0.350	0.150	0.150	0.435	-0.245	1
dataset	alaska_schema	lodo_weights	0.350	0.150	0.350	0.150	0.439	-0.037	1
dataset	alaska_schema	full_best	0.350	0.600	0.050	0.000	0.434	0.018	1
dataset	alaska_schema	v1	0.350	0.350	0.150	0.150	0.427	-0.055	1
dataset	amazon_google	lodo_weights	0.200	0.250	0.300	0.250	0.439	0.020	1
dataset	amazon_google	full_best	0.350	0.600	0.050	0.000	0.434	0.020	1
dataset	amazon_google	v1	0.350	0.350	0.150	0.150	0.425	0.020	1
dataset	bird	lodo_weights	0.350	0.600	0.050	0.000	0.422	0.418	1
dataset	bird	full_best	0.350	0.600	0.050	0.000	0.422	0.418	1
dataset	bird	v1	0.350	0.350	0.150	0.150	0.404	0.600	1
dataset	dblp_acm	lodo_weights	0.350	0.600	0.050	0.000	0.422	0.497	1
dataset	dblp_acm	full_best	0.350	0.600	0.050	0.000	0.422	0.497	1
dataset	dblp_acm	v1	0.350	0.350	0.150	0.150	0.407	0.540	1
dataset	dblp_scholar	lodo_weights	0.350	0.200	0.050	0.400	0.441	0.000	1
dataset	dblp_scholar	full_best	0.350	0.600	0.050	0.000	0.434	0.019	1
dataset	dblp_scholar	v1	0.350	0.350	0.150	0.150	0.428	0.039	1
dataset	fetaqa	lodo_weights	0.100	0.600	0.200	0.100	0.426	0.404	1
dataset	fetaqa	full_best	0.350	0.600	0.050	0.000	0.421	0.455	1
dataset	fetaqa	v1	0.350	0.350	0.150	0.150	0.422	0.389	1
dataset	fintagging	lodo_weights	0.150	0.250	0.350	0.250	0.426	0.477	1
dataset	fintagging	full_best	0.350	0.600	0.050	0.000	0.415	0.564	1
dataset	fintagging	v1	0.350	0.350	0.150	0.150	0.408	0.491	1
dataset	hitab	lodo_weights	0.200	0.250	0.300	0.250	0.444	-0.183	1
dataset	hitab	full_best	0.350	0.600	0.050	0.000	0.441	-0.164	1
dataset	hitab	v1	0.350	0.350	0.150	0.150	0.431	-0.200	1
dataset	machamp	lodo_weights	0.200	0.250	0.300	0.250	0.414	0.756	1
dataset	machamp	full_best	0.350	0.600	0.050	0.000	0.412	0.806	1
dataset	machamp	v1	0.350	0.350	0.150	0.150	0.398	0.799	1
dataset	magneto_gdc	lodo_weights	0.000	0.150	0.000	0.850	0.418	0.598	1
dataset	magneto_gdc	full_best	0.350	0.600	0.050	0.000	0.417	0.564	1
dataset	magneto_gdc	v1	0.350	0.350	0.150	0.150	0.405	0.564	1
dataset	mmtu	lodo_weights	0.150	0.050	0.600	0.200	0.424	0.404	1
dataset	mmtu	full_best	0.350	0.600	0.050	0.000	0.420	0.491	1
dataset	mmtu	v1	0.350	0.350	0.150	0.150	0.416	0.440	1
dataset	officeqa	lodo_weights	0.300	0.000	0.450	0.250	0.419	0.486	1
dataset	officeqa	full_best	0.350	0.600	0.050	0.000	0.411	0.709	1
dataset	officeqa	v1	0.350	0.350	0.150	0.150	0.402	0.673	1
dataset	officeqa_pro_v2	lodo_weights	0.100	0.750	0.000	0.150	0.415	0.636	1
dataset	officeqa_pro_v2	full_best	0.350	0.600	0.050	0.000	0.414	0.600	1
dataset	officeqa_pro_v2	v1	0.350	0.350	0.150	0.150	0.406	0.661	1
dataset	opensanctions_pairs	lodo_weights	0.150	0.700	0.000	0.150	0.435	0.056	1
dataset	opensanctions_pairs	full_best	0.350	0.600	0.050	0.000	0.433	0.073	1
dataset	opensanctions_pairs	v1	0.350	0.350	0.150	0.150	0.423	0.037	1
dataset	papadakis_dn	lodo_weights	0.200	0.250	0.300	0.250	0.444	-0.083	1
dataset	papadakis_dn	full_best	0.350	0.600	0.050	0.000	0.436	0.000	1
dataset	papadakis_dn	v1	0.350	0.350	0.150	0.150	0.428	-0.042	1
dataset	realhitbench	lodo_weights	0.200	0.250	0.300	0.250	0.416	0.661	1
dataset	realhitbench	full_best	0.350	0.600	0.050	0.000	0.413	0.673	1
dataset	realhitbench	v1	0.350	0.350	0.150	0.150	0.403	0.636	1

dataset	smat	lodo_weights	0.200	0.250	0.300	0.250	0.429	0.309	1
dataset	smat	full_best	0.350	0.600	0.050	0.000	0.422	0.367	1
dataset	smat	v1	0.350	0.350	0.150	0.150	0.416	0.309	1
dataset	suc	lodo_weights	0.150	0.250	0.350	0.250	0.418	0.550	1
dataset	suc	full_best	0.350	0.600	0.050	0.000	0.416	0.587	1
dataset	suc	v1	0.350	0.350	0.150	0.150	0.407	0.514	1
dataset	tab_fact	lodo_weights	0.150	0.250	0.350	0.250	0.439	-0.019	1
dataset	tab_fact	full_best	0.350	0.600	0.050	0.000	0.438	-0.056	1
dataset	tab_fact	v1	0.350	0.350	0.150	0.150	0.427	-0.019	1
dataset	tabis	lodo_weights	0.200	0.000	0.500	0.300	0.422	0.472	1
dataset	tabis	full_best	0.350	0.600	0.050	0.000	0.418	0.486	1
dataset	tabis	v1	0.350	0.350	0.150	0.150	0.410	0.449	1
dataset	tablebench	lodo_weights	0.350	0.200	0.050	0.400	0.427	0.449	1
dataset	tablebench	full_best	0.350	0.600	0.050	0.000	0.422	0.418	1
dataset	tablebench	v1	0.350	0.350	0.150	0.150	0.409	0.455	1
dataset	tableeval	lodo_weights	0.050	0.600	0.250	0.100	0.428	0.236	1
dataset	tableeval	full_best	0.350	0.600	0.050	0.000	0.425	0.273	1
dataset	tableeval	v1	0.350	0.350	0.150	0.150	0.419	0.236	1
dataset	tat_qa	lodo_weights	0.000	1.000	0.000	0.000	0.424	0.164	1
dataset	tat_qa	full_best	0.350	0.600	0.050	0.000	0.416	0.600	1
dataset	tat_qa	v1	0.350	0.350	0.150	0.150	0.410	0.550	1
dataset	tpcdi_cells	lodo_weights	0.150	0.700	0.000	0.150	0.420	0.570	1
dataset	tpcdi_cells	full_best	0.350	0.600	0.050	0.000	0.415	0.623	1
dataset	tpcdi_cells	v1	0.350	0.350	0.150	0.150	0.405	0.623	1
dataset	valentine	lodo_weights	0.200	0.000	0.500	0.300	0.424	0.419	1
dataset	valentine	full_best	0.350	0.600	0.050	0.000	0.420	0.472	1
dataset	valentine	v1	0.350	0.350	0.150	0.150	0.411	0.397	1
dataset	walmart_amazon	lodo_weights	0.000	0.250	0.150	0.600	0.414	0.738	1
dataset	walmart_amazon	full_best	0.350	0.600	0.050	0.000	0.413	0.686	1
dataset	walmart_amazon	v1	0.350	0.350	0.150	0.150	0.401	0.686	1
dataset	wdc_lspc	lodo_weights	0.000	0.250	0.150	0.600	0.420	0.507	1
dataset	wdc_lspc	full_best	0.350	0.600	0.050	0.000	0.415	0.583	1
dataset	wdc_lspc	v1	0.350	0.350	0.150	0.150	0.407	0.543	1
dataset	wdc_products	lodo_weights	0.100	0.800	0.000	0.100	0.425	0.375	1
dataset	wdc_products	full_best	0.350	0.600	0.050	0.000	0.423	0.350	1
dataset	wdc_products	v1	0.350	0.350	0.150	0.150	0.413	0.350	1
dataset	wikitablequestions	lodo_weights	0.200	0.250	0.300	0.250	0.418	0.697	1
dataset	wikitablequestions	full_best	0.350	0.600	0.050	0.000	0.408	0.782	1
dataset	wikitablequestions	v1	0.350	0.350	0.150	0.150	0.401	0.709	1

Table 30: Leave-one-family-out: fitted weights and held-out τ.

kind	held_out	weights_from	w_f1	w_decay	w_precision	w_recall	J_rest	tau_heldout_mean	n_heldout
family	EM	lodo_weights	0.050	0.600	0.250	0.100	0.485	0.327	11
family	EM	full_best	0.350	0.600	0.050	0.000	0.482	0.318	11
family	EM	v1	0.350	0.350	0.150	0.150	0.465	0.324	11
family	SM	lodo_weights	0.100	0.150	0.100	0.650	0.428	0.358	6
family	SM	full_best	0.350	0.600	0.050	0.000	0.410	0.290	6
family	SM	v1	0.350	0.350	0.150	0.150	0.409	0.277	6
family	TQA	lodo_weights	0.300	0.600	0.100	0.000	0.433	0.367	11
family	TQA	full_best	0.350	0.600	0.050	0.000	0.431	0.367	11
family	TQA	v1	0.350	0.350	0.150	0.150	0.420	0.357	11
family	FIN	lodo_weights	0.100	0.850	0.000	0.050	0.405	0.500	3
family	FIN	full_best	0.350	0.600	0.050	0.000	0.393	0.576	3
family	FIN	v1	0.350	0.350	0.150	0.150	0.391	0.563	3

Table 31: Leave-one-family-out: weights re-fitted without the family (F1 / decay / precision / recall) and the held-out family's τ to the pooled rest under those weights, the full-data best and v1.

family	LOFO weights	tau: LOFO weights	full best	v1
EM	0.05 / 0.60 / 0.25 / 0.10	0.327	0.318	0.324
SM	0.10 / 0.15 / 0.10 / 0.65	0.358	0.290	0.277
TQA	0.30 / 0.60 / 0.10 / 0.00	0.367	0.367	0.357
FIN	0.10 / 0.85 / 0.00 / 0.05	0.500	0.576	0.563

Table 32: Majority-class baselines for the boolean datasets.

dataset	majority-class accuracy	n
abt_buy	0.510	100
alaska_camera	0.530	100
amazon_google	0.550	100
dblp_acm	0.510	100
dblp_scholar	0.510	100
machamp	0.530	100
opensanctions_pairs	0.510	100
papadakis_dn	0.850	100
tab_fact	0.540	100
walmart_amazon	0.590	100
wdc_lspc	0.540	100
wdc_products	0.540	100

Table 33: Every quantity saved by the nested item bootstrap.

quantity	value
B	1000
n_candidates	1771
in_bag_J_of_selected_mean	0.3788396423051148
out_of_bag_J_of_selected_mean	0.3323686045648369
optimism_gap	0.0464710377402789
out_of_bag_J_v1_mean	0.33856895267995957
out_of_bag_J_full_best_mean	0.3353216406578332
selected_vs_v1_out_of_bag	"mean_diff": -0.006200348115122659, "ci95_lo": -0.03179926569512316, "ci95_hi": 0.011292872130425633, "p_boot_one_sided": 0.709, "share_a_ahead": 0.291
full_best_vs_v1_out_of_bag	"mean_diff": -0.003247312022126434, "ci95_lo": -0.0191724986130256, "ci95_hi": 0.013382479290899968, "p_boot_one_sided": 0.674, "share_a_ahead": 0.326
full_best_vs_v1_in_bag	"mean_diff": -0.003467710584402084, "ci95_lo": -0.019269110262393953, "ci95_hi": 0.012504931539297099, "p_boot_one_sided": 0.663, "share_a_ahead": 0.337
share_resamples_v1_in_top_5pct	0.079
selected_weights_mean	"f1": 0.255, "decay": 0.313, "precision": 0.173, "recall": 0.26
selected_weights_sd	"f1": 0.235, "decay": 0.213, "precision": 0.168, "recall": 0.228
n_distinct_selected	617
plateau_size	1754
plateau_share	0.9904009034443817
v1_in_plateau	True
v1_p_vs_best	0.663
practical_plateau_size_within_0.02	1728
v1_in_practical_plateau	True
practical_plateau_weight_ranges	"f1": [0.0, 1.0], "decay": [0.0, 1.0], "precision": [0.0, 0.85], "recall": [0.0, 0.95]
plateau_weight_ranges	"f1": [0.0, 1.0], "decay": [0.0, 1.0], "precision": [0.0, 0.95], "recall": [0.0, 1.0]

Table 34: Official-split estimates under v1 averaged over the ten models.

dataset	suite_mean	official_estimate	suite_f1_pos	official_precision_pos	official_recall_pos	official_f1_pos	official_positive_share
abt_buy	0.971	0.987	0.969	0.933	0.950	0.940	0.108
alaska_camera	0.945	0.945	0.939	1.000	0.895	0.939	0.500
alaska_schema	0.929	0.929	–	–	–	–	–
amazon_google	0.935	0.932	0.928	0.609	0.941	0.738	0.102
bird	0.191	0.218	–	–	–	–	–
dblp_acm	0.987	0.992	0.986	0.981	0.980	0.978	0.180
dblp_scholar	0.948	0.966	0.947	0.901	0.922	0.909	0.186
fetaqa	0.387	0.387	–	–	–	–	–
fintagging	0.884	0.882	–	–	–	–	–
hitab	0.772	0.776	–	–	–	–	–
machamp	0.887	0.891	0.892	0.649	0.852	0.735	0.175
magneto_gdc	0.937	0.940	–	–	–	–	–
mmtu	0.874	0.837	–	–	–	–	–
officeqa	0.135	0.135	–	–	–	–	–
officeqa_pro_v2	0.151	0.151	–	–	–	–	–
opensanctions_pairs	0.835	0.748	0.790	0.993	0.677	0.795	0.769
papadakis_dn	0.940	0.987	0.831	0.527	0.967	0.662	0.018
realhitbench	0.677	0.676	–	–	–	–	–
smat	0.916	0.914	–	–	–	–	–
suc	0.917	0.915	–	–	–	–	–
tab_fact	0.802	0.792	0.761	0.903	0.674	0.767	0.500
tabis	0.918	0.923	–	–	–	–	–
tablebench	0.824	0.822	–	–	–	–	–
tableeval	0.755	0.758	–	–	–	–	–
tat_qa	0.804	0.759	–	–	–	–	–
tpcdi_cells	0.456	0.462	–	–	–	–	–
valentine	0.985	0.981	–	–	–	–	–
walmart_amazon	0.959	0.974	0.949	0.824	0.931	0.872	0.094
wdc_lspc	0.982	0.984	0.983	0.968	0.978	0.972	0.264
wdc_products	0.978	0.986	0.979	0.982	0.968	0.975	0.280
wikitablequestions	0.848	0.846	–	–	–	–	–

Table 35: Official-split estimates under v1 for every dataset and model.

dataset	model	suite_mean	official_ estimate	official_lo95	official_hi95	effective_n	suite_f1_pos	official_f1_ pos	official_f1_ lo95	official_f1_ hi95	official_ positive_ share
abt_buy	Claude Opus 5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	Claude Sonnet 5	0.970	0.993	0.985	1.000	63.075	0.968	0.968	0.925	1.000	0.108
abt_buy	DeepSeek-chat	0.960	0.976	0.933	0.998	63.075	0.958	0.893	0.735	0.990	0.108
abt_buy	DeepSeek-R1	0.970	0.978	0.931	1.000	63.075	0.969	0.904	0.755	1.000	0.108
abt_buy	Gemini 3.1 Pro	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.945	1.000	0.108
abt_buy	Gemini 3.8 Flash	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.944	1.000	0.108
abt_buy	GPT-5.4-mini	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.736	0.991	0.108
abt_buy	GPT-5.5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	gpt-oss-120b	0.940	0.987	0.974	0.996	63.075	0.935	0.935	0.871	0.978	0.108
abt_buy	Grok 4	0.960	0.976	0.935	0.998	63.075	0.958	0.893	0.739	0.990	0.108
abt_buy	Llama 3.3 70B	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.737	0.990	0.108
alaska_camera	Claude Opus 5	0.820	0.820	0.740	0.890	100.000	0.795	0.795	0.700	0.882	0.500
alaska_camera	Claude Sonnet 5	0.980	0.980	0.950	1.000	100.000	0.981	0.981	0.948	1.000	0.500
alaska_camera	DeepSeek-chat	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	DeepSeek-R1	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.968	1.000	0.500
alaska_camera	Gemini 3.1 Pro	0.810	0.810	0.730	0.880	100.000	0.782	0.782	0.667	0.869	0.500
alaska_camera	Gemini 3.8 Flash	0.890	0.890	0.830	0.950	100.000	0.884	0.884	0.804	0.945	0.500
alaska_camera	GPT-5.4-mini	0.970	0.970	0.930	1.000	100.000	0.971	0.971	0.932	1.000	0.500
alaska_camera	GPT-5.5	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	gpt-oss-120b	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	Grok 4	0.940	0.940	0.890	0.980	100.000	0.940	0.940	0.889	0.981	0.500
alaska_camera	Llama 3.3 70B	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.968	1.000	0.500
alaska_schema	Claude Opus 5	0.945	0.945	0.903	0.982	93.000	–	–	–	–	–
alaska_schema	Claude Sonnet 5	0.929	0.929	0.880	0.972	93.000	–	–	–	–	–
alaska_schema	DeepSeek-chat	0.951	0.951	0.910	0.985	93.000	–	–	–	–	–
alaska_schema	DeepSeek-R1	0.957	0.957	0.923	0.988	93.000	–	–	–	–	–
alaska_schema	Gemini 3.1 Pro	0.928	0.928	0.875	0.970	93.000	–	–	–	–	–
alaska_schema	Gemini 3.8 Flash	0.918	0.918	0.863	0.962	93.000	–	–	–	–	–
alaska_schema	GPT-5.4-mini	0.958	0.958	0.920	0.989	93.000	–	–	–	–	–
alaska_schema	GPT-5.5	0.928	0.928	0.880	0.972	93.000	–	–	–	–	–
alaska_schema	gpt-oss-120b	0.934	0.934	0.889	0.972	93.000	–	–	–	–	–
alaska_schema	Grok 4	0.929	0.929	0.880	0.973	93.000	–	–	–	–	–
alaska_schema	Llama 3.3 70B	0.840	0.840	0.780	0.894	93.000	–	–	–	–	–
amazon_google	Claude Opus 5	0.940	0.916	0.839	0.981	67.152	0.936	0.704	0.526	0.919	0.102
amazon_google	Claude Sonnet 5	0.960	0.935	0.868	0.984	67.152	0.957	0.758	0.588	0.935	0.102
amazon_google	DeepSeek-chat	0.960	0.949	0.893	0.998	67.152	0.957	0.796	0.614	0.990	0.102

amazon_google	DeepSeek-R1	0.950	0.932	0.866	0.984	67.152	0.946	0.747	0.579	0.932	0.102
amazon_google	Gemini 3.1 Pro	0.930	0.928	0.862	0.981	67.152	0.923	0.725	0.551	0.914	0.102
amazon_google	Gemini 3.8 Flash	0.930	0.928	0.859	0.981	67.152	0.923	0.725	0.548	0.913	0.102
amazon_google	GPT-5.4-mini	0.930	0.942	0.880	0.990	67.152	0.921	0.762	0.583	0.959	0.102
amazon_google	GPT-5.5	0.940	0.916	0.844	0.982	67.152	0.936	0.704	0.535	0.915	0.102
amazon_google	gpt-oss-120b	0.860	0.926	0.861	0.973	67.152	0.829	0.676	0.507	0.874	0.102
amazon_google	Grok 4	0.940	0.944	0.875	0.992	67.152	0.933	0.773	0.607	0.966	0.102
amazon_google	Llama 3.3 70B	0.950	0.932	0.861	0.982	67.152	0.946	0.747	0.561	0.941	0.102
bird	Claude Opus 5	0.293	0.323	0.233	0.419	74.082	–	–	–	–	–
bird	Claude Sonnet 5	0.233	0.260	0.174	0.349	74.082	–	–	–	–	–
bird	DeepSeek-chat	0.187	0.203	0.132	0.289	74.082	–	–	–	–	–
bird	DeepSeek-R1	0.216	0.253	0.170	0.343	74.082	–	–	–	–	–
bird	Gemini 3.1 Pro	0.183	0.228	0.143	0.315	74.082	–	–	–	–	–
bird	Gemini 3.8 Flash	0.293	0.313	0.227	0.402	74.082	–	–	–	–	–
bird	GPT-5.4-mini	0.178	0.218	0.131	0.308	74.082	–	–	–	–	–
bird	GPT-5.5	0.185	0.204	0.118	0.293	74.082	–	–	–	–	–
bird	gpt-oss-120b	0.084	0.109	0.051	0.181	74.082	–	–	–	–	–
bird	Grok 4	0.122	0.144	0.073	0.228	74.082	–	–	–	–	–
bird	Llama 3.3 70B	0.122	0.144	0.082	0.218	74.082	–	–	–	–	–
dblp_acm	Claude Opus 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Claude Sonnet 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-chat	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-R1	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.965	1.000	0.180
dblp_acm	Gemini 3.1 Pro	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Gemini 3.8 Flash	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	GPT-5.4-mini	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.965	1.000	0.180
dblp_acm	GPT-5.5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	gpt-oss-120b	0.910	0.967	0.945	0.987	72.166	0.899	0.899	0.829	0.960	0.180
dblp_acm	Grok 4	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Llama 3.3 70B	0.970	0.952	0.893	1.000	72.166	0.970	0.882	0.750	1.000	0.180
dblp_scholar	Claude Opus 5	0.950	0.956	0.901	0.996	70.463	0.950	0.888	0.752	0.981	0.186
dblp_scholar	Claude Sonnet 5	0.970	0.976	0.932	1.000	70.463	0.970	0.937	0.835	1.000	0.186
dblp_scholar	DeepSeek-chat	0.960	0.959	0.907	0.997	70.463	0.961	0.898	0.780	0.991	0.186
dblp_scholar	DeepSeek-R1	0.950	0.969	0.931	0.996	70.463	0.949	0.917	0.818	0.987	0.186
dblp_scholar	Gemini 3.1 Pro	0.940	0.965	0.927	0.993	70.463	0.939	0.906	0.812	0.980	0.186
dblp_scholar	Gemini 3.8 Flash	0.930	0.961	0.914	0.990	70.463	0.928	0.895	0.791	0.971	0.186
dblp_scholar	GPT-5.4-mini	0.940	0.965	0.925	0.992	70.463	0.939	0.906	0.799	0.980	0.186
dblp_scholar	GPT-5.5	0.960	0.959	0.905	0.996	70.463	0.961	0.898	0.784	0.991	0.186
dblp_scholar	gpt-oss-120b	0.910	0.967	0.943	0.986	70.463	0.903	0.903	0.829	0.961	0.186
dblp_scholar	Grok 4	0.950	0.969	0.928	0.996	70.463	0.949	0.917	0.812	0.988	0.186
dblp_scholar	Llama 3.3 70B	0.970	0.976	0.936	1.000	70.463	0.970	0.937	0.847	1.000	0.186

fetaqa	Claude Opus 5	0.401	0.401	0.373	0.430	100.000	–	–	–	–	–
fetaqa	Claude Sonnet 5	0.373	0.373	0.345	0.400	100.000	–	–	–	–	–
fetaqa	DeepSeek-chat	0.434	0.434	0.411	0.462	100.000	–	–	–	–	–
fetaqa	DeepSeek-R1	0.413	0.413	0.383	0.445	100.000	–	–	–	–	–
fetaqa	Gemini 3.1 Pro	0.433	0.433	0.409	0.456	100.000	–	–	–	–	–
fetaqa	Gemini 3.8 Flash	0.445	0.445	0.422	0.470	100.000	–	–	–	–	–
fetaqa	GPT-5.4-mini	0.312	0.312	0.289	0.336	100.000	–	–	–	–	–
fetaqa	GPT-5.5	0.438	0.438	0.412	0.466	100.000	–	–	–	–	–
fetaqa	gpt-oss-120b	0.322	0.322	0.301	0.341	100.000	–	–	–	–	–
fetaqa	Grok 4	0.428	0.428	0.405	0.451	100.000	–	–	–	–	–
fetaqa	Llama 3.3 70B	0.253	0.253	0.226	0.279	100.000	–	–	–	–	–
fintagging	Claude Opus 5	0.909	0.932	0.876	0.978	51.769	–	–	–	–	–
fintagging	Claude Sonnet 5	0.898	0.897	0.832	0.956	51.769	–	–	–	–	–
fintagging	DeepSeek-chat	0.858	0.847	0.774	0.912	51.769	–	–	–	–	–
fintagging	DeepSeek-R1	0.878	0.894	0.828	0.949	51.769	–	–	–	–	–
fintagging	Gemini 3.1 Pro	0.887	0.907	0.841	0.960	51.769	–	–	–	–	–
fintagging	Gemini 3.8 Flash	0.897	0.908	0.845	0.957	51.769	–	–	–	–	–
fintagging	GPT-5.4-mini	0.892	0.863	0.794	0.922	51.769	–	–	–	–	–
fintagging	GPT-5.5	0.893	0.899	0.834	0.957	51.769	–	–	–	–	–
fintagging	gpt-oss-120b	0.863	0.847	0.770	0.916	51.769	–	–	–	–	–
fintagging	Grok 4	0.887	0.876	0.811	0.943	51.769	–	–	–	–	–
fintagging	Llama 3.3 70B	0.858	0.836	0.762	0.910	51.769	–	–	–	–	–
hitab	Claude Opus 5	0.781	0.790	0.708	0.859	94.537	–	–	–	–	–
hitab	Claude Sonnet 5	0.763	0.758	0.671	0.835	94.537	–	–	–	–	–
hitab	DeepSeek-chat	0.814	0.822	0.749	0.889	94.537	–	–	–	–	–
hitab	DeepSeek-R1	0.792	0.787	0.715	0.860	94.537	–	–	–	–	–
hitab	Gemini 3.1 Pro	0.778	0.786	0.704	0.858	94.537	–	–	–	–	–
hitab	Gemini 3.8 Flash	0.748	0.757	0.677	0.830	94.537	–	–	–	–	–
hitab	GPT-5.4-mini	0.784	0.783	0.708	0.855	94.537	–	–	–	–	–
hitab	GPT-5.5	0.757	0.764	0.682	0.839	94.537	–	–	–	–	–
hitab	gpt-oss-120b	0.750	0.752	0.670	0.827	94.537	–	–	–	–	–
hitab	Grok 4	0.746	0.749	0.671	0.825	94.537	–	–	–	–	–
hitab	Llama 3.3 70B	0.784	0.783	0.700	0.853	94.537	–	–	–	–	–
machamp	Claude Opus 5	0.900	0.912	0.826	0.970	32.491	0.904	0.774	0.617	0.908	0.175
machamp	Claude Sonnet 5	0.900	0.923	0.852	0.975	32.491	0.902	0.793	0.642	0.924	0.175
machamp	DeepSeek-chat	0.890	0.892	0.801	0.962	32.491	0.897	0.742	0.604	0.895	0.175
machamp	DeepSeek-R1	0.900	0.897	0.808	0.963	32.491	0.906	0.751	0.594	0.895	0.175
machamp	Gemini 3.1 Pro	0.890	0.908	0.832	0.965	32.491	0.893	0.761	0.618	0.898	0.175
machamp	Gemini 3.8 Flash	0.900	0.912	0.836	0.973	32.491	0.904	0.774	0.608	0.920	0.175
machamp	GPT-5.4-mini	0.890	0.893	0.804	0.962	32.491	0.895	0.739	0.568	0.898	0.175

machamp	GPT-5.5	0.900	0.897	0.812	0.964	32.491	0.906	0.751	0.591	0.895	0.175
machamp	gpt-oss-120b	0.830	0.794	0.664	0.913	32.491	0.835	0.558	0.364	0.761	0.175
machamp	Grok 4	0.890	0.896	0.806	0.962	32.491	0.895	0.747	0.594	0.906	0.175
machamp	Llama 3.3 70B	0.870	0.880	0.791	0.947	32.491	0.874	0.698	0.529	0.846	0.175
magneto__gdc	Claude Opus 5	0.971	0.974	0.946	0.995	93.027	–	–	–	–	–
magneto__gdc	Claude Sonnet 5	0.955	0.955	0.916	0.987	93.027	–	–	–	–	–
magneto__gdc	DeepSeek-chat	0.953	0.956	0.919	0.986	93.027	–	–	–	–	–
magneto__gdc	DeepSeek-R1	0.978	0.981	0.959	1.000	93.027	–	–	–	–	–
magneto__gdc	Gemini 3.1 Pro	0.956	0.957	0.923	0.983	93.027	–	–	–	–	–
magneto__gdc	Gemini 3.8 Flash	0.963	0.967	0.936	0.993	93.027	–	–	–	–	–
magneto__gdc	GPT-5.4-mini	0.912	0.914	0.862	0.962	93.027	–	–	–	–	–
magneto__gdc	GPT-5.5	0.967	0.972	0.940	0.994	93.027	–	–	–	–	–
magneto__gdc	gpt-oss-120b	0.961	0.965	0.933	0.989	93.027	–	–	–	–	–
magneto__gdc	Grok 4	0.939	0.944	0.901	0.979	93.027	–	–	–	–	–
magneto__gdc	Llama 3.3 70B	0.753	0.759	0.694	0.820	93.027	–	–	–	–	–
mmtu	Claude Opus 5	0.882	0.846	0.760	0.930	86.538	–	–	–	–	–
mmtu	Claude Sonnet 5	0.872	0.838	0.750	0.921	86.538	–	–	–	–	–
mmtu	DeepSeek-chat	0.865	0.831	0.743	0.911	86.538	–	–	–	–	–
mmtu	DeepSeek-R1	0.850	0.802	0.710	0.888	86.538	–	–	–	–	–
mmtu	Gemini 3.1 Pro	0.914	0.881	0.794	0.952	86.538	–	–	–	–	–
mmtu	Gemini 3.8 Flash	0.894	0.856	0.773	0.932	86.538	–	–	–	–	–
mmtu	GPT-5.4-mini	0.882	0.851	0.769	0.926	86.538	–	–	–	–	–
mmtu	GPT-5.5	0.917	0.882	0.804	0.950	86.538	–	–	–	–	–
mmtu	gpt-oss-120b	0.830	0.774	0.676	0.868	86.538	–	–	–	–	–
mmtu	Grok 4	0.901	0.869	0.793	0.945	86.538	–	–	–	–	–
mmtu	Llama 3.3 70B	0.807	0.773	0.683	0.852	86.538	–	–	–	–	–
officeqa	Claude Opus 5	0.199	0.199	0.158	0.247	100.000	–	–	–	–	–
officeqa	Claude Sonnet 5	0.171	0.171	0.129	0.211	100.000	–	–	–	–	–
officeqa	DeepSeek-chat	0.075	0.075	0.041	0.116	100.000	–	–	–	–	–
officeqa	DeepSeek-R1	0.147	0.147	0.113	0.189	100.000	–	–	–	–	–
officeqa	Gemini 3.1 Pro	0.119	0.119	0.088	0.157	100.000	–	–	–	–	–
officeqa	Gemini 3.8 Flash	0.194	0.194	0.158	0.235	100.000	–	–	–	–	–
officeqa	GPT-5.4-mini	0.107	0.107	0.082	0.137	100.000	–	–	–	–	–
officeqa	GPT-5.5	0.168	0.168	0.133	0.210	100.000	–	–	–	–	–
officeqa	gpt-oss-120b	0.079	0.079	0.051	0.113	100.000	–	–	–	–	–
officeqa	Grok 4	0.129	0.129	0.095	0.168	100.000	–	–	–	–	–
officeqa	Llama 3.3 70B	0.097	0.097	0.072	0.124	100.000	–	–	–	–	–
officeqa__pro__v2	Claude Opus 5	0.270	0.270	0.210	0.337	90.000	–	–	–	–	–
officeqa__pro__v2	Claude Sonnet 5	0.157	0.157	0.112	0.208	90.000	–	–	–	–	–
officeqa__pro__v2	DeepSeek-chat	0.121	0.121	0.085	0.157	90.000	–	–	–	–	–
officeqa__pro__v2	DeepSeek-R1	0.148	0.148	0.097	0.202	90.000	–	–	–	–	–

officeqa_pro_v2	Gemini 3.1 Pro	0.120	0.120	0.082	0.162	90.000	–	–	–	–	–
officeqa_pro_v2	Gemini 3.8 Flash	0.198	0.198	0.153	0.252	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.4-mini	0.123	0.123	0.096	0.155	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.5	0.188	0.188	0.128	0.248	90.000	–	–	–	–	–
officeqa_pro_v2	gpt-oss-120b	0.104	0.104	0.065	0.147	90.000	–	–	–	–	–
officeqa_pro_v2	Grok 4	0.119	0.119	0.082	0.164	90.000	–	–	–	–	–
officeqa_pro_v2	Llama 3.3 70B	0.115	0.115	0.082	0.152	90.000	–	–	–	–	–
opensanctions_pairs	Claude Opus 5	0.940	0.906	0.831	0.969	76.249	0.935	0.935	0.878	0.979	0.769
opensanctions_pairs	Claude Sonnet 5	0.730	0.587	0.479	0.701	76.249	0.630	0.636	0.500	0.755	0.769
opensanctions_pairs	DeepSeek-chat	0.760	0.635	0.529	0.741	76.249	0.684	0.691	0.543	0.800	0.769
opensanctions_pairs	DeepSeek-R1	0.850	0.765	0.670	0.863	76.249	0.819	0.819	0.714	0.900	0.769
opensanctions_pairs	Gemini 3.1 Pro	0.970	0.964	0.917	1.000	76.249	0.969	0.976	0.943	1.000	0.769
opensanctions_pairs	Gemini 3.8 Flash	0.860	0.780	0.681	0.882	76.249	0.833	0.833	0.732	0.911	0.769
opensanctions_pairs	GPT-5.4-mini	0.900	0.865	0.781	0.940	76.249	0.891	0.905	0.833	0.960	0.769
opensanctions_pairs	GPT-5.5	0.750	0.608	0.503	0.724	76.249	0.658	0.658	0.525	0.778	0.769
opensanctions_pairs	gpt-oss-120b	0.780	0.666	0.561	0.780	76.249	0.718	0.725	0.588	0.815	0.769
opensanctions_pairs	Grok 4	0.850	0.776	0.676	0.868	76.249	0.824	0.830	0.734	0.913	0.769
opensanctions_pairs	Llama 3.3 70B	0.790	0.682	0.567	0.796	76.249	0.734	0.741	0.619	0.845	0.769
papadakis_dn	Claude Opus 5	0.920	0.955	0.894	0.995	32.601	0.789	0.322	0.131	0.814	0.018
papadakis_dn	Claude Sonnet 5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.559	0.939	0.018
papadakis_dn	DeepSeek-chat	0.940	0.993	0.986	0.998	32.601	0.833	0.757	0.534	0.923	0.018
papadakis_dn	DeepSeek-R1	0.940	0.993	0.986	0.998	32.601	0.833	0.757	0.527	0.918	0.018
papadakis_dn	Gemini 3.1 Pro	0.930	0.974	0.927	0.997	32.601	0.811	0.452	0.196	0.886	0.018
papadakis_dn	Gemini 3.8 Flash	0.940	0.992	0.984	0.998	32.601	0.824	0.702	0.442	0.893	0.018
papadakis_dn	GPT-5.4-mini	0.950	0.994	0.988	0.999	32.601	0.857	0.786	0.575	0.949	0.018
papadakis_dn	GPT-5.5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.564	0.944	0.018
papadakis_dn	gpt-oss-120b	0.930	0.992	0.984	0.998	32.601	0.788	0.682	0.431	0.890	0.018
papadakis_dn	Grok 4	0.950	0.994	0.988	0.999	32.601	0.857	0.786	0.562	0.951	0.018
papadakis_dn	Llama 3.3 70B	0.940	0.975	0.926	0.998	32.601	0.833	0.463	0.185	0.905	0.018
realhitbench	Claude Opus 5	0.743	0.745	0.662	0.813	98.080	–	–	–	–	–
realhitbench	Claude Sonnet 5	0.681	0.684	0.606	0.769	98.080	–	–	–	–	–
realhitbench	DeepSeek-chat	0.662	0.656	0.577	0.745	98.080	–	–	–	–	–
realhitbench	DeepSeek-R1	0.672	0.673	0.590	0.755	98.080	–	–	–	–	–
realhitbench	Gemini 3.1 Pro	0.699	0.698	0.614	0.780	98.080	–	–	–	–	–
realhitbench	Gemini 3.8 Flash	0.746	0.747	0.668	0.823	98.080	–	–	–	–	–
realhitbench	GPT-5.4-mini	0.661	0.658	0.574	0.742	98.080	–	–	–	–	–
realhitbench	GPT-5.5	0.724	0.721	0.644	0.799	98.080	–	–	–	–	–
realhitbench	gpt-oss-120b	0.603	0.602	0.514	0.691	98.080	–	–	–	–	–
realhitbench	Grok 4	0.640	0.638	0.549	0.725	98.080	–	–	–	–	–
realhitbench	Llama 3.3 70B	0.611	0.616	0.538	0.694	98.080	–	–	–	–	–
smat	Claude Opus 5	0.933	0.931	0.886	0.968	98.100	–	–	–	–	–

smat	Claude Sonnet 5	0.938	0.936	0.894	0.973	98.100	–	–	–	–	–
smat	DeepSeek-chat	0.918	0.916	0.863	0.958	98.100	–	–	–	–	–
smat	DeepSeek-R1	0.930	0.928	0.882	0.968	98.100	–	–	–	–	–
smat	Gemini 3.1 Pro	0.883	0.881	0.822	0.930	98.100	–	–	–	–	–
smat	Gemini 3.8 Flash	0.921	0.919	0.871	0.960	98.100	–	–	–	–	–
smat	GPT-5.4-mini	0.956	0.958	0.922	0.988	98.100	–	–	–	–	–
smat	GPT-5.5	0.940	0.939	0.899	0.975	98.100	–	–	–	–	–
smat	gpt-oss-120b	0.916	0.913	0.862	0.955	98.100	–	–	–	–	–
smat	Grok 4	0.936	0.933	0.887	0.973	98.100	–	–	–	–	–
smat	Llama 3.3 70B	0.806	0.803	0.737	0.856	98.100	–	–	–	–	–
suc	Claude Opus 5	0.986	0.988	0.965	1.000	98.920	–	–	–	–	–
suc	Claude Sonnet 5	0.911	0.908	0.857	0.954	98.920	–	–	–	–	–
suc	DeepSeek-chat	0.858	0.858	0.795	0.910	98.920	–	–	–	–	–
suc	DeepSeek-R1	0.942	0.939	0.893	0.976	98.920	–	–	–	–	–
suc	Gemini 3.1 Pro	0.945	0.939	0.898	0.976	98.920	–	–	–	–	–
suc	Gemini 3.8 Flash	0.976	0.978	0.949	1.000	98.920	–	–	–	–	–
suc	GPT-5.4-mini	0.914	0.910	0.855	0.957	98.920	–	–	–	–	–
suc	GPT-5.5	0.943	0.938	0.895	0.977	98.920	–	–	–	–	–
suc	gpt-oss-120b	0.932	0.930	0.882	0.972	98.920	–	–	–	–	–
suc	Grok 4	0.943	0.938	0.897	0.976	98.920	–	–	–	–	–
suc	Llama 3.3 70B	0.740	0.738	0.656	0.812	98.920	–	–	–	–	–
tab_fact	Claude Opus 5	0.660	0.667	0.571	0.762	99.360	0.673	0.696	0.586	0.781	0.500
tab_fact	Claude Sonnet 5	0.740	0.733	0.641	0.820	99.360	0.698	0.710	0.590	0.806	0.500
tab_fact	DeepSeek-chat	0.840	0.826	0.744	0.901	99.360	0.789	0.789	0.677	0.884	0.500
tab_fact	DeepSeek-R1	0.770	0.763	0.672	0.846	99.360	0.729	0.740	0.617	0.836	0.500
tab_fact	Gemini 3.1 Pro	0.830	0.818	0.746	0.901	99.360	0.785	0.788	0.672	0.876	0.500
tab_fact	Gemini 3.8 Flash	0.860	0.848	0.773	0.913	99.360	0.821	0.821	0.720	0.900	0.500
tab_fact	GPT-5.4-mini	0.820	0.808	0.734	0.891	99.360	0.769	0.772	0.654	0.859	0.500
tab_fact	GPT-5.5	0.840	0.829	0.748	0.903	99.360	0.800	0.803	0.700	0.892	0.500
tab_fact	gpt-oss-120b	0.830	0.815	0.739	0.890	99.360	0.773	0.773	0.647	0.868	0.500
tab_fact	Grok 4	0.850	0.839	0.762	0.912	99.360	0.810	0.812	0.707	0.897	0.500
tab_fact	Llama 3.3 70B	0.780	0.769	0.681	0.853	99.360	0.725	0.732	0.615	0.832	0.500
tabis	Claude Opus 5	0.960	0.964	0.915	0.994	77.069	–	–	–	–	–
tabis	Claude Sonnet 5	0.920	0.917	0.846	0.971	77.069	–	–	–	–	–
tabis	DeepSeek-chat	0.770	0.710	0.602	0.807	77.069	–	–	–	–	–
tabis	DeepSeek-R1	0.970	0.982	0.957	1.000	77.069	–	–	–	–	–
tabis	Gemini 3.1 Pro	0.950	0.970	0.940	0.994	77.069	–	–	–	–	–
tabis	Gemini 3.8 Flash	0.950	0.970	0.940	0.994	77.069	–	–	–	–	–
tabis	GPT-5.4-mini	0.900	0.905	0.835	0.962	77.069	–	–	–	–	–
tabis	GPT-5.5	0.960	0.976	0.946	0.994	77.069	–	–	–	–	–

tabis	gpt-oss-120b	0.920	0.951	0.915	0.983	77.069	–	–	–	–	–
tabis	Grok 4	0.980	0.988	0.966	1.000	77.069	–	–	–	–	–
tabis	Llama 3.3 70B	0.820	0.821	0.734	0.901	77.069	–	–	–	–	–
tablebench	Claude Opus 5	0.846	0.846	0.776	0.912	89.187	–	–	–	–	–
tablebench	Claude Sonnet 5	0.877	0.875	0.813	0.931	89.187	–	–	–	–	–
tablebench	DeepSeek-chat	0.833	0.844	0.782	0.904	89.187	–	–	–	–	–
tablebench	DeepSeek-R1	0.803	0.809	0.733	0.876	89.187	–	–	–	–	–
tablebench	Gemini 3.1 Pro	0.858	0.853	0.781	0.917	89.187	–	–	–	–	–
tablebench	Gemini 3.8 Flash	0.845	0.843	0.767	0.906	89.187	–	–	–	–	–
tablebench	GPT-5.4-mini	0.841	0.835	0.764	0.898	89.187	–	–	–	–	–
tablebench	GPT-5.5	0.852	0.850	0.781	0.916	89.187	–	–	–	–	–
tablebench	gpt-oss-120b	0.813	0.808	0.738	0.875	89.187	–	–	–	–	–
tablebench	Grok 4	0.839	0.837	0.760	0.903	89.187	–	–	–	–	–
tablebench	Llama 3.3 70B	0.660	0.641	0.554	0.725	89.187	–	–	–	–	–
tableeval	Claude Opus 5	0.765	0.784	0.689	0.874	69.689	–	–	–	–	–
tableeval	Claude Sonnet 5	0.733	0.764	0.665	0.852	69.689	–	–	–	–	–
tableeval	DeepSeek-chat	0.711	0.708	0.609	0.802	69.689	–	–	–	–	–
tableeval	DeepSeek-R1	0.773	0.760	0.662	0.856	69.689	–	–	–	–	–
tableeval	Gemini 3.1 Pro	0.790	0.794	0.702	0.884	69.689	–	–	–	–	–
tableeval	Gemini 3.8 Flash	0.791	0.805	0.709	0.894	69.689	–	–	–	–	–
tableeval	GPT-5.4-mini	0.742	0.751	0.654	0.842	69.689	–	–	–	–	–
tableeval	GPT-5.5	0.784	0.794	0.693	0.882	69.689	–	–	–	–	–
tableeval	gpt-oss-120b	0.773	0.774	0.674	0.861	69.689	–	–	–	–	–
tableeval	Grok 4	0.802	0.785	0.685	0.877	69.689	–	–	–	–	–
tableeval	Llama 3.3 70B	0.636	0.618	0.506	0.722	69.689	–	–	–	–	–
tat_qa	Claude Opus 5	0.861	0.846	0.769	0.913	60.725	–	–	–	–	–
tat_qa	Claude Sonnet 5	0.845	0.845	0.777	0.906	60.725	–	–	–	–	–
tat_qa	DeepSeek-chat	0.806	0.774	0.684	0.864	60.725	–	–	–	–	–
tat_qa	DeepSeek-R1	0.815	0.780	0.705	0.862	60.725	–	–	–	–	–
tat_qa	Gemini 3.1 Pro	0.802	0.745	0.652	0.833	60.725	–	–	–	–	–
tat_qa	Gemini 3.8 Flash	0.829	0.815	0.729	0.888	60.725	–	–	–	–	–
tat_qa	GPT-5.4-mini	0.789	0.719	0.630	0.805	60.725	–	–	–	–	–
tat_qa	GPT-5.5	0.789	0.748	0.669	0.827	60.725	–	–	–	–	–
tat_qa	gpt-oss-120b	0.765	0.684	0.586	0.778	60.725	–	–	–	–	–
tat_qa	Grok 4	0.791	0.710	0.619	0.798	60.725	–	–	–	–	–
tat_qa	Llama 3.3 70B	0.752	0.679	0.573	0.778	60.725	–	–	–	–	–
tpcdi_cells	Claude Opus 5	0.462	0.465	0.378	0.549	95.091	–	–	–	–	–
tpcdi_cells	Claude Sonnet 5	0.462	0.465	0.379	0.560	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-chat	0.462	0.465	0.381	0.561	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-R1	0.462	0.466	0.370	0.552	95.091	–	–	–	–	–
tpcdi_cells	Gemini 3.1 Pro	0.462	0.465	0.372	0.548	95.091	–	–	–	–	–

tpcdi_cells	Gemini 3.8 Flash	0.462	0.465	0.373	0.555	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.4-mini	0.455	0.458	0.373	0.542	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.5	0.462	0.466	0.376	0.559	95.091	–	–	–	–	–
tpcdi_cells	gpt-oss-120b	0.442	0.453	0.362	0.534	95.091	–	–	–	–	–
tpcdi_cells	Grok 4	0.447	0.456	0.368	0.543	95.091	–	–	–	–	–
tpcdi_cells	Llama 3.3 70B	0.442	0.453	0.366	0.544	95.091	–	–	–	–	–
valentine	Claude Opus 5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Claude Sonnet 5	0.994	0.989	0.966	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-chat	0.979	0.967	0.925	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-R1	0.990	0.986	0.957	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.1 Pro	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.8 Flash	0.970	0.988	0.973	1.000	81.736	–	–	–	–	–
valentine	GPT-5.4-mini	0.985	0.978	0.943	1.000	81.736	–	–	–	–	–
valentine	GPT-5.5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	gpt-oss-120b	0.979	0.971	0.934	1.000	81.736	–	–	–	–	–
valentine	Grok 4	0.973	0.957	0.910	1.000	81.736	–	–	–	–	–
valentine	Llama 3.3 70B	0.970	0.958	0.915	0.992	81.736	–	–	–	–	–
walmart_amazon	Claude Opus 5	0.980	0.982	0.947	1.000	70.807	0.976	0.912	0.770	1.000	0.094
walmart_amazon	Claude Sonnet 5	0.950	0.962	0.914	0.996	70.807	0.938	0.823	0.644	0.977	0.094
walmart_amazon	DeepSeek-chat	0.950	0.962	0.914	0.996	70.807	0.938	0.823	0.661	0.977	0.094
walmart_amazon	DeepSeek-R1	0.960	0.978	0.940	0.998	70.807	0.950	0.887	0.733	0.989	0.094
walmart_amazon	Gemini 3.1 Pro	0.970	0.967	0.920	1.000	70.807	0.964	0.848	0.669	1.000	0.094
walmart_amazon	Gemini 3.8 Flash	0.990	0.998	0.993	1.000	70.807	0.988	0.988	0.955	1.000	0.094
walmart_amazon	GPT-5.4-mini	0.950	0.975	0.941	0.998	70.807	0.937	0.874	0.708	0.978	0.094
walmart_amazon	GPT-5.5	0.970	0.967	0.920	1.000	70.807	0.964	0.848	0.676	1.000	0.094
walmart_amazon	gpt-oss-120b	0.940	0.973	0.936	0.996	70.807	0.923	0.860	0.690	0.974	0.094
walmart_amazon	Grok 4	0.960	0.978	0.943	0.998	70.807	0.950	0.887	0.733	0.989	0.094
walmart_amazon	Llama 3.3 70B	0.930	0.971	0.935	0.994	70.807	0.909	0.847	0.703	0.966	0.094
wdc_lspc	Claude Opus 5	0.990	0.996	0.986	1.000	75.082	0.991	0.992	0.974	1.000	0.264
wdc_lspc	Claude Sonnet 5	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.957	1.000	0.264
wdc_lspc	DeepSeek-chat	0.960	0.940	0.876	0.995	75.082	0.964	0.895	0.785	0.989	0.264
wdc_lspc	DeepSeek-R1	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Gemini 3.1 Pro	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.959	1.000	0.264
wdc_lspc	Gemini 3.8 Flash	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.959	1.000	0.264
wdc_lspc	GPT-5.4-mini	0.970	0.986	0.967	1.000	75.082	0.971	0.973	0.937	1.000	0.264
wdc_lspc	GPT-5.5	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	gpt-oss-120b	0.980	0.990	0.976	1.000	75.082	0.981	0.981	0.951	1.000	0.264
wdc_lspc	Grok 4	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Llama 3.3 70B	0.960	0.940	0.875	0.994	75.082	0.964	0.895	0.784	0.990	0.264
wdc_products	Claude Opus 5	0.980	0.990	0.975	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Claude Sonnet 5	0.970	0.986	0.967	1.000	77.046	0.971	0.974	0.938	1.000	0.280

wdc_products	DeepSeek-chat	0.970	0.977	0.941	1.000	77.046	0.972	0.958	0.905	1.000	0.280
wdc_products	DeepSeek-R1	0.980	0.990	0.975	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.1 Pro	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.8 Flash	0.990	0.995	0.984	1.000	77.046	0.991	0.991	0.967	1.000	0.280
wdc_products	GPT-5.4-mini	0.970	0.986	0.965	1.000	77.046	0.971	0.974	0.935	1.000	0.280
wdc_products	GPT-5.5	0.980	0.979	0.939	1.000	77.046	0.981	0.964	0.895	1.000	0.280
wdc_products	gpt-oss-120b	0.980	0.979	0.939	1.000	77.046	0.981	0.964	0.899	1.000	0.280
wdc_products	Grok 4	0.990	0.995	0.983	1.000	77.046	0.991	0.991	0.969	1.000	0.280
wdc_products	Llama 3.3 70B	0.970	0.977	0.942	1.000	77.046	0.972	0.958	0.899	1.000	0.280
wikitablequestions	Claude Opus 5	0.879	0.878	0.812	0.934	97.686	–	–	–	–	–
wikitablequestions	Claude Sonnet 5	0.863	0.863	0.800	0.926	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-chat	0.843	0.842	0.772	0.901	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-R1	0.857	0.856	0.787	0.917	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.1 Pro	0.855	0.855	0.788	0.914	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.8 Flash	0.881	0.881	0.822	0.936	97.686	–	–	–	–	–
wikitablequestions	GPT-5.4-mini	0.842	0.839	0.767	0.901	97.686	–	–	–	–	–
wikitablequestions	GPT-5.5	0.865	0.862	0.798	0.918	97.686	–	–	–	–	–
wikitablequestions	gpt-oss-120b	0.824	0.821	0.750	0.887	97.686	–	–	–	–	–
wikitablequestions	Grok 4	0.867	0.864	0.801	0.923	97.686	–	–	–	–	–
wikitablequestions	Llama 3.3 70B	0.749	0.747	0.673	0.821	97.686	–	–	–	–	–

Table 36: Cross-dataset pairwise tests under rule best-rho: Friedman $\chi^2 = 84.3$ ($p = 7.2\text{e-}14$), Iman–Davenport $F = 11.2$, Nemenyi CD = 2.71, Kendall's $W = 0.263$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3226	21	8	2	0.0729	1.0000	no	no
Claude Opus 5	DeepSeek-chat	-2.6935	21	8	2	0.0187	0.6346	no	no
Claude Opus 5	DeepSeek-R1	-0.9194	16	12	3	0.1468	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5161	20	7	4	0.0533	1.0000	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.4677	19	9	3	0.2060	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.5645	23	8	0	0.0187	0.6346	no	no
Claude Opus 5	GPT-5.5	0.1452	13	11	7	0.5172	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5806	26	4	1	0.0004	0.0171	yes	yes
Claude Opus 5	Grok 4	-1.8710	18	10	3	0.1017	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.9677	24	7	0	0.0006	0.0257	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.3710	20	7	4	0.0318	0.8904	no	no
Claude Sonnet 5	DeepSeek-R1	0.4032	15	14	2	0.5238	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.1935	14	15	2	0.9844	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8548	11	17	3	0.0997	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.2419	19	9	3	0.1036	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.4677	9	19	3	0.0334	0.9012	no	no
Claude Sonnet 5	gpt-oss-120b	-3.2581	21	8	2	0.0038	0.1422	yes	no
Claude Sonnet 5	Grok 4	-0.5484	19	10	2	0.3620	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6452	24	5	2	0.0002	0.0090	yes	yes
DeepSeek-chat	DeepSeek-R1	1.7742	9	21	1	0.0246	0.7366	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.1774	11	17	3	0.1363	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.2258	7	21	3	0.0081	0.2928	no	no
DeepSeek-chat	GPT-5.4-mini	0.1290	12	15	4	0.3777	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.8387	5	22	4	0.0016	0.0638	yes	no
DeepSeek-chat	gpt-oss-120b	-1.8871	20	9	2	0.0334	0.9012	no	no
DeepSeek-chat	Grok 4	0.8226	13	15	3	0.3993	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.2742	23	4	4	0.0004	0.0173	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5968	17	13	1	0.6309	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.4516	11	18	2	0.1337	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.6452	22	8	1	0.0995	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.0645	9	18	4	0.1116	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.6613	26	4	1	0.0000	0.0021	yes	yes
DeepSeek-R1	Grok 4	-0.9516	16	11	4	0.2091	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0005	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.0484	8	17	6	0.0465	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.0484	20	8	3	0.0197	0.6346	no	no
Gemini 3.1 Pro	GPT-5.5	1.6613	8	19	4	0.0196	0.6346	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.0645	23	4	4	0.0004	0.0173	yes	yes
Gemini 3.1 Pro	Grok 4	-0.3548	15	14	2	0.4865	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.4516	26	5	0	0.0001	0.0057	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.0968	21	9	1	0.0254	0.7375	no	no
Gemini 3.8 Flash	GPT-5.5	0.6129	13	16	2	0.8140	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.1129	27	3	1	0.0000	0.0024	yes	yes
Gemini 3.8 Flash	Grok 4	-1.4032	18	11	2	0.0559	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5000	26	4	1	0.0001	0.0052	yes	yes
GPT-5.4-mini	GPT-5.5	2.7097	6	24	1	0.0023	0.0876	no	no
GPT-5.4-mini	gpt-oss-120b	-2.0161	23	8	0	0.0021	0.0814	no	no
GPT-5.4-mini	Grok 4	0.6935	14	14	3	0.8446	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4032	25	4	2	0.0001	0.0038	no	yes
GPT-5.5	gpt-oss-120b	-4.7258	27	2	2	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-2.0161	19	7	5	0.0131	0.4581	no	no
GPT-5.5	Llama 3.3 70B	-5.1129	27	4	0	0.0000	0.0016	yes	yes
gpt-oss-120b	Grok 4	2.7097	5	26	0	0.0001	0.0045	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.3871	18	13	0	0.3081	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.0968	25	5	1	0.0003	0.0130	yes	yes

Table 37: Cross-dataset pairwise tests under rule exact match: Friedman $\chi^2 = 91.5$ ($p = 2.7\text{e-}15$), Iman–Davenport $F = 12.6$, Nemenyi CD = 2.71, Kendall's $W = 0.277$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5000	21	7	3	0.0266	0.7453	no	no
Claude Opus 5	DeepSeek-chat	-3.0161	22	6	3	0.0049	0.1955	yes	no
Claude Opus 5	DeepSeek-R1	-1.2258	15	9	7	0.0873	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.0161	21	6	4	0.0133	0.4573	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0484	19	8	4	0.0550	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.2742	25	5	1	0.0057	0.2158	yes	no
Claude Opus 5	GPT-5.5	-0.3387	13	9	9	0.2972	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.7419	26	3	2	0.0004	0.0157	yes	yes
Claude Opus 5	Grok 4	-1.9677	18	9	4	0.0559	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.5323	26	5	0	0.0001	0.0060	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5161	19	5	7	0.0153	0.4892	no	no
Claude Sonnet 5	DeepSeek-R1	0.2742	12	14	5	0.3506	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.5161	14	13	4	0.8598	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.4516	11	13	7	0.3876	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.7742	19	7	5	0.1052	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.1613	7	15	9	0.0397	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.2419	22	7	2	0.0089	0.3190	yes	no
Claude Sonnet 5	Grok 4	-0.4677	18	9	4	0.4680	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-4.0323	25	4	2	0.0001	0.0045	yes	yes
DeepSeek-chat	DeepSeek-R1	1.7903	8	19	4	0.0202	0.6265	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0000	10	15	6	0.1854	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.9677	6	19	6	0.0131	0.4573	no	no
DeepSeek-chat	GPT-5.4-mini	-0.2581	12	14	5	0.4859	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.6774	6	19	6	0.0052	0.2018	no	no
DeepSeek-chat	gpt-oss-120b	-1.7258	19	9	3	0.0827	1.0000	no	no
DeepSeek-chat	Grok 4	1.0484	12	15	4	0.2936	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.5161	22	3	6	0.0001	0.0065	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.7903	16	12	3	0.6024	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.1774	12	15	4	0.4211	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-2.0484	20	8	3	0.1157	1.0000	no	no
DeepSeek-R1	GPT-5.5	0.8871	8	16	7	0.1471	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5161	26	3	2	0.0001	0.0041	yes	yes
DeepSeek-R1	Grok 4	-0.7419	14	10	7	0.3978	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.3065	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	0.9677	7	17	7	0.0608	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2581	20	8	3	0.0516	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	1.6774	5	16	10	0.0253	0.7351	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.7258	21	3	7	0.0008	0.0336	yes	yes
Gemini 3.1 Pro	Grok 4	0.0484	11	14	6	0.9922	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5161	27	4	0	0.0001	0.0044	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.2258	21	8	2	0.0385	1.0000	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	10	15	6	0.6092	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.6935	24	2	5	0.0001	0.0040	yes	yes
Gemini 3.8 Flash	Grok 4	-0.9194	15	10	6	0.2232	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.4839	26	4	1	0.0001	0.0045	yes	yes
GPT-5.4-mini	GPT-5.5	2.9355	5	24	2	0.0064	0.2363	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.4677	20	10	1	0.0134	0.4573	no	no
GPT-5.4-mini	Grok 4	1.3065	9	16	6	0.3613	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2581	23	4	4	0.0001	0.0053	no	yes
GPT-5.5	gpt-oss-120b	-4.4032	26	1	4	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.6290	16	6	9	0.0480	1.0000	no	no
GPT-5.5	Llama 3.3 70B	-5.1935	27	4	0	0.0000	0.0010	yes	yes
gpt-oss-120b	Grok 4	2.7742	3	24	4	0.0002	0.0066	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.7903	20	8	3	0.0217	0.6516	no	no

Grok 4	Llama 3.3 70B	-3.5645	24	4	3	0.0001	0.0045	yes	yes

Table 38: Cross-dataset pairwise tests under rule knee: Friedman $\chi^2 = 89.3$ ($p = 7.5\text{e-}15$), Iman–Davenport $F = 12.1$, Nemenyi CD = 2.71, Kendall's $W = 0.279$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2581	22	7	2	0.0275	0.7416	no	no
Claude Opus 5	DeepSeek-chat	-2.8871	21	8	2	0.0102	0.3482	yes	no
Claude Opus 5	DeepSeek-R1	-1.0484	16	12	3	0.1079	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5806	20	7	4	0.0367	0.9334	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.2742	16	12	3	0.2513	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.6613	24	7	0	0.0057	0.2057	no	no
Claude Opus 5	GPT-5.5	0.1774	13	11	7	0.4922	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5806	26	4	1	0.0003	0.0134	yes	yes
Claude Opus 5	Grok 4	-1.5806	18	10	3	0.0612	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.0645	25	6	0	0.0002	0.0083	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.6290	20	7	4	0.0235	0.6827	no	no
Claude Sonnet 5	DeepSeek-R1	0.2097	15	14	2	0.6590	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.3226	15	14	2	0.8446	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9839	11	17	3	0.1169	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4032	21	7	3	0.0359	0.9334	no	no
Claude Sonnet 5	GPT-5.5	1.4355	8	20	3	0.0260	0.7293	no	no
Claude Sonnet 5	gpt-oss-120b	-3.3226	22	7	2	0.0030	0.1135	yes	no
Claude Sonnet 5	Grok 4	-0.3226	18	11	2	0.4504	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.8065	25	4	2	0.0001	0.0031	yes	yes
DeepSeek-chat	DeepSeek-R1	1.8387	9	21	1	0.0210	0.6733	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.3065	10	18	3	0.0957	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.6129	6	22	3	0.0061	0.2121	no	no
DeepSeek-chat	GPT-5.4-mini	0.2258	10	17	4	0.2096	1.0000	no	no
DeepSeek-chat	GPT-5.5	3.0645	6	21	4	0.0018	0.0730	yes	no
DeepSeek-chat	gpt-oss-120b	-1.6935	20	9	2	0.0560	1.0000	no	no
DeepSeek-chat	Grok 4	1.3065	11	17	3	0.1700	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1774	22	5	4	0.0006	0.0238	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5323	17	13	1	0.6309	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.7742	11	18	2	0.1610	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.6129	21	9	1	0.0712	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.2258	8	19	4	0.0950	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5323	26	4	1	0.0001	0.0032	yes	yes
DeepSeek-R1	Grok 4	-0.5323	14	13	4	0.3987	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0161	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.3065	8	17	6	0.0465	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.0806	21	7	3	0.0359	0.9334	no	no
Gemini 3.1 Pro	GPT-5.5	1.7581	7	20	4	0.0151	0.4969	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.0000	23	4	4	0.0004	0.0182	yes	yes
Gemini 3.1 Pro	Grok 4	0.0000	14	15	2	0.8369	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.4839	25	6	0	0.0001	0.0048	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.3871	22	8	1	0.0218	0.6762	no	no
Gemini 3.8 Flash	GPT-5.5	0.4516	14	15	2	0.7837	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.3065	27	3	1	0.0000	0.0024	yes	yes
Gemini 3.8 Flash	Grok 4	-1.3065	18	11	2	0.1190	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.7903	26	4	1	0.0001	0.0031	yes	yes
GPT-5.4-mini	GPT-5.5	2.8387	6	24	1	0.0034	0.1253	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.9194	23	8	0	0.0024	0.0928	no	no
GPT-5.4-mini	Grok 4	1.0806	12	16	3	0.5305	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4032	26	3	2	0.0000	0.0017	no	yes
GPT-5.5	gpt-oss-120b	-4.7581	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7581	19	7	5	0.0223	0.6762	no	no
GPT-5.5	Llama 3.3 70B	-5.2419	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	3.0000	4	27	0	0.0001	0.0043	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4839	20	11	0	0.1470	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.4839	25	5	1	0.0001	0.0065	yes	yes

Table 39: Cross-dataset pairwise tests under rule top1: Friedman $\chi^2 = 89.3$ ($p = 7.5\text{e-}15$), Iman–Davenport $F = 12.1$, Nemenyi CD = 2.71, Kendall's $W = 0.279$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2581	22	7	2	0.0275	0.7416	no	no
Claude Opus 5	DeepSeek-chat	-2.8871	21	8	2	0.0102	0.3482	yes	no
Claude Opus 5	DeepSeek-R1	-1.0484	16	12	3	0.1079	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5806	20	7	4	0.0367	0.9334	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.2742	16	12	3	0.2513	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.6613	24	7	0	0.0057	0.2057	no	no
Claude Opus 5	GPT-5.5	0.1774	13	11	7	0.4922	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5806	26	4	1	0.0003	0.0134	yes	yes
Claude Opus 5	Grok 4	-1.5806	18	10	3	0.0612	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.0645	25	6	0	0.0002	0.0083	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.6290	20	7	4	0.0235	0.6827	no	no
Claude Sonnet 5	DeepSeek-R1	0.2097	15	14	2	0.6590	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.3226	15	14	2	0.8446	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9839	11	17	3	0.1169	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4032	21	7	3	0.0359	0.9334	no	no
Claude Sonnet 5	GPT-5.5	1.4355	8	20	3	0.0260	0.7293	no	no
Claude Sonnet 5	gpt-oss-120b	-3.3226	22	7	2	0.0030	0.1135	yes	no
Claude Sonnet 5	Grok 4	-0.3226	18	11	2	0.4504	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.8065	25	4	2	0.0001	0.0031	yes	yes
DeepSeek-chat	DeepSeek-R1	1.8387	9	21	1	0.0210	0.6733	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.3065	10	18	3	0.0957	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.6129	6	22	3	0.0061	0.2121	no	no
DeepSeek-chat	GPT-5.4-mini	0.2258	10	17	4	0.2096	1.0000	no	no
DeepSeek-chat	GPT-5.5	3.0645	6	21	4	0.0018	0.0730	yes	no
DeepSeek-chat	gpt-oss-120b	-1.6935	20	9	2	0.0560	1.0000	no	no
DeepSeek-chat	Grok 4	1.3065	11	17	3	0.1700	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1774	22	5	4	0.0006	0.0238	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5323	17	13	1	0.6309	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.7742	11	18	2	0.1610	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.6129	21	9	1	0.0712	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.2258	8	19	4	0.0950	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5323	26	4	1	0.0001	0.0032	yes	yes
DeepSeek-R1	Grok 4	-0.5323	14	13	4	0.3987	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0161	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.3065	8	17	6	0.0465	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.0806	21	7	3	0.0359	0.9334	no	no
Gemini 3.1 Pro	GPT-5.5	1.7581	7	20	4	0.0151	0.4969	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.0000	23	4	4	0.0004	0.0182	yes	yes
Gemini 3.1 Pro	Grok 4	0.0000	14	15	2	0.8369	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.4839	25	6	0	0.0001	0.0048	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.3871	22	8	1	0.0218	0.6762	no	no
Gemini 3.8 Flash	GPT-5.5	0.4516	14	15	2	0.7837	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.3065	27	3	1	0.0000	0.0024	yes	yes
Gemini 3.8 Flash	Grok 4	-1.3065	18	11	2	0.1190	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.7903	26	4	1	0.0001	0.0031	yes	yes
GPT-5.4-mini	GPT-5.5	2.8387	6	24	1	0.0034	0.1253	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.9194	23	8	0	0.0024	0.0928	no	no
GPT-5.4-mini	Grok 4	1.0806	12	16	3	0.5305	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4032	26	3	2	0.0000	0.0017	no	yes
GPT-5.5	gpt-oss-120b	-4.7581	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7581	19	7	5	0.0223	0.6762	no	no
GPT-5.5	Llama 3.3 70B	-5.2419	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	3.0000	4	27	0	0.0001	0.0043	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4839	20	11	0	0.1470	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.4839	25	5	1	0.0001	0.0065	yes	yes

Table 40: Cross-dataset pairwise tests under rule top2: Friedman $\chi^2 = 89.1$ ($p = 8.0\text{e-}15$), Iman–Davenport $F = 12.1$, Nemenyi CD = 2.71, Kendall's $W = 0.278$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2581	22	7	2	0.0275	0.7416	no	no
Claude Opus 5	DeepSeek-chat	-2.8871	21	8	2	0.0102	0.3482	yes	no
Claude Opus 5	DeepSeek-R1	-1.0161	16	12	3	0.1167	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5806	20	7	4	0.0367	0.9334	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.3065	16	12	3	0.2513	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.6613	24	7	0	0.0057	0.2057	no	no
Claude Opus 5	GPT-5.5	0.1774	13	11	7	0.5046	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5806	26	4	1	0.0003	0.0134	yes	yes
Claude Opus 5	Grok 4	-1.5806	18	10	3	0.0640	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.0645	25	6	0	0.0002	0.0083	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.6290	20	7	4	0.0235	0.6827	no	no
Claude Sonnet 5	DeepSeek-R1	0.2419	15	14	2	0.6876	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.3226	15	14	2	0.8446	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9516	11	17	3	0.1169	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4032	21	7	3	0.0359	0.9334	no	no
Claude Sonnet 5	GPT-5.5	1.4355	8	20	3	0.0260	0.7293	no	no
Claude Sonnet 5	gpt-oss-120b	-3.3226	22	7	2	0.0028	0.1065	yes	no
Claude Sonnet 5	Grok 4	-0.3226	18	11	2	0.4504	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.8065	25	4	2	0.0001	0.0031	yes	yes
DeepSeek-chat	DeepSeek-R1	1.8710	9	21	1	0.0170	0.5454	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.3065	10	18	3	0.0957	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.5806	6	22	3	0.0061	0.2121	no	no
DeepSeek-chat	GPT-5.4-mini	0.2258	10	17	4	0.2026	1.0000	no	no
DeepSeek-chat	GPT-5.5	3.0645	6	21	4	0.0019	0.0760	yes	no
DeepSeek-chat	gpt-oss-120b	-1.6935	20	9	2	0.0489	1.0000	no	no
DeepSeek-chat	Grok 4	1.3065	11	17	3	0.1825	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1774	22	5	4	0.0006	0.0238	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5645	17	13	1	0.6309	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.7097	12	17	2	0.1857	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.6452	21	9	1	0.0712	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.1935	8	19	4	0.0950	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5645	26	4	1	0.0001	0.0032	yes	yes
DeepSeek-R1	Grok 4	-0.5645	14	13	4	0.3666	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.2742	8	17	6	0.0465	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.0806	21	7	3	0.0359	0.9334	no	no
Gemini 3.1 Pro	GPT-5.5	1.7581	7	20	4	0.0159	0.5244	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.0000	23	4	4	0.0004	0.0182	yes	yes
Gemini 3.1 Pro	Grok 4	0.0000	14	15	2	0.8369	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.4839	25	6	0	0.0001	0.0048	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.3548	22	8	1	0.0207	0.6420	no	no
Gemini 3.8 Flash	GPT-5.5	0.4839	14	15	2	0.7988	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.2742	27	3	1	0.0000	0.0024	yes	yes
Gemini 3.8 Flash	Grok 4	-1.2742	18	11	2	0.1190	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.7581	26	4	1	0.0001	0.0031	yes	yes
GPT-5.4-mini	GPT-5.5	2.8387	6	24	1	0.0034	0.1253	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.9194	23	8	0	0.0018	0.0732	no	no
GPT-5.4-mini	Grok 4	1.0806	12	16	3	0.5434	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.4032	26	3	2	0.0000	0.0017	no	yes
GPT-5.5	gpt-oss-120b	-4.7581	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7581	19	7	5	0.0223	0.6690	no	no
GPT-5.5	Llama 3.3 70B	-5.2419	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	3.0000	4	27	0	0.0001	0.0040	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4839	20	11	0	0.1701	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.4839	25	5	1	0.0001	0.0065	yes	yes

Table 41: Cross-dataset pairwise tests under rule top3: Friedman $\chi^2 = 89.8$ ($p = 5.9\text{e-}15$), Iman–Davenport $F = 12.2$, Nemenyi CD = 2.71, Kendall's $W = 0.280$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5806	22	7	2	0.0224	0.6501	no	no
Claude Opus 5	DeepSeek-chat	-2.6613	21	8	2	0.0151	0.4827	no	no
Claude Opus 5	DeepSeek-R1	-1.1129	17	11	3	0.0880	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5806	20	7	4	0.0333	0.8144	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5968	19	9	3	0.2204	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.2097	24	7	0	0.0042	0.1601	yes	no
Claude Opus 5	GPT-5.5	0.1452	13	11	7	0.4922	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.6774	26	4	1	0.0004	0.0148	yes	yes
Claude Opus 5	Grok 4	-1.7419	18	10	3	0.0977	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.1613	25	6	0	0.0002	0.0086	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.0806	19	8	4	0.0793	1.0000	no	no
Claude Sonnet 5	DeepSeek-R1	0.4677	14	15	2	0.4620	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	0.0000	13	16	2	0.7989	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9839	12	16	3	0.1470	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.6290	20	8	3	0.0326	0.8144	no	no
Claude Sonnet 5	GPT-5.5	1.7258	8	20	3	0.0260	0.7032	no	no
Claude Sonnet 5	gpt-oss-120b	-3.0968	21	8	2	0.0052	0.1934	yes	no
Claude Sonnet 5	Grok 4	-0.1613	17	12	2	0.7169	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.5806	25	4	2	0.0001	0.0061	yes	yes
DeepSeek-chat	DeepSeek-R1	1.5484	9	21	1	0.0608	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0806	12	16	3	0.2318	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.0645	6	22	3	0.0072	0.2533	no	no
DeepSeek-chat	GPT-5.4-mini	-0.5484	15	12	4	0.6663	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.8065	5	22	4	0.0029	0.1123	yes	no
DeepSeek-chat	gpt-oss-120b	-2.0161	21	8	2	0.0172	0.5319	no	no
DeepSeek-chat	Grok 4	0.9194	13	15	3	0.4330	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.5000	23	4	4	0.0001	0.0067	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.4677	16	14	1	0.6449	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.5161	11	18	2	0.1442	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-2.0968	23	7	1	0.0295	0.7664	no	no
DeepSeek-R1	GPT-5.5	1.2581	8	19	4	0.1116	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5645	27	3	1	0.0000	0.0013	yes	yes
DeepSeek-R1	Grok 4	-0.6290	14	13	4	0.4558	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	0.9839	7	18	6	0.0423	0.9740	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.6290	22	6	3	0.0064	0.2316	no	no
Gemini 3.1 Pro	GPT-5.5	1.7258	7	20	4	0.0081	0.2764	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.0968	24	3	4	0.0002	0.0069	yes	yes
Gemini 3.1 Pro	Grok 4	-0.1613	15	14	2	0.6451	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5806	26	5	0	0.0001	0.0053	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.6129	22	8	1	0.0091	0.3012	no	no
Gemini 3.8 Flash	GPT-5.5	0.7419	13	16	2	0.9375	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.0806	26	4	1	0.0001	0.0043	yes	yes
Gemini 3.8 Flash	Grok 4	-1.1452	18	11	2	0.0845	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5645	25	5	1	0.0001	0.0049	yes	yes
GPT-5.4-mini	GPT-5.5	3.3548	5	25	1	0.0009	0.0357	yes	yes
GPT-5.4-mini	gpt-oss-120b	-1.4677	21	10	0	0.0242	0.6771	no	no
GPT-5.4-mini	Grok 4	1.4677	11	17	3	0.2318	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-1.9516	24	5	2	0.0001	0.0050	no	yes
GPT-5.5	gpt-oss-120b	-4.8226	27	2	2	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.8871	19	7	5	0.0172	0.5319	no	no
GPT-5.5	Llama 3.3 70B	-5.3065	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.9355	5	26	0	0.0001	0.0034	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4839	19	12	0	0.0919	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.4194	26	4	1	0.0002	0.0068	yes	yes

Table 42: Cross-dataset pairwise tests under rule top4: Friedman $\chi^2 = 89.8$ ($p = 5.9\text{e-}15$), Iman–Davenport $F = 12.2$, Nemenyi CD = 2.71, Kendall's $W = 0.280$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5806	22	7	2	0.0224	0.6501	no	no
Claude Opus 5	DeepSeek-chat	-2.6613	21	8	2	0.0151	0.4827	no	no
Claude Opus 5	DeepSeek-R1	-1.1129	17	11	3	0.0810	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5806	20	7	4	0.0317	0.7925	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5968	19	9	3	0.2204	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.2097	24	7	0	0.0042	0.1601	yes	no
Claude Opus 5	GPT-5.5	0.1452	13	11	7	0.4922	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.6774	26	4	1	0.0004	0.0148	yes	yes
Claude Opus 5	Grok 4	-1.7419	18	10	3	0.0977	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.1613	25	6	0	0.0002	0.0079	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.0806	19	8	4	0.0793	1.0000	no	no
Claude Sonnet 5	DeepSeek-R1	0.4677	14	15	2	0.4620	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	0.0000	13	16	2	0.7989	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9839	12	16	3	0.1416	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.6290	20	8	3	0.0326	0.7925	no	no
Claude Sonnet 5	GPT-5.5	1.7258	8	20	3	0.0260	0.7032	no	no
Claude Sonnet 5	gpt-oss-120b	-3.0968	21	8	2	0.0052	0.1934	yes	no
Claude Sonnet 5	Grok 4	-0.1613	17	12	2	0.6878	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.5806	25	4	2	0.0001	0.0061	yes	yes
DeepSeek-chat	DeepSeek-R1	1.5484	9	21	1	0.0608	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0806	12	16	3	0.2395	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.0645	6	22	3	0.0072	0.2533	no	no
DeepSeek-chat	GPT-5.4-mini	-0.5484	15	12	4	0.6950	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.8065	5	22	4	0.0029	0.1123	yes	no
DeepSeek-chat	gpt-oss-120b	-2.0161	21	8	2	0.0172	0.5173	no	no
DeepSeek-chat	Grok 4	0.9194	13	15	3	0.4563	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.5000	23	4	4	0.0001	0.0067	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.4677	16	14	1	0.6449	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.5161	11	18	2	0.1388	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-2.0968	23	7	1	0.0295	0.7664	no	no
DeepSeek-R1	GPT-5.5	1.2581	8	19	4	0.1116	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5645	27	3	1	0.0000	0.0013	yes	yes
DeepSeek-R1	Grok 4	-0.6290	14	13	4	0.4558	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	0.9839	7	18	6	0.0385	0.8859	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.6290	22	6	3	0.0068	0.2457	no	no
Gemini 3.1 Pro	GPT-5.5	1.7258	7	20	4	0.0086	0.2928	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.0968	24	3	4	0.0002	0.0075	yes	yes
Gemini 3.1 Pro	Grok 4	-0.1613	15	14	2	0.6172	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5806	26	5	0	0.0001	0.0053	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.6129	22	8	1	0.0091	0.3012	no	no
Gemini 3.8 Flash	GPT-5.5	0.7419	13	16	2	0.9375	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.0806	26	4	1	0.0001	0.0043	yes	yes
Gemini 3.8 Flash	Grok 4	-1.1452	18	11	2	0.0862	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5645	25	5	1	0.0001	0.0049	yes	yes
GPT-5.4-mini	GPT-5.5	3.3548	5	25	1	0.0009	0.0357	yes	yes
GPT-5.4-mini	gpt-oss-120b	-1.4677	21	10	0	0.0242	0.6771	no	no
GPT-5.4-mini	Grok 4	1.4677	11	17	3	0.2318	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-1.9516	24	5	2	0.0001	0.0050	no	yes
GPT-5.5	gpt-oss-120b	-4.8226	27	2	2	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.8871	19	7	5	0.0163	0.5043	no	no
GPT-5.5	Llama 3.3 70B	-5.3065	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.9355	5	26	0	0.0001	0.0034	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4839	19	12	0	0.0882	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.4194	26	4	1	0.0002	0.0068	yes	yes

Table 43: Cross-dataset pairwise tests under rule top5: Friedman $\chi^2 = 89.7$ ($p = 6.1\text{e-}15$), Iman–Davenport $F = 12.2$, Nemenyi CD = 2.71, Kendall's $W = 0.280$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5161	22	7	2	0.0289	0.7796	no	no
Claude Opus 5	DeepSeek-chat	-2.5645	21	8	2	0.0218	0.6115	no	no
Claude Opus 5	DeepSeek-R1	-0.9839	16	12	3	0.1414	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5161	20	7	4	0.0404	0.9483	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5000	18	10	3	0.2513	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.1129	24	7	0	0.0031	0.1201	yes	no
Claude Opus 5	GPT-5.5	0.2097	13	11	7	0.5046	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.6452	26	4	1	0.0004	0.0160	yes	yes
Claude Opus 5	Grok 4	-1.7419	18	10	3	0.1192	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.0968	25	6	0	0.0003	0.0119	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.0484	19	8	4	0.0827	1.0000	no	no
Claude Sonnet 5	DeepSeek-R1	0.5323	14	15	2	0.4270	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	0.0000	14	15	2	0.8293	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	1.0161	12	16	3	0.1263	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.5968	20	8	3	0.0395	0.9483	no	no
Claude Sonnet 5	GPT-5.5	1.7258	7	21	3	0.0163	0.4891	no	no
Claude Sonnet 5	gpt-oss-120b	-3.1290	21	8	2	0.0044	0.1610	yes	no
Claude Sonnet 5	Grok 4	-0.2258	17	12	2	0.6592	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.5806	25	4	2	0.0001	0.0059	yes	yes
DeepSeek-chat	DeepSeek-R1	1.5806	10	20	1	0.0608	1.0000	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0484	11	17	3	0.1957	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.0645	6	22	3	0.0086	0.2930	no	no
DeepSeek-chat	GPT-5.4-mini	-0.5484	15	12	4	0.7538	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.7742	5	22	4	0.0033	0.1243	yes	no
DeepSeek-chat	gpt-oss-120b	-2.0806	21	8	2	0.0163	0.4891	no	no
DeepSeek-chat	Grok 4	0.8226	13	15	3	0.5434	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.5323	24	3	4	0.0001	0.0050	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5323	16	14	1	0.6876	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.4839	11	18	2	0.1191	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-2.1290	23	7	1	0.0295	0.7796	no	no
DeepSeek-R1	GPT-5.5	1.1935	9	18	4	0.1355	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.6613	27	3	1	0.0000	0.0015	yes	yes
DeepSeek-R1	Grok 4	-0.7581	15	12	4	0.3074	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.1129	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.0161	7	18	6	0.0367	0.9178	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.5968	22	6	3	0.0077	0.2686	no	no
Gemini 3.1 Pro	GPT-5.5	1.7258	7	20	4	0.0135	0.4186	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.1290	23	4	4	0.0003	0.0146	yes	yes
Gemini 3.1 Pro	Grok 4	-0.2258	16	13	2	0.4743	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5806	26	5	0	0.0001	0.0056	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.6129	22	8	1	0.0091	0.3012	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	13	16	2	0.9687	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.1452	26	4	1	0.0001	0.0037	yes	yes
Gemini 3.8 Flash	Grok 4	-1.2419	18	11	2	0.0760	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5968	25	5	1	0.0001	0.0055	yes	yes
GPT-5.4-mini	GPT-5.5	3.3226	5	25	1	0.0010	0.0383	yes	yes
GPT-5.4-mini	gpt-oss-120b	-1.5323	22	9	0	0.0048	0.1715	no	no
GPT-5.4-mini	Grok 4	1.3710	11	17	3	0.2242	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-1.9839	24	5	2	0.0001	0.0050	no	yes
GPT-5.5	gpt-oss-120b	-4.8548	27	2	2	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.9516	19	7	5	0.0124	0.3963	no	no
GPT-5.5	Llama 3.3 70B	-5.3065	27	4	0	0.0000	0.0011	yes	yes
gpt-oss-120b	Grok 4	2.9032	5	26	0	0.0001	0.0050	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4516	18	13	0	0.2027	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.3548	26	4	1	0.0001	0.0063	yes	yes

Table 44: Cross-dataset pairwise tests under rule v1: Friedman $\chi^2 = 88.1$ ($p = 1.3\text{e-}14$), Iman–Davenport $F = 11.9$, Nemenyi CD = 2.71, Kendall's $W = 0.275$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4194	22	7	2	0.0275	0.7965	no	no
Claude Opus 5	DeepSeek-chat	-2.7581	21	8	2	0.0151	0.4676	yes	no
Claude Opus 5	DeepSeek-R1	-1.1129	16	12	3	0.1167	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.7097	20	7	4	0.0465	1.0000	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.4355	18	10	3	0.2279	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.9839	25	6	0	0.0048	0.1715	yes	no
Claude Opus 5	GPT-5.5	0.0806	13	11	7	0.5172	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5806	26	4	1	0.0003	0.0141	yes	yes
Claude Opus 5	Grok 4	-1.7419	18	10	3	0.0900	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.1613	25	6	0	0.0002	0.0097	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.3387	20	7	4	0.0488	1.0000	no	no
Claude Sonnet 5	DeepSeek-R1	0.3065	15	14	2	0.4620	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.2903	14	15	2	0.9844	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9839	11	17	3	0.0957	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.5645	20	8	3	0.0395	0.9491	no	no
Claude Sonnet 5	GPT-5.5	1.5000	9	19	3	0.0368	0.9491	no	no
Claude Sonnet 5	gpt-oss-120b	-3.1613	21	8	2	0.0038	0.1461	yes	no
Claude Sonnet 5	Grok 4	-0.3226	18	11	2	0.5898	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.7419	25	4	2	0.0001	0.0039	yes	yes
DeepSeek-chat	DeepSeek-R1	1.6452	9	21	1	0.0365	0.9491	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0484	12	16	3	0.1524	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.3226	6	22	3	0.0064	0.2251	no	no
DeepSeek-chat	GPT-5.4-mini	-0.2258	13	14	4	0.6241	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.8387	6	21	4	0.0025	0.0988	yes	no
DeepSeek-chat	gpt-oss-120b	-1.8226	20	9	2	0.0351	0.9480	no	no
DeepSeek-chat	Grok 4	1.0161	13	15	3	0.3270	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.4032	22	5	4	0.0004	0.0165	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5968	17	13	1	0.4986	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.6774	11	18	2	0.1287	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.8710	22	8	1	0.0454	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.1935	8	19	4	0.1073	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.4677	26	4	1	0.0000	0.0018	yes	yes
DeepSeek-R1	Grok 4	-0.6290	14	13	4	0.3987	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.2742	7	18	6	0.0317	0.8885	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2742	21	7	3	0.0114	0.3776	no	no
Gemini 3.1 Pro	GPT-5.5	1.7903	6	21	4	0.0086	0.2928	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.8710	23	4	4	0.0003	0.0138	yes	yes
Gemini 3.1 Pro	Grok 4	-0.0323	15	14	2	0.6592	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.4516	26	5	0	0.0001	0.0052	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.5484	22	8	1	0.0114	0.3776	no	no
Gemini 3.8 Flash	GPT-5.5	0.5161	14	15	2	0.8140	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.1452	26	4	1	0.0001	0.0029	yes	yes
Gemini 3.8 Flash	Grok 4	-1.3065	18	11	2	0.0937	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.7258	26	4	1	0.0001	0.0040	yes	yes
GPT-5.4-mini	GPT-5.5	3.0645	5	25	1	0.0013	0.0505	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.5968	22	9	0	0.0042	0.1558	no	no
GPT-5.4-mini	Grok 4	1.2419	12	16	3	0.3468	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1774	25	4	2	0.0001	0.0029	no	yes
GPT-5.5	gpt-oss-120b	-4.6613	27	2	2	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.8226	18	8	5	0.0212	0.6353	no	no
GPT-5.5	Llama 3.3 70B	-5.2419	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.8387	5	26	0	0.0001	0.0038	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5806	19	12	0	0.1169	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.4194	26	4	1	0.0001	0.0052	yes	yes

Table 45: The (J, ρ) Pareto front.

rank	w__f1	w__decay	w__precision	w__recall	J	rho	auc	wrong__mean	hedge__gap
1	0.350	0.600	0.050	0.000	0.425	0.876	0.994	0.036	-0.560
2	0.300	0.600	0.100	0.000	0.424	0.876	0.994	0.036	-0.564
5	0.300	0.150	0.400	0.150	0.424	0.877	0.994	0.032	-0.510
26	0.200	0.150	0.500	0.150	0.422	0.877	0.994	0.032	-0.519
51	0.750	0.100	0.050	0.100	0.421	0.877	0.994	0.031	-0.486
90	0.150	0.550	0.150	0.150	0.420	0.879	0.994	0.036	-0.526
117	0.350	0.300	0.300	0.050	0.419	0.879	0.994	0.033	-0.540
146	0.350	0.500	0.150	0.000	0.419	0.881	0.994	0.035	-0.559
239	0.250	0.500	0.150	0.100	0.418	0.881	0.994	0.035	-0.534
472	0.500	0.400	0.100	0.000	0.416	0.882	0.994	0.034	-0.545
574	0.400	0.450	0.150	0.000	0.415	0.882	0.994	0.035	-0.554
657	0.450	0.400	0.150	0.000	0.414	0.882	0.994	0.034	-0.549
1132	0.300	0.400	0.250	0.050	0.411	0.882	0.994	0.034	-0.546
1366	0.250	0.400	0.300	0.050	0.409	0.882	0.994	0.034	-0.550
1372	0.050	0.350	0.550	0.050	0.409	0.882	0.994	0.034	-0.568
1446	0.150	0.350	0.500	0.000	0.407	0.882	0.994	0.034	-0.576
1622	0.000	0.350	0.600	0.050	0.405	0.883	0.994	0.034	-0.573

Table 46: Mean score per dataset and model under rule best-rho.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Son-net 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.916	0.946	0.925	0.958	0.928	0.923	0.926	0.957	0.953	0.931	0.820
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.297	0.297	0.186	0.216	0.234	0.181	0.125	0.177	0.184	0.082	0.119
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.445	0.390	0.450	0.444	0.431	0.455	0.429	0.455	0.450	0.283	0.443
fintagging	0.897	0.908	0.893	0.876	0.898	0.887	0.887	0.892	0.857	0.863	0.856
hitab	0.748	0.775	0.751	0.783	0.757	0.775	0.742	0.778	0.809	0.744	0.782
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.963	0.970	0.968	0.978	0.953	0.953	0.939	0.916	0.952	0.963	0.714
mmtu	0.899	0.882	0.915	0.851	0.874	0.916	0.902	0.885	0.866	0.830	0.802
officeqa	0.194	0.199	0.167	0.145	0.171	0.119	0.129	0.107	0.075	0.079	0.097
officeqa_pro_v2	0.198	0.272	0.188	0.143	0.157	0.120	0.119	0.123	0.121	0.104	0.114
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.752	0.744	0.728	0.669	0.676	0.706	0.643	0.662	0.663	0.611	0.621
smat	0.919	0.932	0.940	0.929	0.938	0.871	0.934	0.956	0.917	0.912	0.783
suc	0.976	0.986	0.943	0.946	0.911	0.949	0.943	0.914	0.858	0.932	0.741
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.848	0.849	0.855	0.806	0.880	0.863	0.841	0.843	0.835	0.820	0.655
tableeval	0.792	0.767	0.786	0.771	0.732	0.790	0.802	0.742	0.704	0.774	0.632
tat_qa	0.821	0.855	0.786	0.815	0.835	0.798	0.787	0.790	0.802	0.760	0.758
tpcdi_cells	0.516	0.516	0.517	0.517	0.516	0.516	0.499	0.510	0.516	0.497	0.497
valentine	0.970	1.000	1.000	0.990	0.994	1.000	0.974	0.984	0.980	0.980	0.969
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.883	0.884	0.870	0.862	0.863	0.860	0.870	0.846	0.849	0.829	0.748

Table 47: Mean score per dataset and model under rule exact match.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.892	0.925	0.903	0.935	0.914	0.903	0.903	0.946	0.925	0.903	0.731
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.190	0.180	0.130	0.130	0.130	0.140	0.080	0.130	0.110	0.050	0.050
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.010	0.020	0.010	0.020	0.010	0.010	0.010	0.000	0.010	0.000	0.000
fintagging	0.820	0.840	0.820	0.790	0.820	0.800	0.810	0.810	0.750	0.760	0.750
hitab	0.680	0.730	0.700	0.740	0.700	0.720	0.680	0.730	0.760	0.680	0.710
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.950	0.960	0.960	0.970	0.940	0.930	0.920	0.890	0.930	0.950	0.620
mmtu	0.870	0.860	0.900	0.830	0.840	0.900	0.880	0.850	0.830	0.790	0.750
officeqa	0.030	0.050	0.030	0.030	0.040	0.020	0.030	0.010	0.040	0.020	0.010
officeqa_pro_v2	0.056	0.133	0.100	0.067	0.056	0.033	0.033	0.011	0.022	0.033	0.022
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.650	0.660	0.630	0.590	0.610	0.620	0.560	0.580	0.580	0.530	0.480
smat	0.880	0.900	0.910	0.900	0.910	0.830	0.910	0.930	0.880	0.880	0.680
suc	0.970	0.980	0.920	0.920	0.880	0.920	0.920	0.880	0.810	0.910	0.670
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.790	0.800	0.810	0.730	0.840	0.810	0.800	0.790	0.780	0.760	0.550
tableeval	0.770	0.740	0.750	0.740	0.660	0.760	0.770	0.690	0.620	0.740	0.540
tat_qa	0.740	0.810	0.680	0.730	0.750	0.700	0.710	0.710	0.730	0.680	0.670
tpcdi_cells	0.350	0.350	0.350	0.350	0.350	0.350	0.330	0.340	0.350	0.330	0.330
valentine	0.970	1.000	1.000	0.990	0.990	1.000	0.960	0.980	0.970	0.970	0.960
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.840	0.840	0.820	0.810	0.830	0.800	0.810	0.810	0.800	0.770	0.670

Table 48: Mean score per dataset and model under rule knee.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.907	0.937	0.918	0.948	0.923	0.917	0.918	0.953	0.941	0.921	0.791
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.319	0.318	0.185	0.237	0.258	0.200	0.128	0.193	0.202	0.082	0.144
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.275	0.247	0.271	0.255	0.228	0.269	0.264	0.186	0.267	0.185	0.147
fintagging	0.868	0.882	0.865	0.843	0.868	0.853	0.858	0.861	0.816	0.823	0.816
hitab	0.745	0.780	0.754	0.790	0.768	0.776	0.747	0.785	0.811	0.744	0.782
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.958	0.967	0.964	0.975	0.949	0.945	0.931	0.903	0.944	0.957	0.689
mmtu	0.896	0.892	0.921	0.854	0.886	0.917	0.904	0.888	0.877	0.833	0.807
officeqa	0.308	0.301	0.260	0.215	0.255	0.190	0.200	0.175	0.100	0.121	0.152
officeqa_pro_v2	0.270	0.343	0.231	0.183	0.210	0.162	0.165	0.189	0.164	0.139	0.165
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.753	0.746	0.731	0.680	0.686	0.709	0.645	0.669	0.660	0.606	0.599
smat	0.904	0.920	0.928	0.918	0.927	0.859	0.925	0.946	0.903	0.901	0.752
suc	0.974	0.984	0.934	0.934	0.901	0.935	0.934	0.901	0.843	0.927	0.718
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.848	0.844	0.857	0.802	0.882	0.860	0.843	0.845	0.836	0.823	0.670
tableeval	0.784	0.756	0.779	0.764	0.728	0.787	0.794	0.741	0.694	0.769	0.623
tat_qa	0.801	0.843	0.777	0.800	0.816	0.791	0.790	0.789	0.787	0.767	0.749
tpcdi_cells	0.443	0.443	0.444	0.444	0.443	0.443	0.426	0.439	0.443	0.424	0.424
valentine	0.970	1.000	1.000	0.990	0.992	1.000	0.968	0.983	0.976	0.976	0.966
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.874	0.874	0.862	0.851	0.856	0.848	0.862	0.841	0.838	0.817	0.740

Table 49: Mean score per dataset and model under rule top1.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.907	0.937	0.918	0.948	0.923	0.917	0.918	0.953	0.941	0.921	0.791
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.319	0.318	0.185	0.237	0.258	0.200	0.128	0.193	0.202	0.082	0.144
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.275	0.247	0.271	0.255	0.228	0.269	0.264	0.186	0.267	0.185	0.147
fintagging	0.868	0.882	0.865	0.843	0.868	0.853	0.858	0.861	0.816	0.823	0.816
hitab	0.745	0.780	0.754	0.790	0.768	0.776	0.747	0.785	0.811	0.744	0.782
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.958	0.967	0.964	0.975	0.949	0.945	0.931	0.903	0.944	0.957	0.689
mmtu	0.896	0.892	0.921	0.854	0.886	0.917	0.904	0.888	0.877	0.833	0.807
officeqa	0.308	0.301	0.260	0.215	0.255	0.190	0.200	0.175	0.100	0.121	0.152
officeqa_pro_v2	0.270	0.343	0.231	0.183	0.210	0.162	0.165	0.189	0.164	0.139	0.165
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.753	0.746	0.731	0.680	0.686	0.709	0.645	0.669	0.660	0.606	0.599
smat	0.904	0.920	0.928	0.918	0.927	0.859	0.925	0.946	0.903	0.901	0.752
suc	0.974	0.984	0.934	0.934	0.901	0.935	0.934	0.901	0.843	0.927	0.718
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.848	0.844	0.857	0.802	0.882	0.860	0.843	0.845	0.836	0.823	0.670
tableeval	0.784	0.756	0.779	0.764	0.728	0.787	0.794	0.741	0.694	0.769	0.623
tat_qa	0.801	0.843	0.777	0.800	0.816	0.791	0.790	0.789	0.787	0.767	0.749
tpcdi_cells	0.443	0.443	0.444	0.444	0.443	0.443	0.426	0.439	0.443	0.424	0.424
valentine	0.970	1.000	1.000	0.990	0.992	1.000	0.968	0.983	0.976	0.976	0.966
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.874	0.874	0.862	0.851	0.856	0.848	0.862	0.841	0.838	0.817	0.740

Table 50: Mean score per dataset and model under rule top2.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.907	0.937	0.917	0.949	0.923	0.917	0.918	0.953	0.941	0.921	0.790
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.319	0.319	0.185	0.238	0.259	0.200	0.128	0.193	0.202	0.082	0.144
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.275	0.247	0.273	0.259	0.234	0.271	0.264	0.201	0.269	0.182	0.167
fintagging	0.868	0.882	0.865	0.843	0.868	0.853	0.858	0.861	0.816	0.823	0.816
hitab	0.745	0.779	0.754	0.790	0.768	0.775	0.747	0.785	0.811	0.744	0.782
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.958	0.967	0.964	0.975	0.949	0.945	0.931	0.904	0.944	0.957	0.686
mmtu	0.896	0.892	0.921	0.854	0.886	0.917	0.904	0.889	0.877	0.833	0.807
officeqa	0.308	0.301	0.260	0.215	0.255	0.190	0.200	0.175	0.100	0.121	0.152
officeqa_pro_v2	0.270	0.343	0.231	0.183	0.210	0.162	0.165	0.189	0.164	0.139	0.164
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.753	0.746	0.731	0.680	0.685	0.709	0.645	0.669	0.660	0.606	0.600
smat	0.904	0.920	0.928	0.918	0.927	0.858	0.925	0.946	0.903	0.901	0.751
suc	0.974	0.984	0.934	0.934	0.901	0.935	0.934	0.901	0.843	0.927	0.718
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.848	0.845	0.857	0.802	0.882	0.861	0.844	0.845	0.836	0.824	0.670
tableeval	0.784	0.756	0.779	0.764	0.728	0.787	0.795	0.741	0.693	0.769	0.623
tat_qa	0.801	0.842	0.777	0.801	0.815	0.791	0.790	0.790	0.787	0.767	0.750
tpcdi_cells	0.449	0.449	0.450	0.450	0.449	0.449	0.431	0.445	0.449	0.430	0.430
valentine	0.970	1.000	1.000	0.990	0.992	1.000	0.968	0.983	0.976	0.976	0.966
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.875	0.875	0.862	0.852	0.856	0.849	0.862	0.841	0.839	0.818	0.741

Table 51: Mean score per dataset and model under rule top3.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.927	0.951	0.937	0.963	0.935	0.938	0.939	0.962	0.959	0.944	0.883
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.270	0.271	0.185	0.199	0.212	0.168	0.116	0.166	0.175	0.086	0.104
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.579	0.524	0.564	0.520	0.460	0.553	0.557	0.345	0.559	0.439	0.247
fintagging	0.921	0.930	0.916	0.905	0.922	0.913	0.911	0.918	0.891	0.895	0.892
hitab	0.750	0.783	0.760	0.793	0.760	0.780	0.744	0.784	0.817	0.756	0.786
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.967	0.975	0.968	0.981	0.960	0.965	0.945	0.916	0.960	0.964	0.813
mmtu	0.890	0.875	0.913	0.847	0.859	0.912	0.898	0.875	0.854	0.826	0.807
officeqa	0.103	0.117	0.093	0.093	0.105	0.063	0.073	0.054	0.055	0.045	0.053
officeqa_pro_v2	0.140	0.211	0.154	0.119	0.114	0.086	0.083	0.071	0.087	0.077	0.075
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.737	0.739	0.717	0.666	0.678	0.689	0.634	0.654	0.663	0.596	0.615
smat	0.934	0.943	0.950	0.939	0.947	0.905	0.945	0.964	0.930	0.929	0.856
suc	0.978	0.988	0.950	0.948	0.919	0.951	0.950	0.924	0.870	0.937	0.758
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.841	0.846	0.847	0.802	0.872	0.855	0.835	0.837	0.829	0.802	0.652
tableeval	0.796	0.771	0.787	0.778	0.736	0.793	0.807	0.743	0.726	0.775	0.646
tat_qa	0.853	0.878	0.798	0.825	0.871	0.812	0.792	0.787	0.822	0.765	0.750
tpcdi_cells	0.451	0.451	0.451	0.451	0.451	0.451	0.438	0.442	0.451	0.431	0.431
valentine	0.970	1.000	1.000	0.990	0.995	1.000	0.977	0.987	0.981	0.981	0.973
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.885	0.881	0.864	0.858	0.869	0.859	0.870	0.841	0.842	0.827	0.754

Table 52: Mean score per dataset and model under rule top4.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.926	0.951	0.937	0.963	0.934	0.937	0.938	0.962	0.959	0.944	0.879
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.270	0.271	0.185	0.198	0.212	0.166	0.116	0.166	0.174	0.085	0.104
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.578	0.520	0.564	0.519	0.460	0.553	0.555	0.349	0.557	0.431	0.252
fintagging	0.921	0.930	0.916	0.905	0.922	0.913	0.911	0.918	0.891	0.895	0.892
hitab	0.749	0.782	0.759	0.791	0.759	0.779	0.743	0.783	0.816	0.755	0.785
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.967	0.975	0.968	0.981	0.960	0.964	0.945	0.917	0.960	0.964	0.805
mmtu	0.890	0.875	0.913	0.847	0.859	0.912	0.898	0.875	0.854	0.826	0.806
officeqa	0.103	0.117	0.093	0.091	0.104	0.063	0.073	0.054	0.055	0.045	0.053
officeqa_pro_v2	0.140	0.211	0.154	0.116	0.114	0.086	0.083	0.071	0.087	0.077	0.075
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.737	0.739	0.717	0.665	0.677	0.689	0.634	0.654	0.662	0.597	0.614
smat	0.934	0.943	0.950	0.939	0.947	0.903	0.945	0.964	0.930	0.928	0.852
suc	0.978	0.988	0.950	0.948	0.919	0.951	0.950	0.924	0.870	0.937	0.758
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.841	0.846	0.847	0.802	0.873	0.855	0.835	0.837	0.829	0.802	0.652
tableeval	0.796	0.771	0.786	0.777	0.735	0.793	0.807	0.742	0.725	0.774	0.645
tat_qa	0.852	0.877	0.796	0.824	0.869	0.811	0.791	0.786	0.821	0.763	0.750
tpcdi_cells	0.453	0.453	0.453	0.453	0.453	0.453	0.439	0.444	0.453	0.432	0.432
valentine	0.970	1.000	1.000	0.990	0.995	1.000	0.977	0.987	0.981	0.981	0.973
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.885	0.880	0.864	0.857	0.868	0.858	0.870	0.840	0.842	0.827	0.753

Table 53: Mean score per dataset and model under rule top5.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.925	0.952	0.934	0.964	0.934	0.933	0.936	0.961	0.960	0.942	0.864
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.274	0.275	0.185	0.198	0.213	0.166	0.119	0.165	0.172	0.084	0.102
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.579	0.513	0.575	0.547	0.511	0.573	0.557	0.473	0.572	0.403	0.417
fintagging	0.921	0.930	0.915	0.904	0.922	0.913	0.911	0.918	0.891	0.894	0.891
hitab	0.750	0.778	0.755	0.785	0.754	0.777	0.741	0.779	0.813	0.750	0.784
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.967	0.974	0.969	0.981	0.958	0.962	0.945	0.921	0.959	0.965	0.776
mmtu	0.894	0.874	0.912	0.848	0.861	0.914	0.899	0.878	0.855	0.827	0.803
officeqa	0.103	0.117	0.093	0.090	0.104	0.063	0.073	0.054	0.055	0.045	0.053
officeqa_pro_v2	0.140	0.213	0.153	0.114	0.114	0.086	0.083	0.070	0.087	0.076	0.074
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.742	0.740	0.721	0.662	0.673	0.694	0.637	0.655	0.664	0.604	0.623
smat	0.932	0.942	0.949	0.938	0.947	0.894	0.943	0.964	0.930	0.925	0.835
suc	0.978	0.988	0.950	0.951	0.919	0.954	0.950	0.924	0.870	0.937	0.758
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.844	0.849	0.849	0.805	0.875	0.858	0.837	0.839	0.831	0.808	0.648
tableeval	0.797	0.772	0.788	0.777	0.735	0.792	0.807	0.742	0.720	0.776	0.643
tat_qa	0.846	0.872	0.795	0.825	0.862	0.808	0.789	0.788	0.818	0.759	0.756
tpcdi_cells	0.500	0.500	0.500	0.500	0.500	0.500	0.485	0.491	0.500	0.480	0.480
valentine	0.970	1.000	1.000	0.990	0.995	1.000	0.978	0.987	0.982	0.982	0.972
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.887	0.885	0.869	0.862	0.868	0.863	0.872	0.844	0.848	0.831	0.753

Table 54: Mean score per dataset and model under rule v1.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Son-net 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.918	0.945	0.928	0.957	0.929	0.928	0.929	0.958	0.951	0.934	0.840
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.293	0.293	0.185	0.216	0.233	0.183	0.122	0.178	0.187	0.084	0.122
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.445	0.401	0.438	0.413	0.373	0.433	0.428	0.312	0.434	0.322	0.253
fintagging	0.897	0.909	0.893	0.878	0.898	0.887	0.887	0.892	0.858	0.863	0.858
hitab	0.748	0.781	0.757	0.792	0.763	0.778	0.746	0.784	0.814	0.750	0.784
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.963	0.971	0.967	0.978	0.955	0.956	0.939	0.912	0.953	0.961	0.753
mmtu	0.894	0.882	0.917	0.850	0.872	0.914	0.901	0.882	0.865	0.830	0.807
officeqa	0.194	0.199	0.168	0.147	0.171	0.119	0.129	0.107	0.075	0.079	0.097
officeqa_pro_v2	0.198	0.270	0.188	0.148	0.157	0.120	0.119	0.123	0.121	0.104	0.115
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.746	0.743	0.724	0.672	0.681	0.699	0.640	0.661	0.662	0.603	0.611
smat	0.921	0.933	0.940	0.930	0.938	0.883	0.936	0.956	0.918	0.916	0.806
suc	0.976	0.986	0.943	0.942	0.911	0.945	0.943	0.914	0.858	0.932	0.740
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.845	0.846	0.852	0.803	0.877	0.858	0.839	0.841	0.833	0.813	0.660
tableeval	0.791	0.765	0.784	0.773	0.733	0.790	0.802	0.742	0.711	0.773	0.636
tat_qa	0.829	0.861	0.789	0.815	0.845	0.802	0.791	0.789	0.806	0.765	0.752
tpcdi_cells	0.462	0.462	0.462	0.462	0.462	0.462	0.447	0.455	0.462	0.442	0.442
valentine	0.970	1.000	1.000	0.990	0.994	1.000	0.973	0.985	0.979	0.979	0.970
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.881	0.879	0.865	0.857	0.863	0.855	0.867	0.842	0.843	0.824	0.749

Table 55: Standard error of the per-dataset mean under v1.

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.014	0.000	0.000	0.017	0.017	0.014	0.020	0.020	0.020	0.024	0.020
alaska_camera	0.031	0.038	0.000	0.010	0.014	0.039	0.024	0.017	0.000	0.000	0.010
alaska_schema	0.026	0.021	0.024	0.017	0.025	0.024	0.023	0.019	0.018	0.022	0.028
amazon_google	0.026	0.024	0.024	0.022	0.020	0.026	0.024	0.026	0.020	0.035	0.022
bird	0.037	0.036	0.034	0.033	0.033	0.034	0.027	0.034	0.031	0.023	0.023
dblp_acm	0.000	0.000	0.000	0.010	0.000	0.000	0.000	0.010	0.000	0.029	0.017
dblp_scholar	0.026	0.022	0.020	0.022	0.017	0.024	0.022	0.024	0.020	0.029	0.017
fetaqa	0.012	0.014	0.013	0.016	0.014	0.013	0.013	0.012	0.014	0.010	0.014
fintagging	0.022	0.021	0.023	0.024	0.022	0.023	0.024	0.022	0.025	0.024	0.025
hitab	0.039	0.037	0.039	0.037	0.038	0.037	0.039	0.037	0.035	0.038	0.036
machamp	0.030	0.030	0.030	0.030	0.030	0.031	0.031	0.031	0.031	0.038	0.034
magneto_gdc	0.017	0.015	0.016	0.012	0.019	0.016	0.021	0.025	0.018	0.017	0.032
mmtu	0.028	0.030	0.026	0.034	0.030	0.026	0.027	0.029	0.031	0.034	0.035
officeqa	0.019	0.022	0.019	0.020	0.022	0.017	0.020	0.014	0.020	0.016	0.014
officeqa_pro_v2	0.025	0.033	0.032	0.027	0.025	0.021	0.021	0.016	0.019	0.021	0.018
opensanctions_pairs	0.035	0.024	0.043	0.036	0.044	0.017	0.036	0.030	0.043	0.041	0.041
papadakis_dn	0.024	0.027	0.022	0.024	0.022	0.026	0.022	0.022	0.024	0.026	0.024
realhitbench	0.037	0.038	0.039	0.041	0.041	0.041	0.043	0.042	0.042	0.044	0.041
smat	0.022	0.021	0.019	0.022	0.020	0.027	0.021	0.016	0.023	0.023	0.029
suc	0.014	0.011	0.020	0.020	0.025	0.019	0.020	0.024	0.031	0.022	0.038
tab_fact	0.035	0.047	0.037	0.042	0.044	0.038	0.036	0.038	0.037	0.038	0.041
tabis	0.022	0.020	0.020	0.017	0.027	0.022	0.014	0.030	0.042	0.027	0.038
tablebench	0.032	0.033	0.032	0.035	0.030	0.031	0.034	0.033	0.034	0.035	0.040
tableeval	0.039	0.040	0.039	0.040	0.040	0.039	0.038	0.041	0.040	0.040	0.042
tat_qa	0.031	0.030	0.033	0.033	0.029	0.033	0.034	0.035	0.034	0.036	0.037
tpcdi_cells	0.042	0.042	0.042	0.042	0.042	0.042	0.041	0.041	0.042	0.041	0.041
valentine	0.017	0.000	0.000	0.010	0.006	0.000	0.014	0.011	0.013	0.013	0.016
walmart_amazon	0.010	0.014	0.017	0.020	0.022	0.017	0.020	0.022	0.022	0.024	0.026
wdc_lspc	0.014	0.010	0.000	0.000	0.014	0.014	0.000	0.017	0.020	0.014	0.020
wdc_products	0.010	0.014	0.014	0.014	0.017	0.014	0.010	0.017	0.017	0.014	0.017
wikitablequestions	0.030	0.030	0.032	0.032	0.032	0.032	0.030	0.034	0.034	0.034	0.039

Table 56: Per-family optimum and the family's J under its own optimum, the global best and v1.

family	datasets	best F1 / decay / prec / rec	J own best	J global best	J v1	L1 to global best
EM	11	0.35 / 0.60 / 0.05 / 0.00	0.358	0.358	0.358	0.000
SM	6	0.00 / 1.00 / 0.00 / 0.00	0.549	0.535	0.512	0.800
TQA	11	0.05 / 0.05 / 0.25 / 0.65	0.494	0.483	0.483	1.700
FIN	3	0.95 / 0.05 / 0.00 / 0.00	0.879	0.772	0.771	1.200

Table 57: The 30 best weightings with their bootstrap statistics (full plateau table in data/selection/plateau.csv).

rank_full	w_f1	w_decay	w_precision	w_recall	J_full	J_boot_mean	J_boot_sd	J_boot_lo	J_boot_hi	p_vs_best	J_oob_mean	times_selected
1	0.350	0.600	0.050	0.000	0.425	0.361	0.026	0.311	0.413	1.000	0.335	4
2	0.300	0.600	0.100	0.000	0.424	0.361	0.026	0.310	0.412	0.582	0.335	3
3	0.600	0.150	0.000	0.250	0.424	0.363	0.025	0.314	0.414	0.573	0.334	1
4	0.650	0.150	0.000	0.200	0.424	0.363	0.025	0.314	0.412	0.573	0.334	0
5	0.300	0.150	0.400	0.150	0.424	0.364	0.025	0.315	0.414	0.581	0.335	1
6	0.000	0.250	0.150	0.600	0.423	0.364	0.025	0.315	0.413	0.587	0.336	0
7	0.250	0.150	0.050	0.550	0.423	0.362	0.025	0.314	0.411	0.510	0.333	1
8	0.150	0.250	0.350	0.250	0.423	0.366	0.026	0.316	0.414	0.660	0.338	4
9	0.100	0.750	0.000	0.150	0.423	0.357	0.026	0.306	0.410	0.281	0.331	0
10	0.000	0.750	0.050	0.200	0.423	0.357	0.026	0.307	0.410	0.310	0.331	1
11	0.500	0.000	0.350	0.150	0.422	0.358	0.026	0.308	0.406	0.416	0.329	3
12	0.550	0.000	0.350	0.100	0.422	0.358	0.026	0.307	0.409	0.423	0.329	3
20	0.400	0.150	0.250	0.200	0.422	0.365	0.025	0.317	0.414	0.636	0.336	1
26	0.200	0.150	0.500	0.150	0.422	0.363	0.026	0.313	0.412	0.540	0.333	1
25	0.550	0.150	0.050	0.250	0.422	0.363	0.025	0.314	0.413	0.580	0.334	0
24	0.500	0.150	0.150	0.200	0.422	0.364	0.025	0.315	0.413	0.613	0.335	0
23	0.450	0.150	0.200	0.200	0.422	0.365	0.025	0.317	0.412	0.614	0.336	1
22	0.400	0.150	0.200	0.250	0.422	0.364	0.025	0.315	0.413	0.615	0.335	0
21	0.350	0.150	0.250	0.250	0.422	0.365	0.025	0.315	0.413	0.602	0.336	1
18	0.300	0.150	0.350	0.200	0.422	0.365	0.025	0.316	0.414	0.627	0.336	1
19	0.350	0.150	0.300	0.200	0.422	0.365	0.025	0.316	0.414	0.643	0.336	1
16	0.250	0.150	0.350	0.250	0.422	0.365	0.025	0.315	0.415	0.630	0.336	3
15	0.200	0.150	0.400	0.250	0.422	0.365	0.025	0.315	0.415	0.605	0.335	1
14	0.450	0.150	0.150	0.250	0.422	0.363	0.025	0.314	0.413	0.583	0.335	0
13	0.150	0.200	0.400	0.250	0.422	0.365	0.025	0.316	0.415	0.643	0.336	2
17	0.300	0.150	0.300	0.250	0.422	0.365	0.025	0.316	0.415	0.633	0.336	1
30	0.100	0.150	0.150	0.600	0.422	0.362	0.025	0.313	0.411	0.516	0.334	0
33	0.150	0.150	0.150	0.550	0.422	0.362	0.025	0.313	0.412	0.526	0.334	0
32	0.050	0.150	0.200	0.600	0.422	0.362	0.025	0.314	0.411	0.532	0.334	0
31	0.200	0.150	0.100	0.550	0.422	0.362	0.025	0.313	0.412	0.516	0.334	0

Table 58: Bradley–Terry rank (Borda rank in brackets) under each reported rule, with the Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
exact match	3 (3)	1 (1)	2 (2)	4 (4)	7 (5)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	2.7e-15	0.28
v1	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	1.3e-14	0.27
top1	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	7.5e-15	0.28
top2	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	8.0e-15	0.28
top3	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (5)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	5.9e-15	0.28
top4	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (5)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	5.9e-15	0.28
top5	1 (3)	2 (2)	4 (1)	3 (4)	5 (5)	6 (6)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	6.1e-15	0.28
best-rho	1 (3)	2 (2)	4 (1)	3 (4)	5 (5)	6 (6)	8 (7)	7 (8)	9 (9)	10 (10)	11 (11)	7.2e-14	0.26
knee	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (6)	8 (8)	9 (9)	10 (10)	11 (11)	7.5e-15	0.28

Table 59: Anchor correctness per dataset under three weightings.

dataset	v1	best J
abt_buy	0.860	0.860
alaska_camera	0.857	0.857
alaska_schema	0.877	0.888
amazon_google	0.863	0.863
bird	0.889	0.874
dblp_acm	0.860	0.860
dblp_scholar	0.858	0.858
fetaqa	0.913	0.911
fintagging	0.874	0.825
hitab	0.906	0.889
machamp	0.857	0.857
magneto_gdc	0.872	0.881
mmtu	0.868	0.868
officeqa	0.894	0.904
officeqa_pro_v2	0.885	0.896
opensanctions_pairs	0.860	0.860
papadakis_dn	0.891	0.891
realhitbench	0.899	0.891
smat	0.914	0.921
suc	0.857	0.854
tab_fact	0.862	0.862
tabis	0.639	0.639
tablebench	0.898	0.886
tableeval	0.866	0.863
tat_qa	0.885	0.896
tpcdi_cells	0.905	0.900
valentine	0.846	0.851
walmart_amazon	0.867	0.867
wdc_lspc	0.856	0.856
wdc_products	0.856	0.856
wikitablequestions	0.877	0.867

Table 60: Kendall τ between the Bradley–Terry rankings of the rules.

rule	exact match	v1	top1	top2	top3	top4	top5	best-rho	knee
exact match	1.000	0.855	0.855	0.855	0.855	0.855	0.818	0.782	0.855
v1	0.855	1.000	1.000	1.000	1.000	1.000	0.964	0.927	1.000
top1	0.855	1.000	1.000	1.000	1.000	1.000	0.964	0.927	1.000
top2	0.855	1.000	1.000	1.000	1.000	1.000	0.964	0.927	1.000
top3	0.855	1.000	1.000	1.000	1.000	1.000	0.964	0.927	1.000
top4	0.855	1.000	1.000	1.000	1.000	1.000	0.964	0.927	1.000
top5	0.818	0.964	0.964	0.964	0.964	0.964	1.000	0.964	0.964
best-rho	0.782	0.927	0.927	0.927	0.927	0.927	0.964	1.000	0.927
knee	0.855	1.000	1.000	1.000	1.000	1.000	0.964	0.927	1.000

Table 61: Run facts per model from the prediction cache.

model	answers	cost_usd	mean_out_tokens	median_latency_s	reasked_2500	reasked_8000	reasked_16000	still_cut
Gemini 3.8 Flash	3083	3.36	262.26	1.28	168	18	0	11
Claude Opus 5	3282	99.75	246.81	2.16	144	38	13	2
GPT-5.5	3283	19.89	169.63	1.91	150	3	0	0
DeepSeek-R1	3083	12.83	1757.11	1.27	243	166	96	125
Claude Sonnet 5	3282	15.01	146.63	1.31	78	36	8	3
Gemini 3.1 Pro	3083	11.80	276.84	3.58	118	1	0	10
Grok 4	3083	8.33	36.95	3.27	0	0	0	0
GPT-5.4-mini	3282	4.22	125.36	0.87	127	4	0	0
DeepSeek-chat	3083	0.62	50.73	0.96	16	1	0	1
gpt-oss-120b	3083	0.70	233.13	1.61	120	12	1	0
Llama 3.3 70B	3083	1.77	82.25	1.40	16	0	0	0

Table 62: One-step sensitivity around v1 and the best weighting.

point	n_neighbours	J_centre	J_min	J_max	J_mean	flatness_range	share_neighbours_better
v1	12.0000	0.4110	0.4063	0.4187	0.4121	0.0124	0.6154
best	9.0000	0.4249	0.4131	0.4249	0.4179	0.0118	0.0000